

TacoCloud – Retos Funcionales

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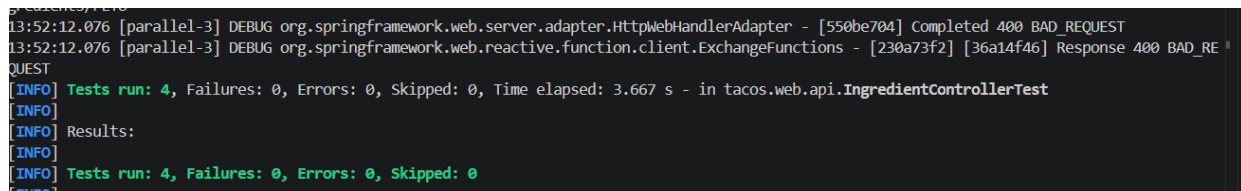
Laboratorio 1 – Cazar operaciones fantasma

TC-01 – Actualizar un ingrediente sin perder el Publisher

Dentro de `tacocloud-api/src/main/java/tacos/web/api/IngredientController.java` se modificó la petición put para cumplir con las pruebas de integración:

```
//TC-01 - Actualizar un ingrediente sin perder el publisher
@PutMapping("/{id}")
public Mono<ResponseEntity<Ingredient>> updateIngredient(@PathVariable String id,
@RequestBody Ingredient ingredient) {
    if (ingredient.getId() != null && !ingredient.getId().equals(id)) {
        return Mono.just(ResponseEntity.badRequest().build());
    }
    ingredient.setId(id);
    return repo.findById(id)
        .flatMap(updatedIngredient -> repo.save(ingredient))
        .map(ResponseEntity::ok)
        .defaultIfEmpty(ResponseEntity.notFound().build());
}
//TC-01 – Fin
```

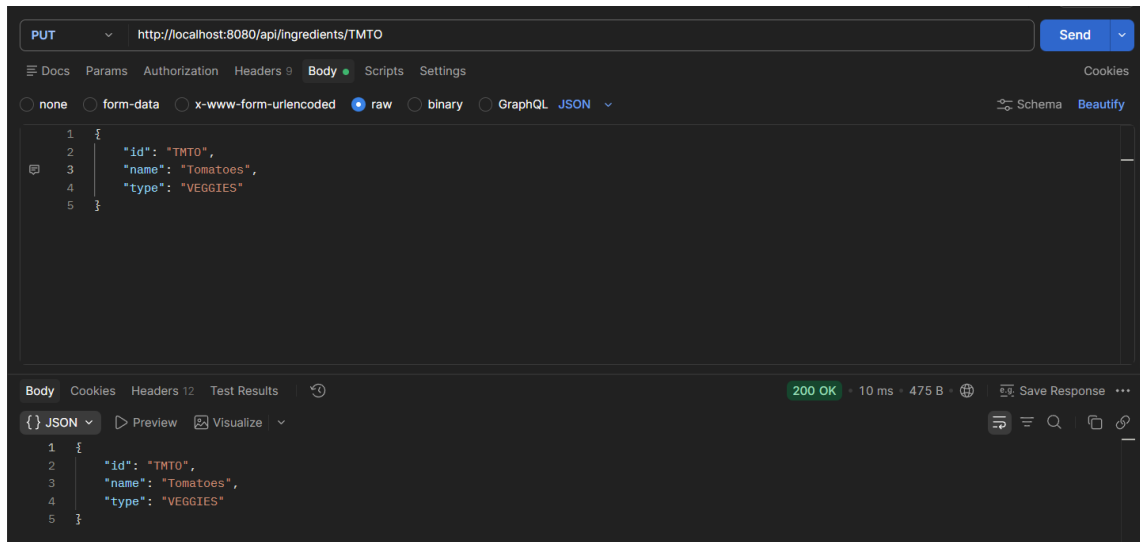
Test



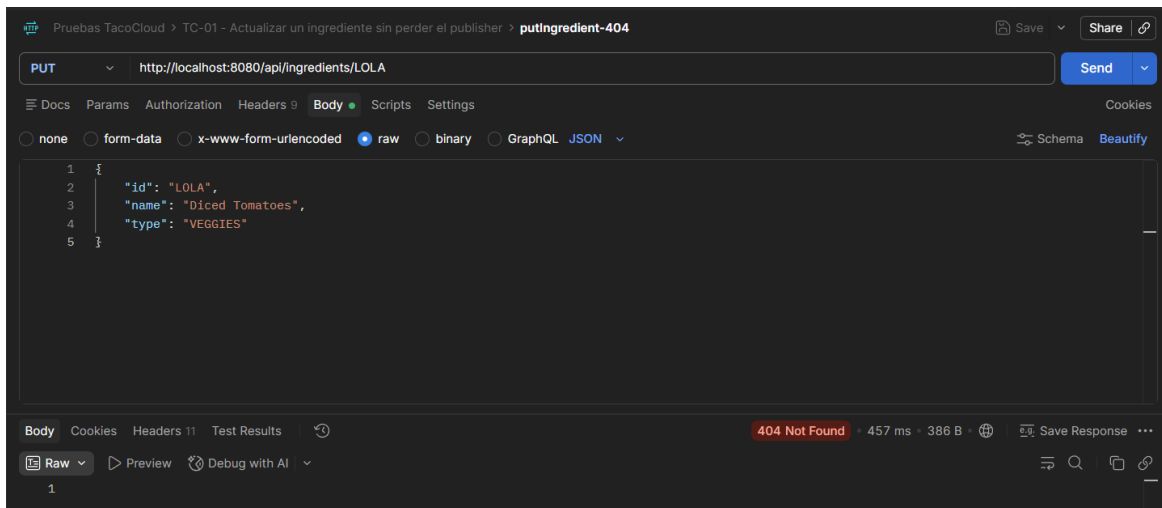
```
13:52:12.076 [parallel-3] DEBUG org.springframework.web.server.adapter.HttpWebHandlerAdapter - [550be704] Completed 400 BAD_REQUEST
13:52:12.076 [parallel-3] DEBUG org.springframework.web.reactive.function.client.ExchangeFunctions - [230a73f2] [36a14f46] Response 400 BAD_RE
QUEST
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.667 s - in tacos.web.api.IngredientControllerTest
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
```

Postman

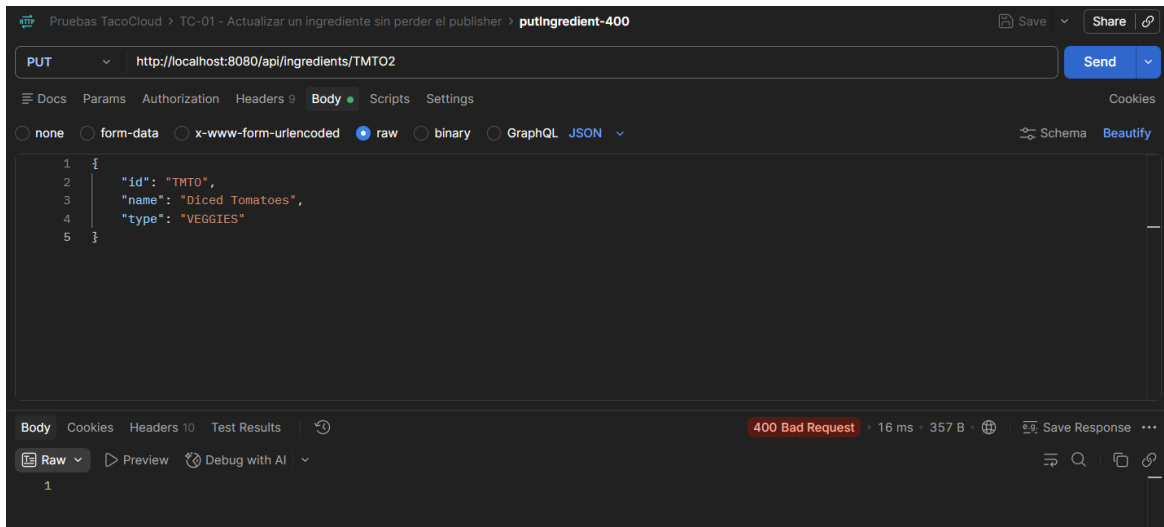
Status 200 con la entidad actualizada



Status 404 cuando el ingrediente no existe



Status 400 cuando el ID es inconsistente



TC-02 – Eliminar de verdad y responder con semántica HTTP

//TC-02 - Eliminar de verdad y responder con semantica HTTP

```
@DeleteMapping("/{id}")
public Mono<ResponseEntity<Void>> deleteIngredient(@PathVariable String id){
    if(id == null || id.isEmpty()){
        return Mono.just(ResponseEntity.badRequest().build());
    }
    return repo.findById(id)
        .flatMap(ingredient -> repo.deleteById(ingredient.getId()))
        .then(Mono.just(ResponseEntity.noContent().<Void>build()))
        .defaultIfEmpty(ResponseEntity.notFound().build());
}
```

//TC-02 - Fin

Test

```
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.294 s - in tacos.web.api.IngredientControllerTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman

Eliminando ingrediente TMTO (mostrando su existencia)

Pruebas TacoCloud > TC-01 - Actualizar un ingrediente sin perder el publisher > **getIngredient** Save Share

GET http://localhost:8080/api/ingredients/TMTO Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body Cookies Headers 12 Test Results 200 OK · 414 ms · 483 B Save Response

{ } JSON Preview Visualize

```
1 {
2   "id": "TMTO",
3   "name": "Diced Tomatoes",
4   "type": "VEGGIES"
5 }
```

Pruebas TacoCloud > TC-02 - Eliminar de verdad con semántica HTTP > **deleteIngredient-204** Save Share

DELETE http://localhost:8080/api/ingredients/TMTO Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body Cookies Headers 10 Test Results 204 No Content · 534 ms · 369 B Save Response

Raw Preview Visualize

```
1
```

Eliminando un ingrediente que no existe

Pruebas TacoCloud > TC-02 - Eliminar de verdad con semántica HTTP > **deleteIngredient-204** Save Share

DELETE http://localhost:8080/api/ingredients/TMTO Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body Cookies Headers 11 Test Results 404 Not Found · 34 ms · 388 B Save Response

Raw Preview Debug with AI

```
1
```

TC-03 – Construir Location sin localhost ni rutas rotas

//TC-03 - Construir Location sin localhost ni rutas rotas

@PostMapping

```
public Mono<ResponseEntity<Ingredient>> postIngredient(@RequestBody @Valid Ingredient
ingredient, UriComponentsBuilder ucb) {
    return Mono.just(ingredient)
        .flatMap(repo::save)
        .map(i -> {
            URI location = ucb.path("/api/ingredients/{id}")
                .buildAndExpand(i.getId()).toUri();
            return ResponseEntity.created(location).body(i);
        });
}
```

//TC-03 – Fin

Test

```
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 4.827 s - in tacos.web.api.IngredientControllerTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0
```

Postman

Creando un ingrediente

Pruebas TacoCloud > TC-03 - Construir Location sin localhost ni rutas rotas > **postingredient-201** Save Share

POST `http://localhost:8080/api/ingredients` Send

Docs Params Authorization Headers 9 **Body** Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beautify

```
1 {
2   "id": "ONIO",
3   "name": "onion",
4   "type": "VEGGIES"
5 }
```

Body Cookies Headers 13 Test Results 201 Created 480 ms 512 B Save Response

JSON Preview Visualize

```
1 {
2   "id": "ONIO",
3   "name": "onion",
4   "type": "VEGGIES"
5 }
```

Pruebas TacoCloud > TC-03 - Construir Location sin localhost ni rutas rotas > **postingredient-201** Save Share

POST `http://localhost:8080/api/ingredients` Send

Docs Params Authorization Headers 9 **Body** Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON Schema Beautify

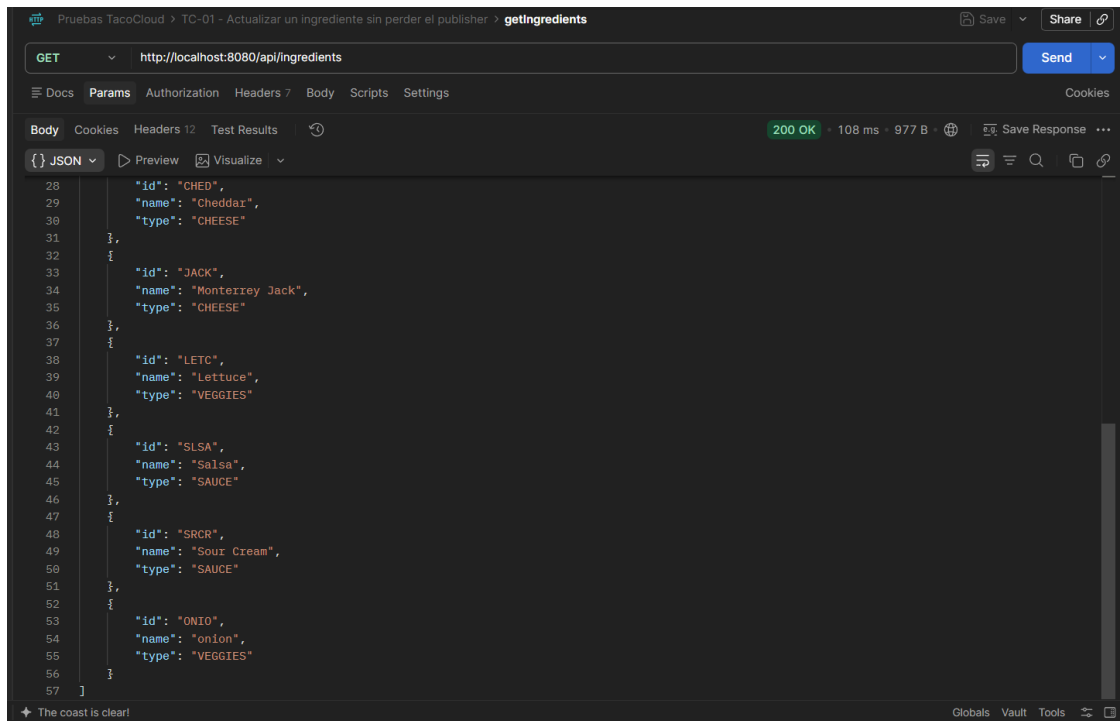
```
1 {
2   "id": "ONIO",
3   "name": "onion",
4   "type": "VEGGIES"
5 }
```

Body Cookies **Headers 12** Test Results 201 Created 333 ms 451 B Save Response

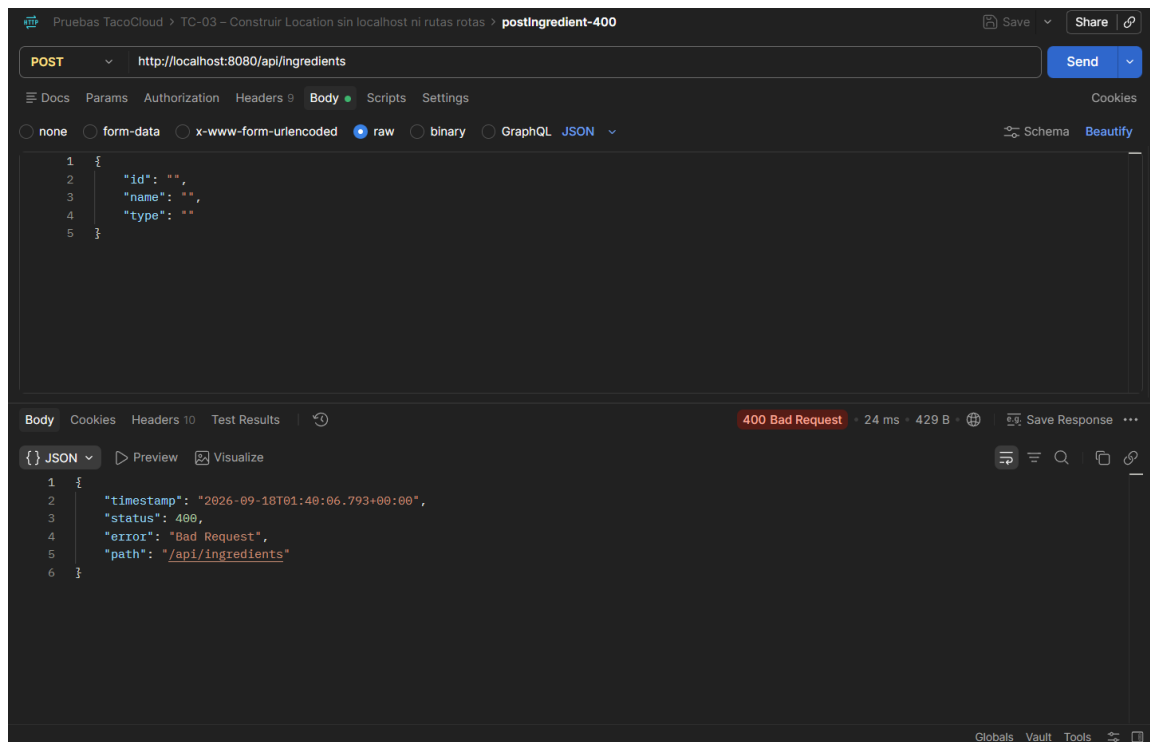
Key	Value
X-Content-Type-Options	nosniff
X-Xss-Protection	1; mode=block
Cache-Control	no-cache, no-store, max-age=0, must-revalidate
Pragma	no-cache
Expires	0
X-Frame-Options	SAMEORIGIN
Location	<code>http://localhost:8080/api/ingredients/ONIO</code>
Content-Type	<code>application/json</code>
Transfer-Encoding	chunked
Date	Fri, 18 Sep 2026 01:44:35 GMT
Keep-Alive	timeout=60
Connection	keep-alive

Globals Vault Tools

Listando el ingrediente para verificar que se creó correctamente:



Cuando se envía un ingrediente inválido:



TC-04 – PATCH de órdenes con lista blanca y sin ZIP mutante

Cambios dentro de *OrderApiController.java*

```
//TC-04 - PATCH de ordenes con lista blanca y sin ZIP mutante
@PatchMapping(path="/{orderId}", consumes="application/json")
public Mono<TacoOrder> patchOrder(@PathVariable("orderId") String orderId,
    @RequestBody OrderPatchRequest patch) {

    return repo.findById(orderId)
        .switchIfEmpty(Mono.error(new ResponseStatusException(HttpStatus.NOT_FOUND)))
        .map(order -> {
            if (patch.getDeliveryName() != null) {
                order.setDeliveryName(patch.getDeliveryName());
            }
            if (patch.getDeliveryStreet() != null) {
                order.setDeliveryStreet(patch.getDeliveryStreet());
            }
            if (patch.getDeliveryCity() != null) {
                order.setDeliveryCity(patch.getDeliveryCity());
            }
            if (patch.getDeliveryState() != null) {
                order.setDeliveryState(patch.getDeliveryState());
            }
            if (patch.getDeliveryZip() != null) {
                order.setDeliveryZip(patch.getDeliveryZip());
            }
            return order;
        })
        .flatMap(repo::save);
}

//TC-04 - Fin
```

Se creó un DTO para guardar los campos necesarios

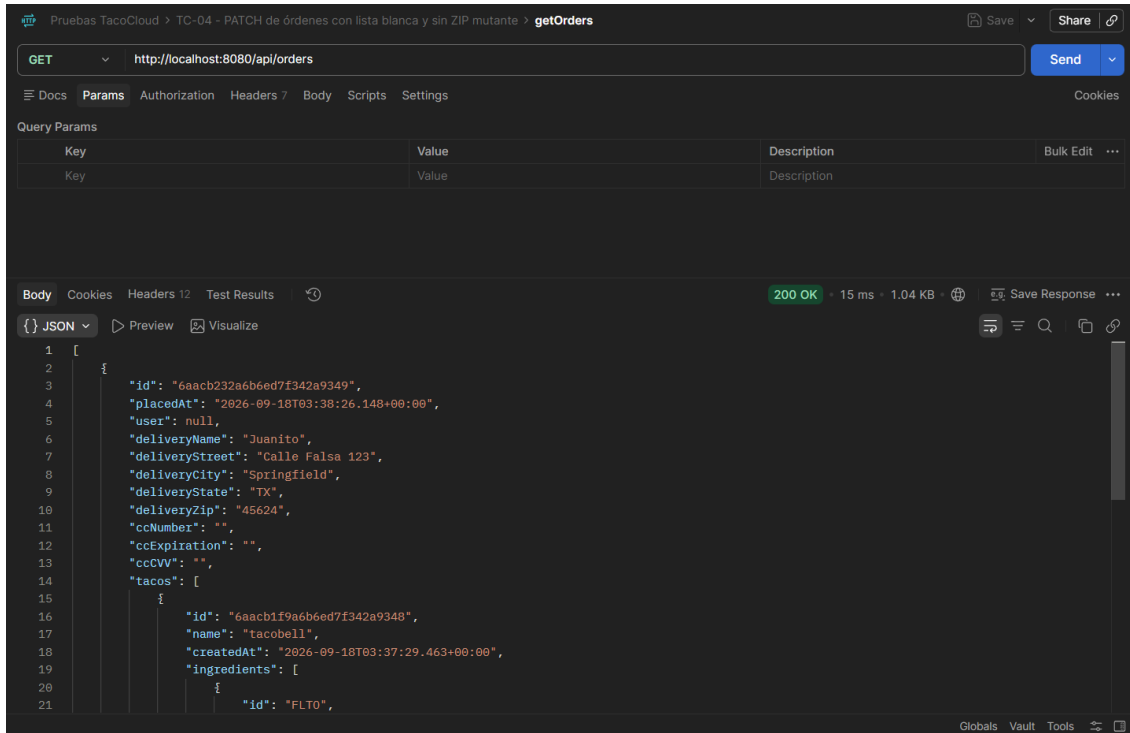
```
@Data
public class OrderPatchRequest {
    private String deliveryName;
    private String deliveryStreet;
    private String deliveryCity;
    private String deliveryState;
    private String deliveryZip;
}
```

Test

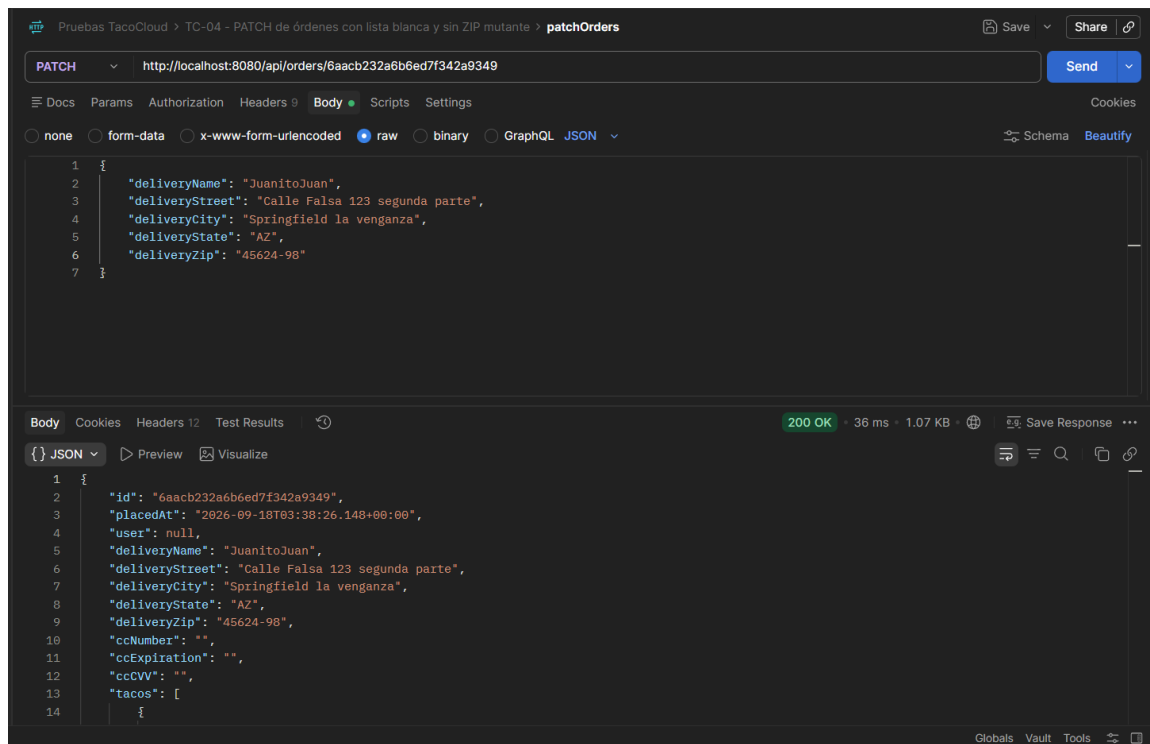
```
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.969 s - in tacos.web.api.OrderApiControllerTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
```

Postman

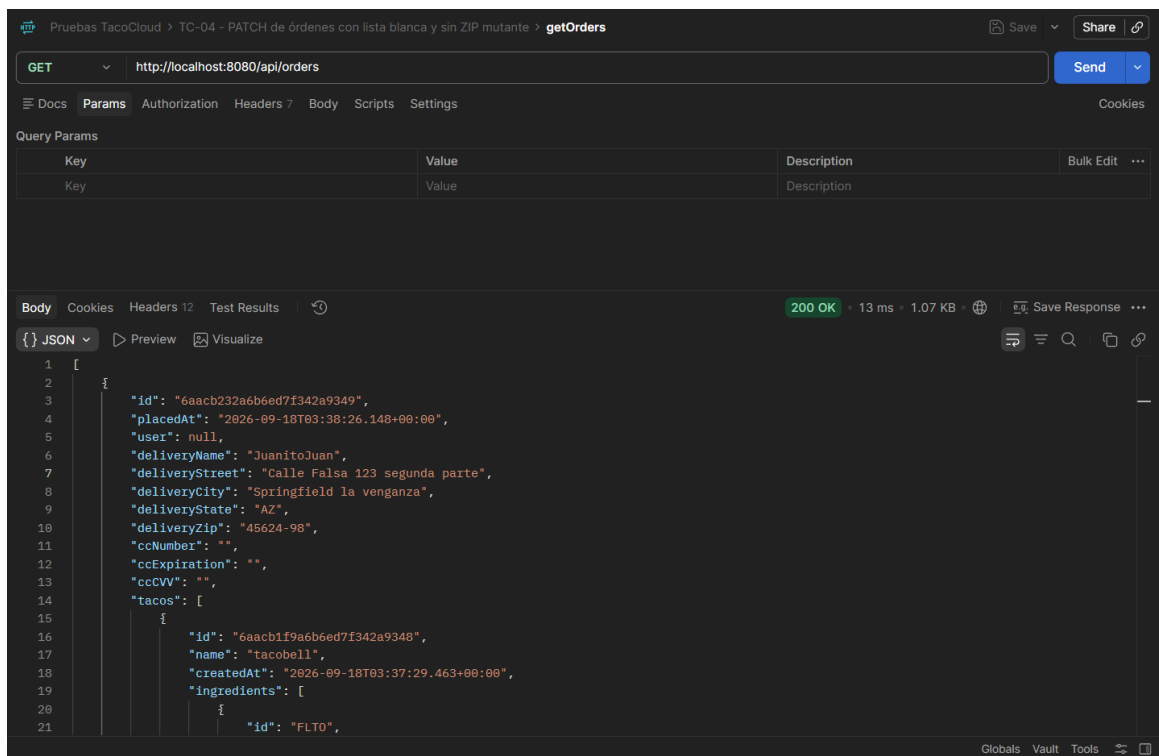
Primero se creó una orden:



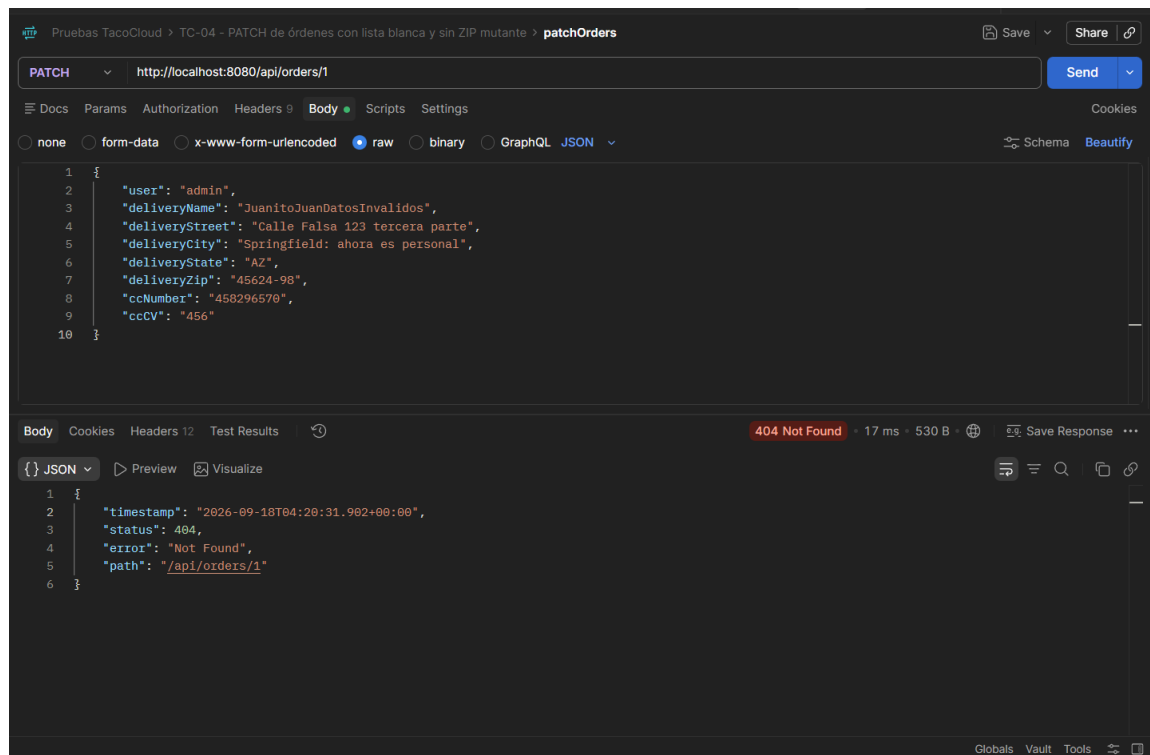
Utilizando el patch correctamente:



Obteniendo la orden para verificar los cambios:



Realizando cambios a campos que NO se deben modificar:



TC-05 – PUT y DELETE de órdenes con identidad consistente

OrderApiController.java

Put

```

/*
 * TC-05 - PUT y DELETE de ordenes con identidad consistente
 * caso PUT
 */
@PutMapping(path =("/{orderId}", consumes = "application/json")
public Mono<ResponseEntity<TacoOrder>> putOrder(@PathVariable("orderId") String orderId,
    @RequestBody TacoOrder order) {
    return repo.findById(orderId)
        .map(existingOrder -> {
            existingOrder.setDeliveryName(order.getDeliveryName());
            existingOrder.setDeliveryStreet(order.getDeliveryStreet());
            existingOrder.setDeliveryCity(order.getDeliveryCity());
            existingOrder.setDeliveryState(order.getDeliveryState());
            existingOrder.setDeliveryZip(order.getDeliveryZip());
            existingOrder.setTacos(order.getTacos());
            return existingOrder;
        })
        .flatMap(repo::save)
        .map(ResponseEntity::ok)

```

```

        .defaultIfEmpty(ResponseEntity.notFound().build());
    }
    // TC-05 - Fin caso PUT

```

Delete

```

/*
 * TC-05 - PUT y DELETE de ordenes con identidad consistente
 * caso DELETE
 */
@DeleteMapping("/{orderId}")
public Mono<ResponseEntity<Void>> deleteOrder(@PathVariable("orderId") String orderId) {
    return repo.findById(orderId)
        .flatMap(order -> repo.deleteById(order.getId()))
        .then(Mono.just(ResponseEntity.noContent().<Void>build()))
        .defaultIfEmpty(ResponseEntity.notFound().build());
}
// TC-05 - Fin caso DELETE

```

Test

```

At=Sat Sep 26 14:58:11 CST 2020, user=null, deliveryname=null, deliveryst (truncated)...]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 4.314 s - in tacos.web.api.OrderApiControllerTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0

```

Postman

Primero se crea una orden:

Pruebas TacoCloud > TC-04 - PATCH de órdenes con lista blanca y sin ZIP mutante > **getOrders** Save Share

GET ▼ Send ▼

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit	...
Key	Value	Description		

Body Cookies Headers 12 Test Results 200 OK 23 ms 1.05 KB Save Response ...

{} JSON ▼ Preview Visualize

```
1 [
2   {
3     "id": "6aacd5515ff9446651333474",
4     "placedAt": "2026-09-18T06:08:17.138+00:00",
5     "user": null,
6     "deliveryName": "LuisMi",
7     "deliveryStreet": "Calle13",
8     "deliveryCity": "Buenos Aires",
9     "deliveryState": "AR",
10    "deliveryZip": "78542",
11    "ccNumber": "123456789",
12    "ccExpiration": "09/28",
13    "ccCVV": "123",
14    "tacos": [
15      {
16        "id": "6aacd50f5ff9446651333473",
17        "name": "tacobell",
18        "createdAt": "2026-09-18T06:07:11.957+00:00",
19        "ingredients": [
20          {
21            "id": "GRBF",
```

Globals Vault Tools

Actualizando la orden con PUT:

Pruebas TacoCloud > TC-05 - PUT y DELETE de ordenes con identidad consistente > **putOrders-200** Save Share

PUT ▼ Send ▼

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL JSON ▼ Schema Beautify

```
1 {
2   "deliveryName": "LuisMiguela",
3   "deliveryStreet": "Calle13",
4   "deliveryCity": "Buenos Aires",
5   "deliveryState": "AR",
6   "deliveryZip": "78542"
7 }
```

Body Cookies Headers 12 Test Results 200 OK 32 ms 717 B Save Response ...

{} JSON ▼ Preview Visualize

```
1 {
2   "id": "6aacd5515ff9446651333474",
3   "placedAt": "2026-09-18T06:08:17.138+00:00",
4   "user": null,
5   "deliveryName": "LuisMiguela",
6   "deliveryStreet": "Calle13",
7   "deliveryCity": "Buenos Aires",
8   "deliveryState": "AR",
9   "deliveryZip": "78542",
10  "ccNumber": "123456789",
11  "ccExpiration": "09/28",
12  "ccCVV": "123",
13  "tacos": []
14 }
```

Globals Vault Tools

Colocando datos que no se pueden modificar:

Pruebas TacoCloud > TC-05 - PUT y DELETE de ordenes con identidad consistente > putOrders

PUT http://localhost:8080/api/orders/6aacd5515ff9446651333474 Send

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "user": "admin",
3   "deliveryName": "LuisMiguela",
4   "deliveryStreet": "Calle13",
5   "deliveryCity": "Buenos Aires",
6   "deliveryState": "AR",
7   "deliveryZip": "78542",
8   "ccNumber": "987654321",
9   "ccExpiration": "12/27",
10  "ccCVV": "456"
11 }
```

Body Cookies Headers 11 Test Results 400 Bad Request · 30 ms · 526 B Save Response

{ JSON Preview Visualize

```
1 {
2   "timestamp": "2026-09-18T06:19:45.997+00:00",
3   "status": 400,
4   "error": "Bad Request",
5   "path": "/api/orders/6aacd5515ff9446651333474"
6 }
```

Borrando la orden anterior:

Pruebas TacoCloud > TC-05 - PUT y DELETE de ordenes con identidad consistente > deleteOrders

DELETE http://localhost:8080/api/orders/6aacd5515ff9446651333474 Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Query Params

Key	Value	Description	Bulk Edit
Key	Value	Description	

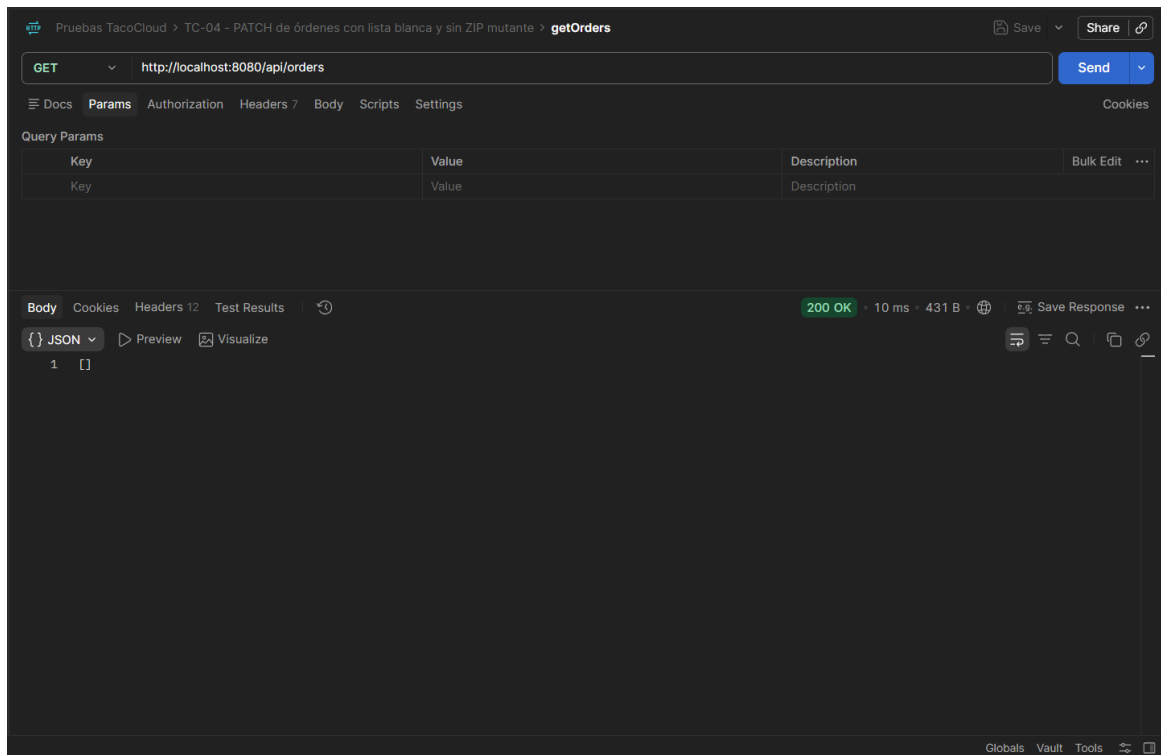
Body Cookies Headers 10 Test Results 204 No Content · 33 ms · 369 B Save Response

Raw Preview Visualize

```
1
```

Globals Vault Tools

Verificando si se eliminó:



TC-06 – Convertir órdenes de correo sin carreras ni nulls sorpresa

Dentro de `EmailOrderService.java`

```
//TC-06 - Convertir ordenes de correo sin carreras ni nulls sorpresa
public Mono<TacoOrder> convertEmailOrderToDomainOrder(Mono<EmailOrder> emailOrder)
{
    return emailOrder
        .switchIfEmpty(Mono.error(new IllegalArgumentException("Email order cannot be empty")))
        .flatMap(eOrder -> userRepo.findByEmail(eOrder.getEmail()))
        .switchIfEmpty(Mono.error(new IllegalArgumentException("User not found for email: " +
eOrder.getEmail())))
        .flatMap(user -> paymentMethodRepo.findById(user.getId()))
        .switchIfEmpty(Mono.error(new IllegalArgumentException("Payment method not found for
user: " + user.getId())))
        .flatMap(paymentMethod -> Mono.justOrEmpty(eOrder.getTacos()))
        .switchIfEmpty(Mono.error(new IllegalArgumentException("Email order has no taco
list")))
        .flatMapMany(Flux::fromIterable)
        .concatMap(emailTaco -> Mono.justOrEmpty(emailTaco.getIngredients()))
        .switchIfEmpty(Mono.error(new IllegalArgumentException("Taco "" +
emailTaco.getName() + "" has no ingredient list")))
        .flatMapMany(Flux::fromIterable)
        .concatMap(ingredientId -> ingredientRepo
            .findById(ingredientId))
```

```

        .switchIfEmpty(Mono.error(new IllegalArgumentException("Ingredient not found for id:
" + ingredientId))))
        .collectList()
        .map(ingredients -> {
            Taco taco = new Taco();
            taco.setName(emailTaco.getName());
            taco.setIngredients(ingredients);
            return taco;
        })
    )
    .collectList()
    .map(tacos -> {
        TacoOrder order = new TacoOrder();
        order.setUser(user);
        order.setCcNumber(paymentMethod.getCcNumber());
        order.setCcCVV(paymentMethod.getCcCVV());
        order.setCcExpiration(paymentMethod.getCcExpiration());
        order.setDeliveryName(user.getFullname());
        order.setDeliveryStreet(user.getStreet());
        order.setDeliveryCity(user.getCity());
        order.setDeliveryState(user.getState());
        order.setDeliveryZip(user.getZip());
        order.setPlacedAt(new Date());
        tacos.forEach(order::addTaco);
        return order;
    })
    )
    );
}
//TC-06 - Fin

```

Dentro de OrderApiController.java

```

//TC-06 - Convertir ordenes de correo sin carreras ni nulls sorpresa
@PostMapping(path = "fromEmail", consumes = "application/json")
@ResponseStatus(HttpStatus.CREATED)
public Mono<TacoOrder> postOrderFromEmail(@RequestBody EmailOrder emailOrder) {
    return emailOrderService.convertEmailOrderToDomainOrder(Mono.just(emailOrder))
        .flatMap(repo::save)
        .flatMap(savedOrder ->
            Mono.fromRunnable(() -> orderMessages.sendOrder(savedOrder))
                .thenReturn(savedOrder)
        );
}

```

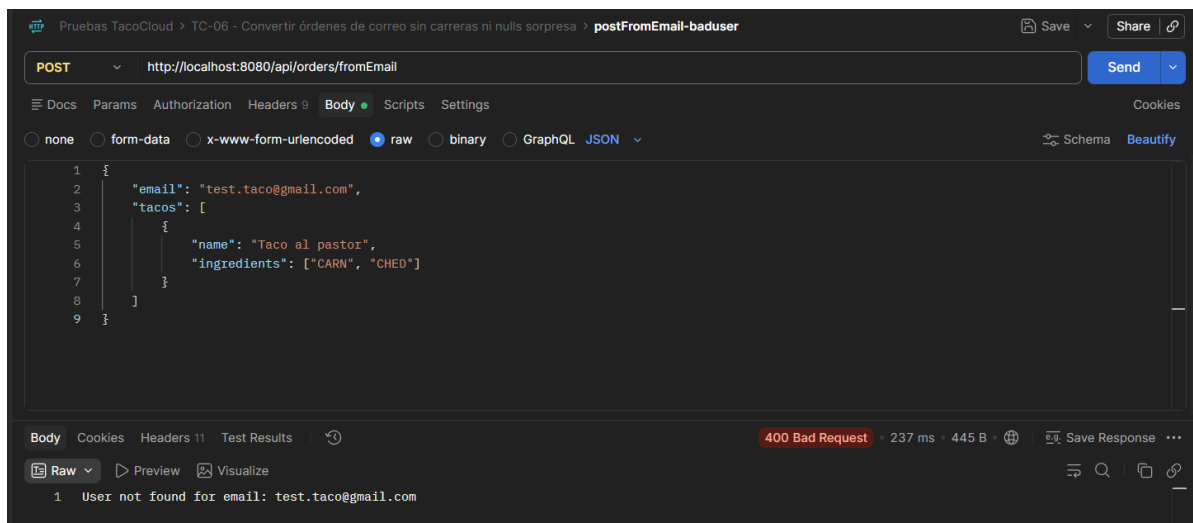
```
@ExceptionHandler(IllegalArgumentException.class)
public ResponseEntity<String> handleValidationExceptions(IllegalArgumentException ex) {
    return ResponseEntity.badRequest().body(ex.getMessage());
}
//TC-06 - Fin
```

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.EmailOrderServiceTest
16:16:59.890 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.31 s - in tacos.web.api.EmailOrderServiceTest
[INFO] Results:
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman

Ingresando un usuario inexistente:



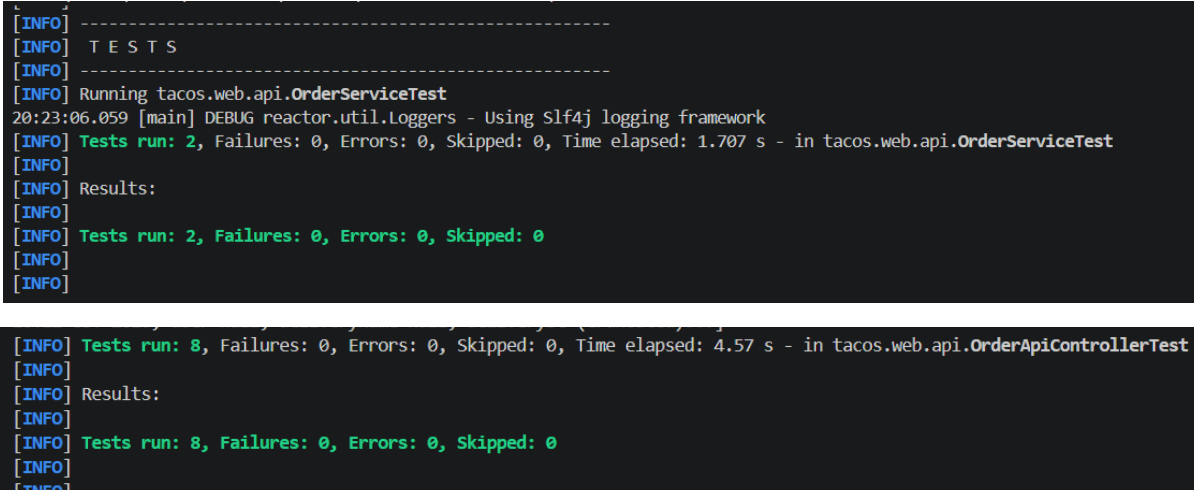
Laboratorio 2 – Contratos y seguridad

TC-07 – Una sola suscripción para guardar y publicar

Se volvió a modificar *postOrderFromEmail* en *OrderApiController.java*

```
@PostMapping(path = "fromEmail", consumes = "application/json")
@ResponseStatus(HttpStatus.CREATED)
public Mono<TacoOrder> postOrderFromEmail(@RequestBody EmailOrder emailOrder) {
    //TC-07 - Una sola suscripcion para guarar y publicar
    return emailOrderService.convertEmailOrderToDomainOrder(Mono.just(emailOrder))
        .flatMap(orderService::saveAndPublish);
    //TC-07 - Fin
}
```

Test



The screenshot shows two terminal windows displaying test results. The top window shows the results for `tacos.web.api.OrderServiceTest`, where 2 tests were run successfully. The bottom window shows the results for `tacos.web.api.OrderApiControllerTest`, where 8 tests were run successfully. Both tests passed without any failures or errors.

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.OrderServiceTest
20:23:06.059 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.707 s - in tacos.web.api.OrderServiceTest
[INFO] Results:
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]

[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 4.57 s - in tacos.web.api.OrderApiControllerTest
[INFO] Results:
[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

TC-08 – Separar DTOs de entrada, respuesta y persistencia

Se creó un paquete *tacos.api.dto* y un paquete *tacos.api.mapper*

Dentro de *dto*:

IngredientRequest.java

```
@Data
@JsonIgnoreProperties(ignoreUnknown = true)
public class IngredientRequest {
    private String id;
    private String name;
    private Ingredient.Type type;
}
```

IngredientResponse.java

```
@Data
public class IngredientResponse {
    private String id;
    private String name;
    private Ingredient.Type type;
}
```

OrderCreateRequest.java

```
@Data
@JsonIgnoreProperties(ignoreUnknown = true)
public class OrderCreateRequest {
    private String deliveryName;
    private String deliveryStreet;
    private String deliveryCity;
    private String deliveryState;
    private String deliveryZip;
    private String ccNumber;
    private String ccExpiration;
    private String ccCVV;
    private List<TacoCreateRequest> tacos = new ArrayList<>();
}
```

OrderPatchRequest.java

```
@Data
public class OrderPatchRequest {
    private String deliveryName;
    private String deliveryStreet;
    private String deliveryCity;
    private String deliveryState;
    private String deliveryZip;
}
```

OrderResponse.java

```
@Data
public class OrderResponse {
    private String deliveryName;
    private String deliveryStreet;
    private String deliveryCity;
    private String deliveryState;
    private String deliveryZip;
    private String ccNumber;
    private String ccExpiration;
    private String ccCVV;
    private List<TacoCreateRequest> tacos = new ArrayList<>();
}
```

OrderUpdateRequest.java

```
@Data
@JsonIgnoreProperties(ignoreUnknown = true)
public class OrderUpdateRequest {
    private String deliveryName;
    private String deliveryStreet;
    private String deliveryCity;
    private String deliveryState;
    private String deliveryZip;
    private List<TacoCreateRequest> tacos = new ArrayList<>();
}
```

TacoCreateRequest.java

```
@Data
@JsonIgnoreProperties(ignoreUnknown = true)
public class TacoCreateRequest {
    private String name;
    private List<String> ingredientIds = new ArrayList<>();
}
```

TacoResponse.java

```
@Data
public class TacoResponse {
    private String name;
    private List<IngredientResponse> ingredients = new ArrayList<>();
}
```

Dentro de mapper

IngredientMapper.java

```
@Component
public class IngredientMapper {
    public Ingredient toEntity(IngredientRequest request){
        return new Ingredient(request.getId(), request.getName(), request.getType());
    }

    public void updateEntity(IngredientRequest request, Ingredient ingredient){
        ingredient.setName(request.getName());
        ingredient.setType(request.getType());
    }

    public IngredientResponse toResponse(Ingredient ingredient){
        IngredientResponse response = new IngredientResponse();
        response.setId(ingredient.getId());
    }
}
```

```

        response.setName(ingredient.getName());
        response.setType(ingredient.getType());
        return response;
    }
}

```

OrderMapper.java

```

@Component
public class OrderMapper {
    private final TacoMapper tacoMapper;

    public OrderMapper(TacoMapper tacoMapper) {
        this.tacoMapper = tacoMapper;
    }

    public TacoOrder toEntity(OrderCreateRequest request, List<Taco> tacos) {
        TacoOrder order = new TacoOrder();
        order.setDeliveryName(request.getDeliveryName());
        order.setDeliveryStreet(request.getDeliveryStreet());
        order.setDeliveryCity(request.getDeliveryCity());
        order.setDeliveryState(request.getDeliveryState());
        order.setDeliveryZip(request.getDeliveryZip());
        order.setCcNumber(request.getCcNumber());
        order.setCcExpiration(request.getCcExpiration());
        order.setCcCVV(request.getCcCVV());
        order.setTacos(tacos);
        return order;
    }

    public void updateEntity(OrderUpdateRequest request, TacoOrder order, List<Taco> tacos) {
        order.setDeliveryName(request.getDeliveryName());
        order.setDeliveryStreet(request.getDeliveryStreet());
        order.setDeliveryCity(request.getDeliveryCity());
        order.setDeliveryState(request.getDeliveryState());
        order.setDeliveryZip(request.getDeliveryZip());
        order.setTacos(tacos);
    }

    public void patchEntity(OrderPatchRequest patch, TacoOrder order) {
        if (patch.getDeliveryName() != null) {
            order.setDeliveryName(patch.getDeliveryName());
        }

        if (patch.getDeliveryStreet() != null) {
            order.setDeliveryStreet(patch.getDeliveryStreet());
        }
    }
}

```

```

        if (patch.getDeliveryCity() != null) {
            order.setDeliveryCity(patch.getDeliveryCity());
        }

        if (patch.getDeliveryState() != null) {
            order.setDeliveryState(patch.getDeliveryState());
        }

        if (patch.getDeliveryZip() != null) {
            order.setDeliveryZip(patch.getDeliveryZip());
        }
    }

    public OrderResponse toResponse(TacoOrder order) {
        OrderResponse response = new OrderResponse();
        response.setId(order.getId());
        response.setPlacedAt(order.getPlacedAt());
        response.setDeliveryName(order.getDeliveryName());
        response.setDeliveryStreet(order.getDeliveryStreet());
        response.setDeliveryCity(order.getDeliveryCity());
        response.setDeliveryState(order.getDeliveryState());
        response.setDeliveryZip(order.getDeliveryZip());
        response.setTacos(order.getTacos().stream()
            .map(tacoMapper::toResponse)
            .collect(Collectors.toList())
        );

        return response;
    }
}

```

TacoMapper.java

```

@Component
public class TacoMapper {
    private final IngredientMapper ingredientMapper;

    public TacoMapper(IngredientMapper ingredientMapper) {
        this.ingredientMapper = ingredientMapper;
    }

    public Taco toEntity(TacoCreateRequest request, List<Ingredient> ingredients) {
        Taco taco = new Taco();
        taco.setName(request.getName());
        taco.setIngredients(ingredients);
        return taco;
    }
}

```

```

public TacoResponse toResponse(Taco taco) {
    TacoResponse response = new TacoResponse();
    response.setName(taco.getName());
    response.setIngredients(taco.getIngredients().stream()
        .map(ingredientMapper::toResponse)
        .collect(Collectors.toList())
    );

    return response;
}
}

```

Con esto se actualizaron las clases *IngredientController.java*, *OrderApiController.java* y *OrderService.java* para integrar los dtos

Test

```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.api.mapper.IngredientMapperTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.054 s - in tacos.api.mapper.IngredientMapperTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]

```

```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.api.mapper.OrderMapperTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.502 s - in tacos.api.mapper.OrderMapperTest
[INFO] Results:
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]

```

Postman

Creando una orden

Pruebas TacoCloud > TC-08 > Separar DTOs de entrada, respuesta y persistencia > postOrders-201

POST http://localhost:8080/api/orders Send

Body Cookies Headers 12 Test Results 5/5

201 Created 24 ms · 781 B Save Response

```
1 {
2   "id": "6ab8c5f2c617a1f9fd63f29",
3   "placedAt": "2026-09-27T07:26:55.345+00:00",
4   "deliveryName": "Test User",
5   "deliveryStreet": "Test Street",
6   "deliveryCity": "Agascalientes",
7   "deliveryState": "AG",
8   "deliveryZip": "20000",
9   "tacos": [
10    {
11      "name": "Taco TC00",
12      "ingredients": [
13        {
14          "id": "CARN",
15          "name": "Carnitas",
16          "type": "PROTEIN"
17        },
18        {
19          "id": "CHED",
20          "name": "Cheddar",
21          "type": "CHEESE"
22        }
23      ]
24    }
25  ]
26 }
```

Pruebas TacoCloud > TC-08 > Separar DTOs de entrada, respuesta y persistencia > postOrders-201

POST http://localhost:8080/api/orders Send

Body Cookies Headers 12 Test Results 5/5

201 Created 24 ms · 781 B Save Response

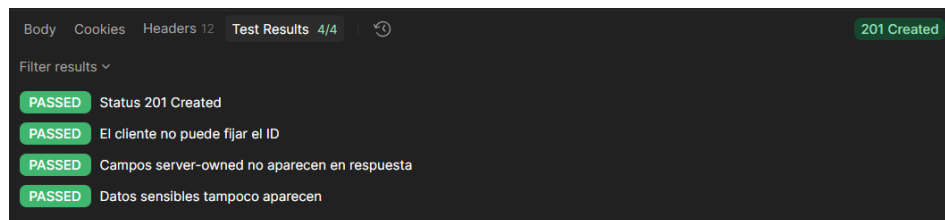
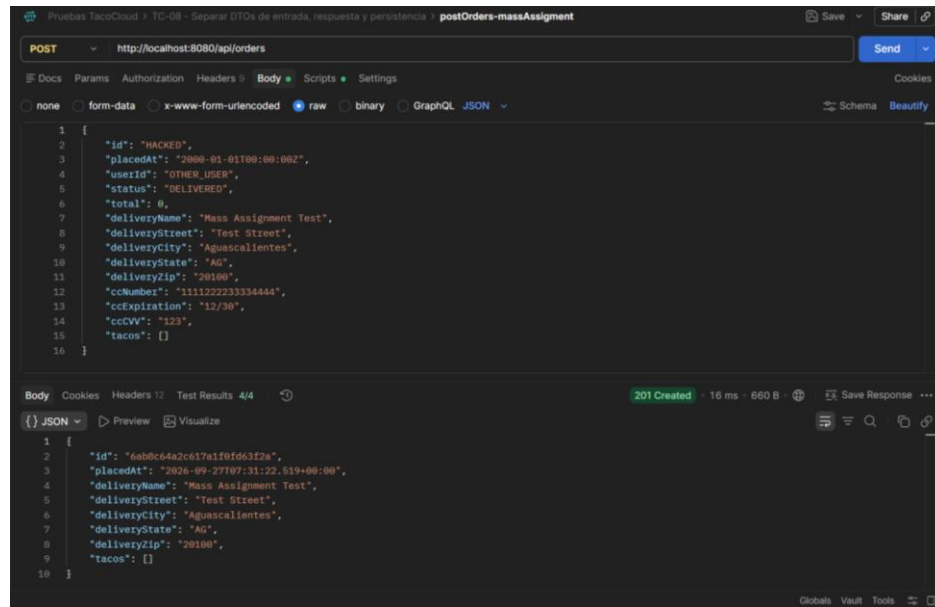
none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "deliveryName": "Test User",
3   "deliveryStreet": "Test Street",
4   "deliveryCity": "Agascalientes",
5   "deliveryState": "AG",
6   "deliveryZip": "20000",
7   "ccNumber": "111122223334444",
8   "ccExpiration": "12/30",
9   "ccCVV": "123",
10  "tacos": [
11    {
12      "name": "Taco TC00",
13      "ingredientIds": [
14        "CARN",
15        "CHED"
16      ]
17    }
18  ]
19 }
```

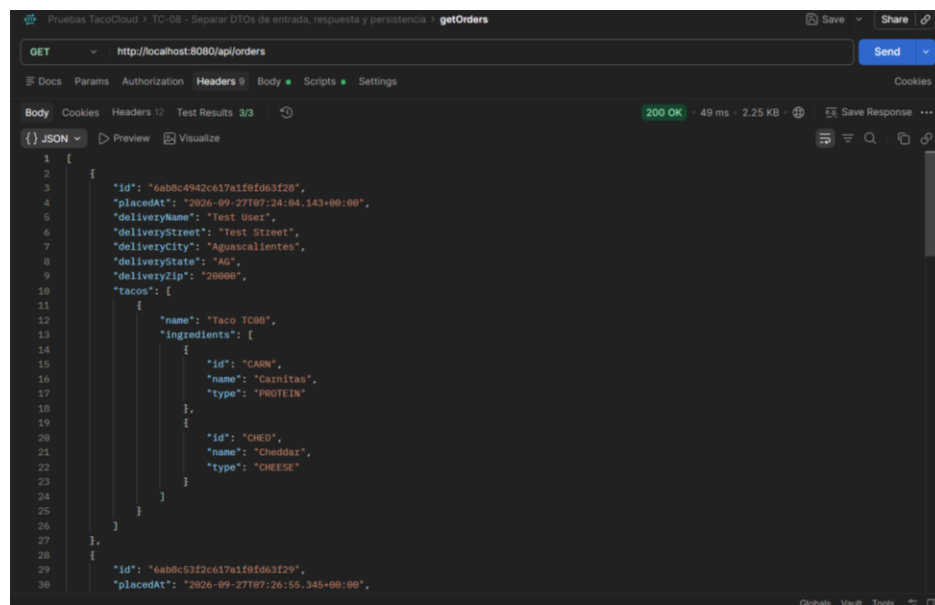
Filter results

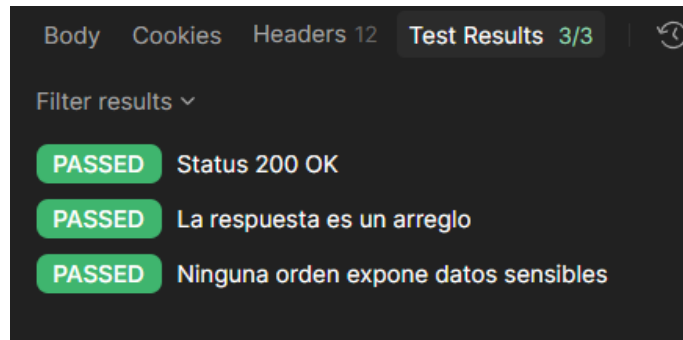
- PASSED Status 201 Created
- PASSED La respuesta contiene id generado
- PASSED La respuesta contiene placedAt
- PASSED La respuesta contiene datos de entrega
- PASSED La respuesta NO expone datos sensibles

Prueba de mass assignment

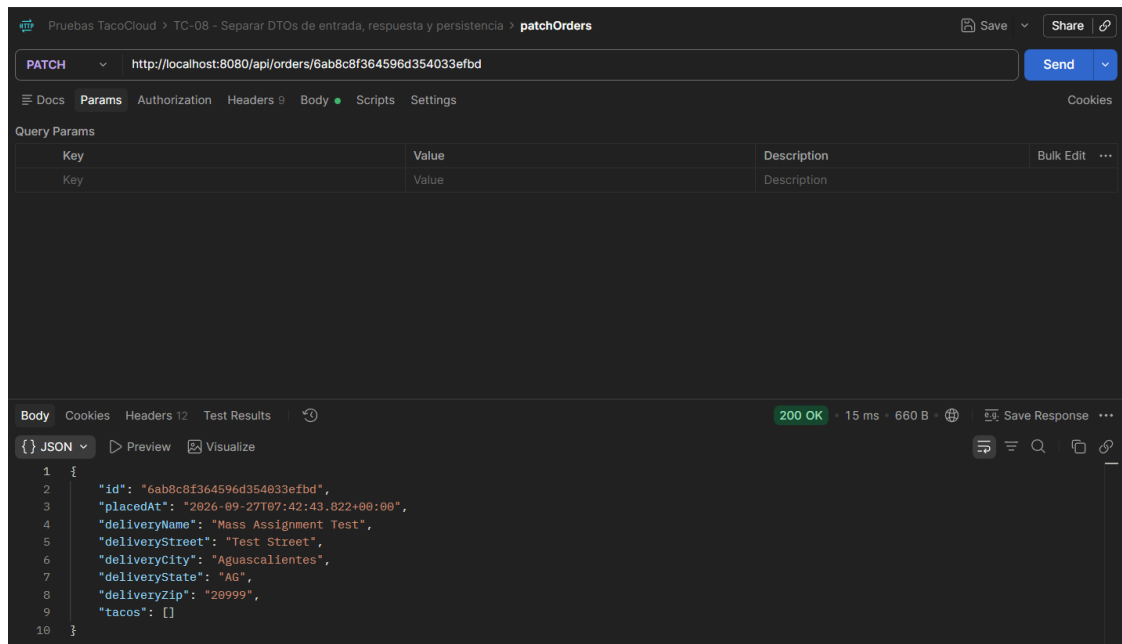


Obteniendo todas las ordenes



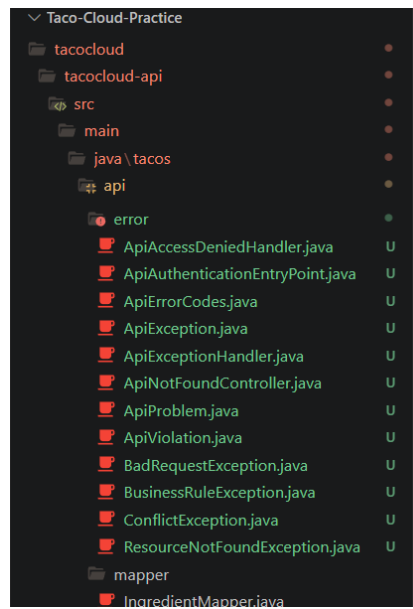


Patch adaptado de TC-04



TC-09 – Validación y errores tipo Problem Details

Para esta implementación, se agregaron nuevos manejadores de errores:

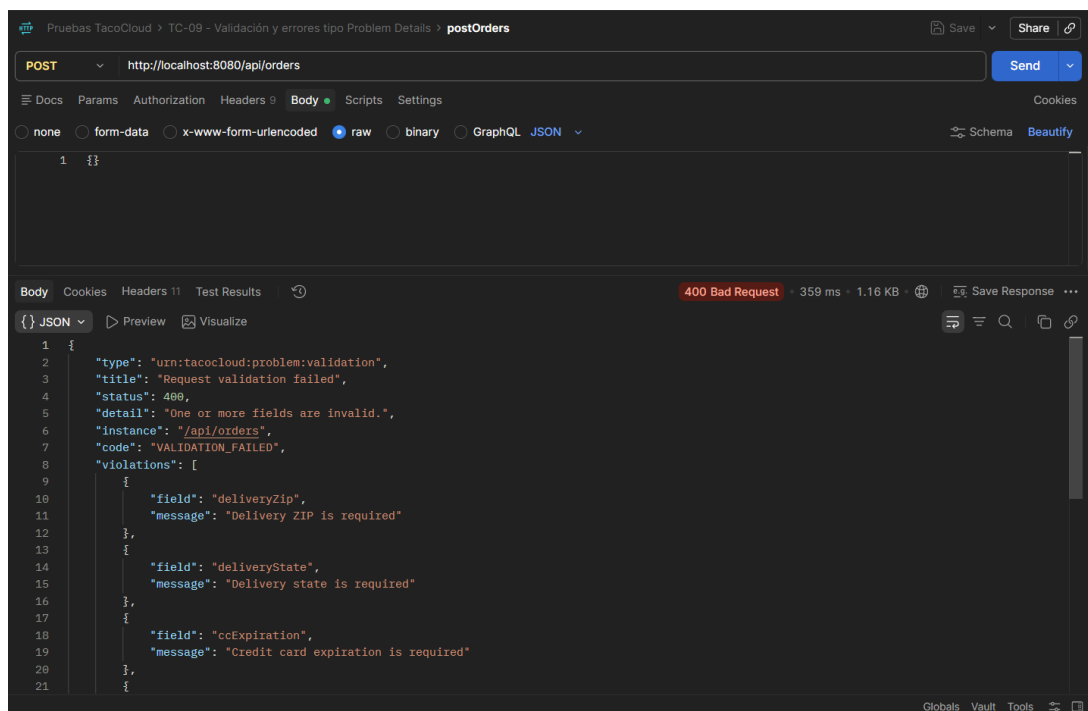


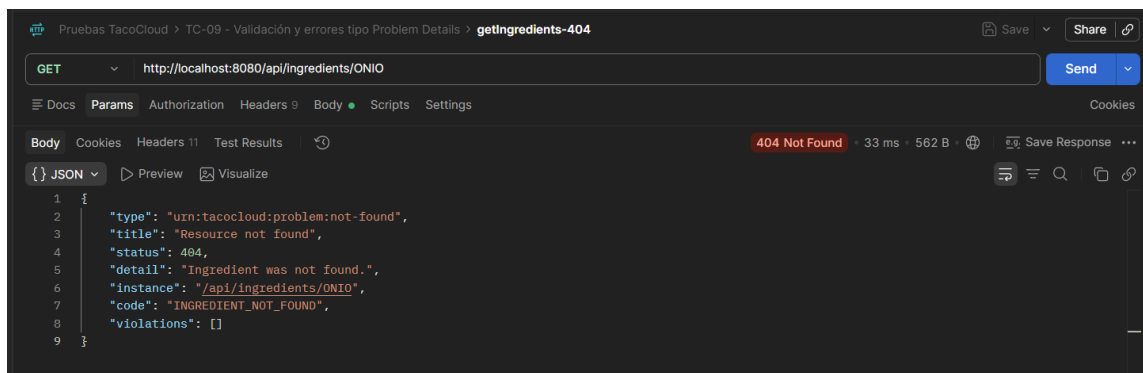
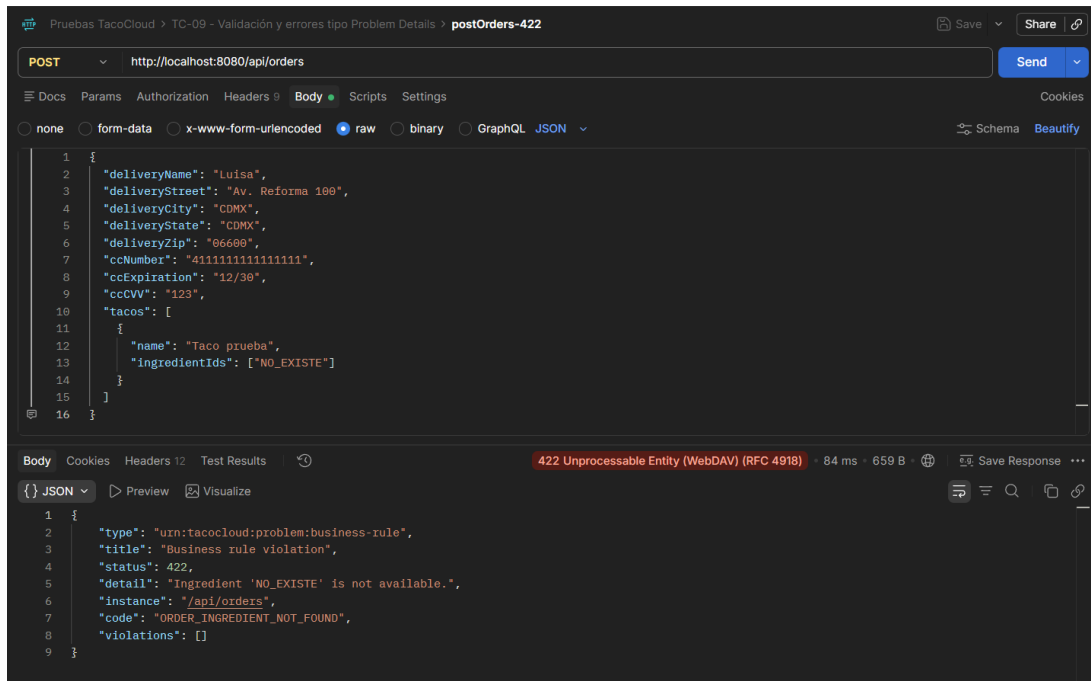
Y se adecuaron los DTOs para permitir las validaciones de requeridos, tamaños, formatos, cantidades y pagos

Test

```
[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.719 s - in tacos.api.error.ApiExceptionHandlerTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman





TC-10 – Registro reactivo con contraseñas protegidas

Se reemplazó *NoOpPasswordEncoder* por *PasswordEncoderFactories.createDelegatingPasswordEncoder()*.

Se añadieron validaciones para username, password, fullname y email.

Se creó *RegistrationService* para componer índices, verificaciones, codificación, guardado y respuesta.

Se añadieron índices únicos para username y email exclusivamente en la colección user.

Se añadieron los códigos USERNAME_ALREADY_EXISTS y EMAIL_ALREADY_EXISTS.

Los conflictos se traducen a 409 mediante el *ApiProblem* de TC-09.

Se creó una respuesta segura sin password, hash ni campos internos.

Se impidió la exposición de *UserRepository* por Spring Data REST.

El password fue excluido de JSON y de toString().

Test

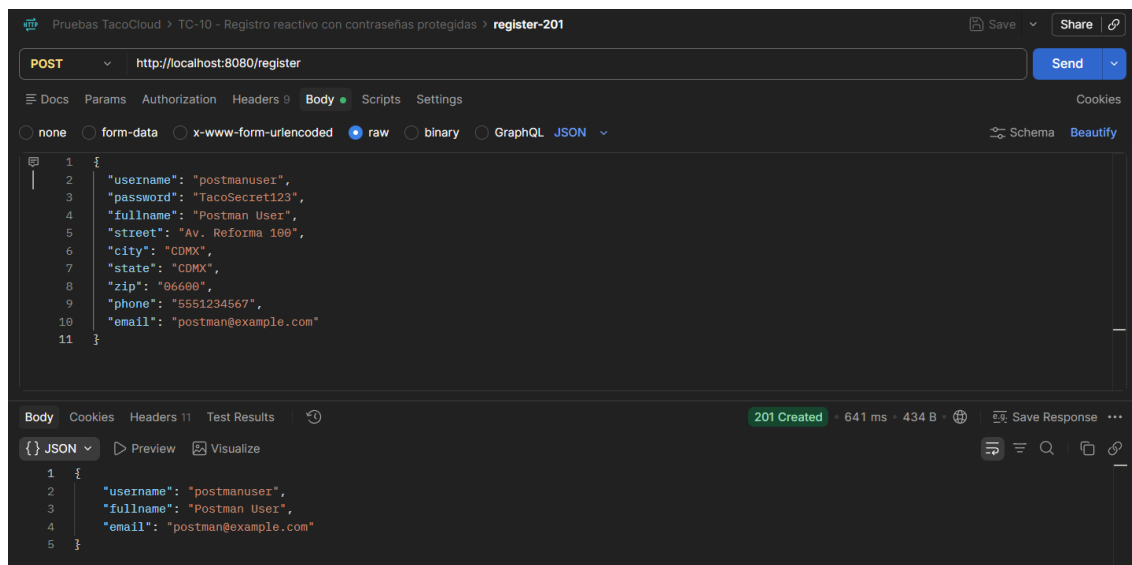
```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.security.SecurityConfigTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.077 s - in tacos.security.SecurityConfigTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.security.RegistrationServiceTest
17:47:02.257 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.91 s - in tacos.security.RegistrationServiceTest
[INFO] Results:
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----

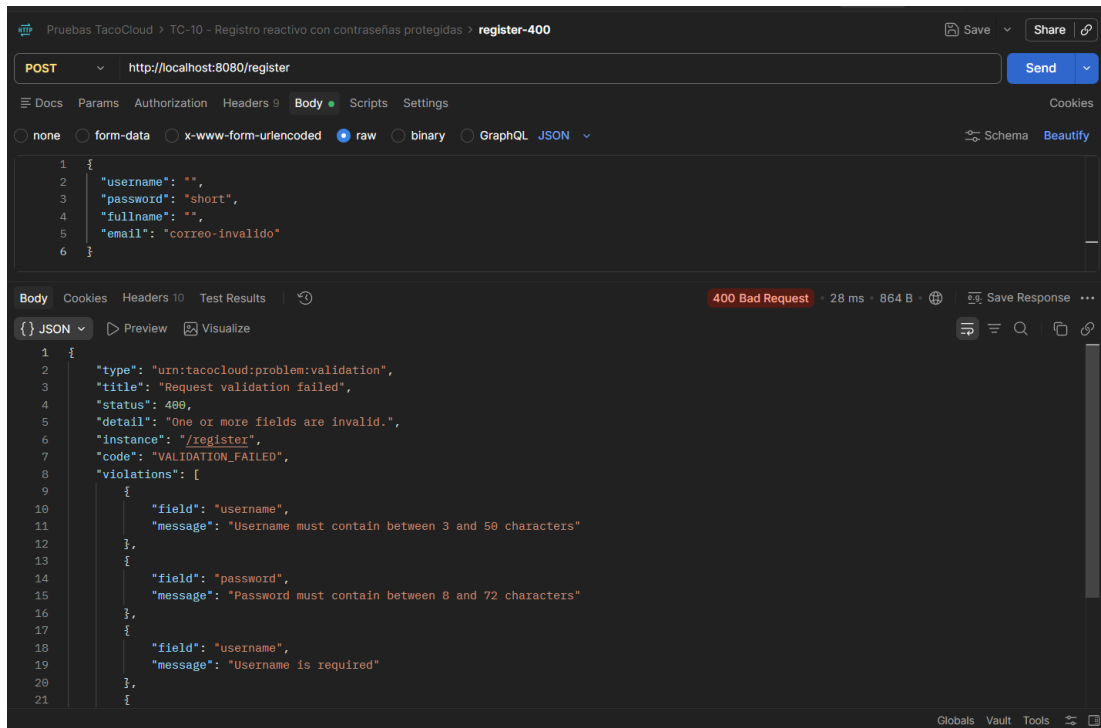
shutting down with code:0
[INFO] Results:
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
```

Postman

Registro exitoso



Validación incorrecta



TC-11 – Autorización deny-by-default y roles útiles

Se realizaron modificaciones en `SecurityConfig`, `User`, `OrderService`, `OrderApiController`, `OrderRepository`, `IngredientRepository`, `TacoController`, `application.yml`, `pom.xml` y tests relacionados.

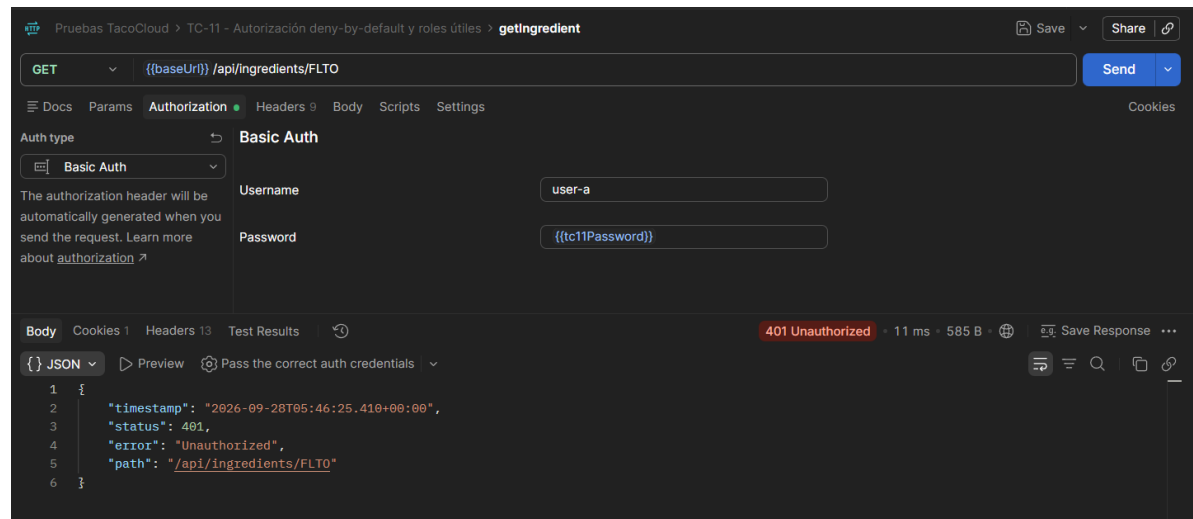
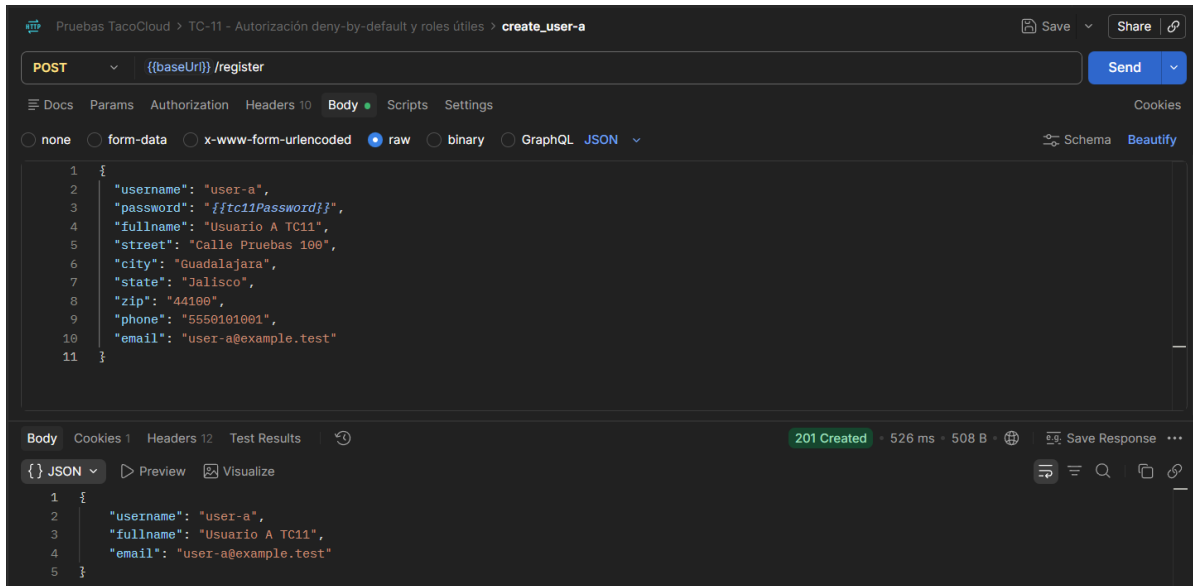
Esto con el fin de implementar una matriz explícita de autorización con roles `USER`, `ADMIN` y `KITCHEN`, ownership de órdenes y protección de endpoints internos.

Test

```
PS C:\Users\luisa\Documents\WTTDATA\Taco-Cloud-Practice\tacocloud> mvn test -Dtest=Tc11AuthorizationIntegrationTest -DfailIfNoTests=false
[INFO] Results:
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0
```

Postman

Creando un usuario con rol `USER` e intentando ver un ingrediente



TC-12 – Tokenizar pago y eliminar PAN/CVV del dominio

TacoOrder almacenaba PAN, expiración y CVV; esos datos podían llegar a Mongo y al evento de cocina.

Ahora la tarjeta llega únicamente a `PaymentMethodTokenizeRequest`, pasa por `PaymentGateway`, genera un `PaymentMethod` tokenizado y la orden conserva solamente `paymentMethodId`.

Entre los cambios principales, se encuentran:

TacoOrder ya no declara `ccNumber`, `ccExpiration` ni `ccCVV`.

`OrderCreateRequest` recibe únicamente `paymentMethodId`.

`PaymentMethod` no almacena PAN ni CVV.

El endpoint de tokenización responde 201 Created.

`PaymentMethodRepository` dejó de exponerse mediante Spring Data REST.

La persistencia se compone con flatMap; no se agregaron block() ni suscripciones manuales en los servicios/controladores reactivos.

Un método inexistente produce 404 PAYMENT_METHOD_NOT_FOUND.

Un método perteneciente a otro usuario produce 403 ACCESS_DENIED.

La interfaz Angular tokeniza primero y posteriormente crea la orden.

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.PaymentDataModelTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.482 s - in tacos.PaymentDataModelTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.payment.FakePaymentGatewayTest
16:07:36.929 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.105 s - in tacos.payment.FakePaymentGatewayTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.OrderServiceTest
16:08:48.466 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 9, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.8 s - in tacos.web.api.OrderServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 9, Failures: 0, Errors: 0, Skipped: 0
[INFO]

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.api.mapper.KitchenOrderEventMapperTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.952 s - in tacos.api.mapper.KitchenOrderEventMapperTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

```

[INFO] TESTS
[INFO] -----
[INFO] Running tacos.data.migration.SensitivePaymentDataMigrationTest
16:10:49.874 [main] DEBUG org.springframework.test.context.BootstrapUtils - Instantiating CacheAwareContextLoaderDelegate from class [org.springframework.test.context.cache.DefaultCacheAwareContextLoaderDelegate]
16:10:49.897 [main] DEBUG org.springframework.test.context.BootstrapUtils - Instantiating BootstrapContext using constructor [public org.springframework.test.context.support.DefaultBootstrapContext(java.lang.Class,org.springframework.test.context.CacheAwareContextLoaderDelegate)]
16:10:49.981 [main] DEBUG org.springframework.test.context.BootstrapUtils - Instantiating TestContextBootstrapper for test class [tacos.data.migration.SensitivePaymentDataMigrationTest] from class [org.springframework.boot.test.autoconfigure.data.mongo.DataMongoTestContextBootstrapper]
16:10:50.012 [main] INFO org.springframework.boot.test.autoconfigure.data.mongo.DataMongoTestContextBootstrapper - Neither @ContextConfiguration nor @ContextHierarchy found for test class [tacos.data.migration.SensitivePaymentDataMigrationTest], using SpringBootContextLoader
16:10:50.021 [main] DEBUG org.springframework.test.context.support.AbstractContextLoader - Did not detect default resource location for test class [tacos.data.migration.SensitivePaymentDataMigrationTest]: class path resource [tacos/data/migration/SensitivePaymentDataMigrationTest-context.xml] does not exist
16:10:50.023 [main] DEBUG org.springframework.test.context.support.AbstractContextLoader - Did not detect default resource location for test class [tacos.data.migration.SensitivePaymentDataMigrationTest]: class path resource [tacos/data/migration/SensitivePaymentDataMigrationTestContext.groovy] does not exist
16:10:50.023 [main] INFO org.springframework.test.context.support.AbstractContextLoader - Could not detect default resource locations for test class [tacos.
Spring Boot-TacoCloudApplication<tacocloud_tc11-tc12> (Taco-Cloud-Practice) Live Share Java: Ready Spaces: 4 UTF-8 CRLF {} JSON with Comments Go Live Prettier

```

```

PS C:\Users\luisa\Documents\WTFDATA\Taco-Cloud-Practice\tacocloud> mvn test -Dtest=SensitivePaymentDataMigrationTest -DfailIfNoTests=false
[INFO] Scanning for projects...
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ tacocloud ---
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]

```

```

[INFO] TESTS
[INFO] -----
[INFO] Running tacos.web.api.Tc12SensitiveDataMigrationControllerTest
16:12:48.752 [main] DEBUG reactor.util.loggers - Using Slf4j logging framework
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.245 s - in tacos.web.api.Tc12SensitiveDataMigrationControllerTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]

```

Postman

Tokenización correcta 201

The screenshot shows a Postman interface for a REST client. The request is a POST to `{{baseUrl}}/api/payment-methods/tokenize` with a raw JSON body:

```

{
  "cardNumber": "9999999999999999",
  "expiration": "12/39",
  "cvv": "999"
}

```

The response is a 201 Created status with a JSON body:

```

{
  "id": "6abadc686371b413c0ca57f7",
  "brand": "LAB_CARD",
  "last4": "9999"
}

```

The interface also shows the 'Test Results' tab with a green '201 Created' message and a 'Save Response' button.

Método de pago inexistente

Pruebas TacoCloud > TC-12 - Tokenizar pago y eliminar PAN/CVV del dominio > order-201

POST {{baseUrl}} /api/orders

Send

Docs Params Authorization Headers 11 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
1 {
2   "deliveryName": "Usuario Laboratorio",
3   "deliveryStreet": "Calle Laboratorio 10",
4   "deliveryCity": "Aguascalientes",
5   "deliveryState": "AG",
6   "deliveryZip": "20000",
7   "paymentMethodId": "6abadf9e6371b413c0ca57f6",
8   "tacos": [
9     {
10      "name": "Taco TC-12",
11      "ingredientIds": ["FLTO", "CARN", "CHED"]
12    }
13  ]
14 }
```

Body Cookies 1 Headers 12 Test Results 1/2 404 Not Found · 29 ms · 637 B Save Response

JSON Preview Visualize

```
1 {
2   "type": "urn:tacocloud:problem:not-found",
3   "title": "Resource not found",
4   "status": 404,
5   "detail": "Payment method was not found.",
6   "instance": "/api/orders",
7   "code": "PAYMENT_METHOD_NOT_FOUND",
8   "violations": []
9 }
```

Laboratorio 3 – Motor de negocio

TC-13 – Catálogo con precio, disponibilidad y stock

El objetivo es ampliar Ingredient para representar precio, disponibilidad comercial e inventario, incluyendo control de concurrencia mediante optimistic locking.

Por lo que se realizaron los siguientes cambios:

- Se agregó unitPrice con BigDecimal.
- Los precios se normalizan a dos decimales con HALF_UP.
- Se agregaron available, stockOnHand, reorderLevel y version.
- Se creó un servicio reactivo para catálogo e inventario.
- Se implementaron PATCH de catálogo y POST de ajustes.
- Se separaron respuestas públicas y administrativas.
- Se reutilizó ApiProblem para errores 400, 404, 409 y 422.
- Los endpoints administrativos reutilizan la seguridad ROLE_ADMIN.
- Se actualizaron los diez ingredientes seed.

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.IngredientCatalogServiceTest
20:28:59.383 [main] DEBUG reactor.util.loggers - Using Slf4j logging framework
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.69 s - in tacos.web.api.IngredientCatalogServiceTest
[INFO] Results:
[INFO] Tests run: 10, Failures: 0, Errors: 0, Skipped: 0
```

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.data.IngredientOptimisticLockingTest
20:34:12.888 [main] DEBUG org.springframework.test.context.BootstrapUtils - Instantiating CacheAwareContextLoaderDelegate from class [org.springframework.test.context.cache.DefaultCacheAwareContextLoaderDelegate]
20:34:12.823 [main] DEBUG org.springframework.test.context.BootstrapUtils - Instantiating BootstrapContext using constructor [public org.springframework.test.context.support.DefaultBootstrapContext(java.lang.Class,org.springframework.test.context.CacheAwareContextLoaderDelegate)]
20:34:12.873 [main] DEBUG org.springframework.test.context.BootstrapUtils - Instantiating TestContextBootstrapper for test class [tacos.data.IngredientOptimisticLockingTest] from class [org.springframework.boot.test.autoconfigure.data.mongo.DataMongoTestContextBootstrapper]
20:34:12.889 [main] INFO org.springframework.boot.test.autoconfigure.data.mongo.DataMongoTestContextBootstrapper - Neither @ContextConfiguration nor @ContextHierarchy found for test class [tacos.data.IngredientOptimisticLockingTest], using SpringBootContextLoader
20:34:12.895 [main] DEBUG org.springframework.test.context.support.AbstractContextLoader - Did not detect default resource location for test class [tacos.data.IngredientOptimisticLockingTest]: class path resource [tacos/data/IngredientOptimisticLockingTest-context.xml] does not exist

Spring Boot-TacoCloudApplication<tacocloud_tc13> (Taco-Cloud-Practice) Live Share Java: Building - 0% UTF-8 CRLF JSON with Comments Go Live Prettier

PS C:\Users\luisa\Documents\WTTDATA\Taco-Cloud-Practice\tacocloud> mvn test -Dtest=IngredientOptimisticLockingTest -DfailIfNoTests=false
2026-09-28 20:34:18.532 INFO 3968 --- [Thread-1] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-28T20:34:18.532-0600 I CONTROL [conn5] now exiting
2026-09-28 20:34:18.532 INFO 3968 --- [Thread-1] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-28T20:34:18.532-0600 I CONTROL [conn5] shutting down with code:0
2026-09-28 20:34:18.532 INFO 3968 --- [Thread-1] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-28T20:34:18.532-0600 I CONTROL [initandlisten] shutting down with code:0
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
```

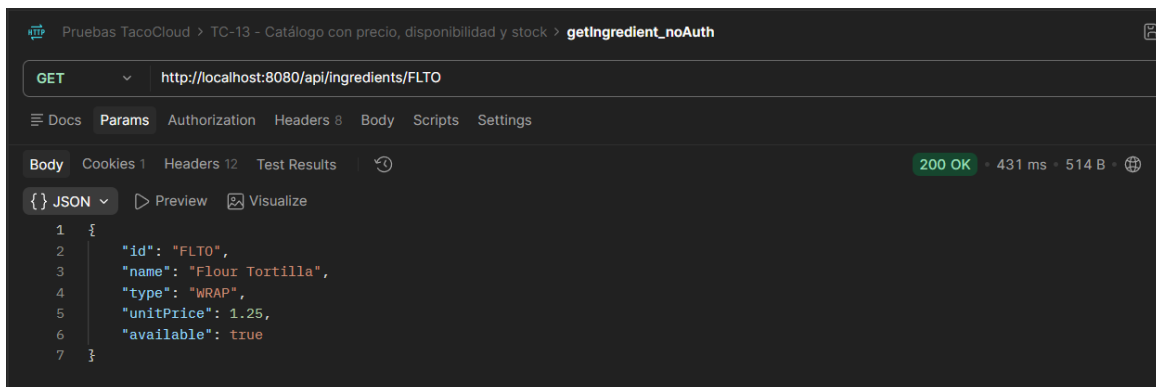
```

[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.Tc13SeedDataTest
20:40:40.724 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.863 s - in tacos.Tc13SeedDataTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]

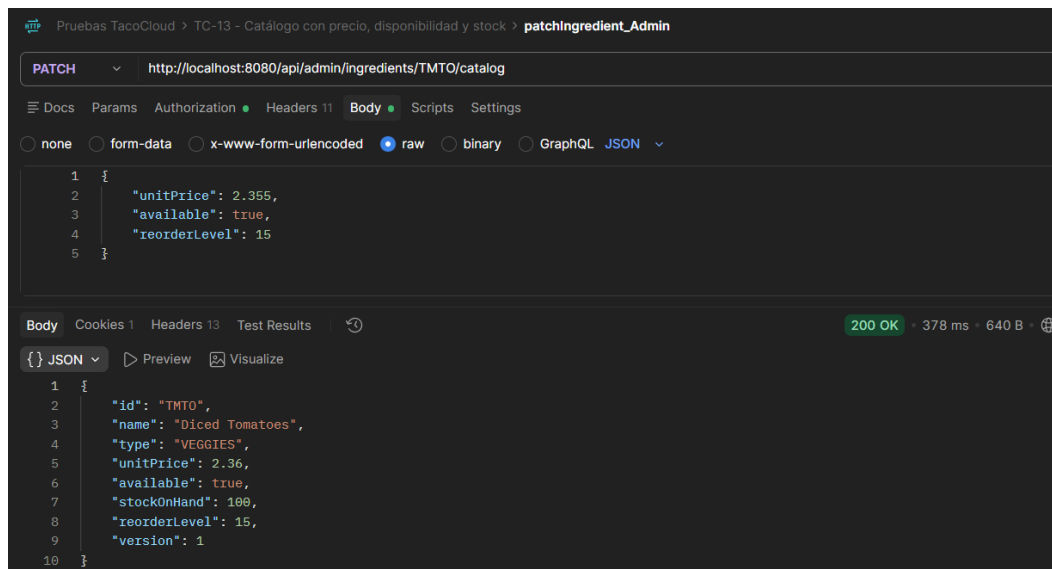
```

Postman

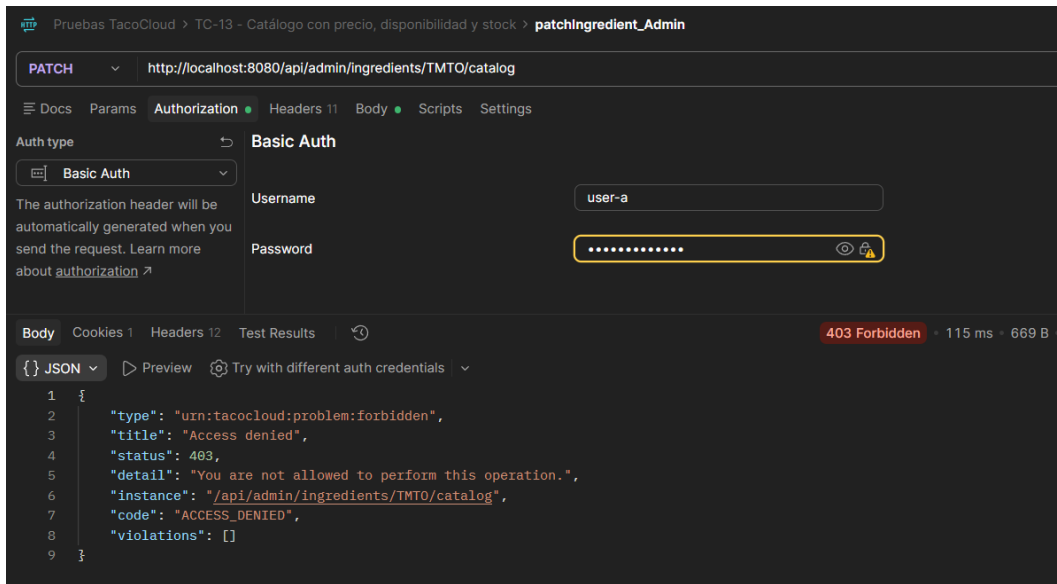
Obteniendo ingrediente de forma segura sin autenticar



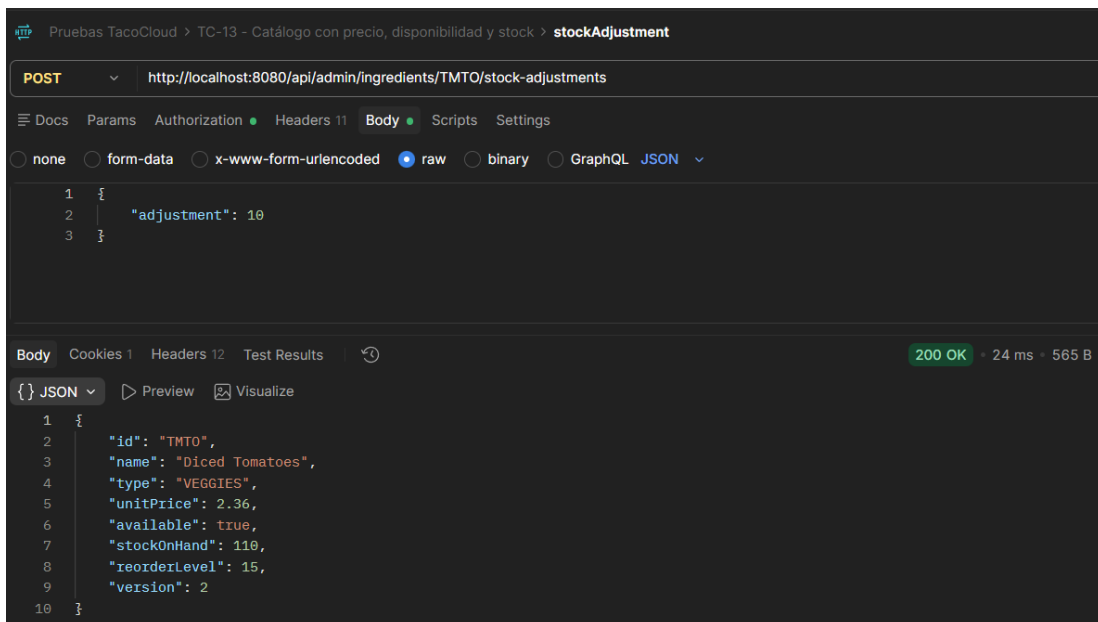
Actualizando ingrediente autenticado



Repitiendo el proceso sin autenticar:



Ajuste de stock



TC-14 – Calcular precios y cantidades del lado del servidor

Este reto consiste en representar cantidades reales y evitar que el cliente controle los valores económicos de una orden.

Para esto el contrato evolucionó a items, cada uno con un taco basado en ingredientIds y una quantity. OrderPricingService obtiene los precios actuales del catálogo TC-13, suma los ingredientes, calcula el subtotal y genera un OrderLine.

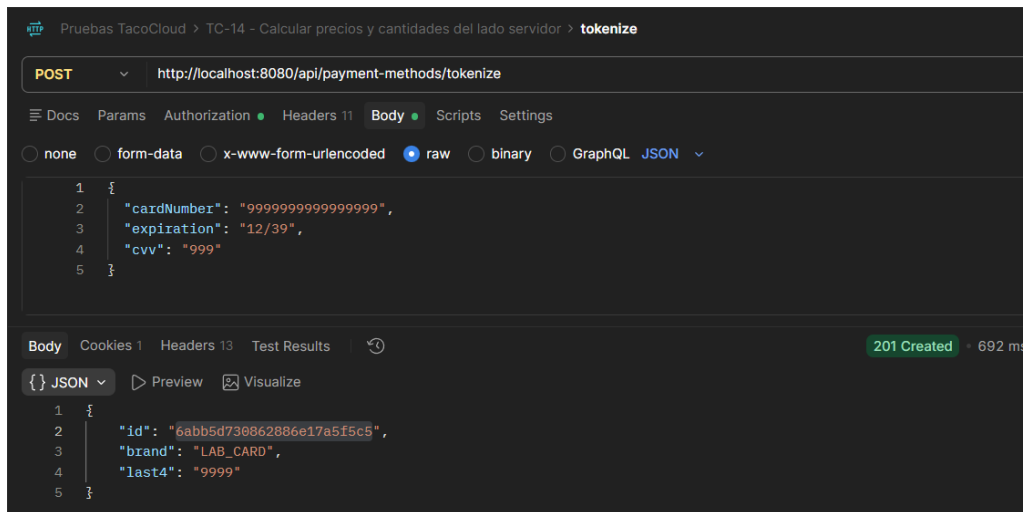
Se realizaron cambios:

- Dominio y DTO: TacoOrder, OrderCreateRequest, OrderUpdateRequest, OrderResponse.
- Backend: OrderService, OrderMapper, ApiErrorCodes, EmailOrderService.
- Cocina: KitchenOrderEventManager, KitchenOrderEvent.
- Configuración: application.yml.
- Angular: CartComponent, CartItem, template del carrito y encabezado.
- Tests existentes de servicio, mapper, controlador, email, evento y regresión TC-11.
- Configuración de Karma y Protractor para ejecución headless estable.

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.OrderPricingServiceTest
00:53:35.949 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.285 s - in tacos.web.api.OrderPricingServiceTest
[INFO] Results:
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman



Con cantidad = 2

Pruebas TacoCloud > TC-14 - Calcular precios y cantidades del lado servidor > quantity=2

POST http://localhost:8080/api/orders

Docs Params Authorization Headers 11 Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
1 {
2   "deliveryName": "Usuario TC14",
3   "deliveryStreet": "Calle Laboratorio 14",
4   "deliveryCity": "Ciudad",
5   "deliveryState": "ST",
6   "deliveryZip": "00000",
7   "paymentMethodId": "6abb5d730862886e17a5f5c5",
8   "items": [
9     {
10      "taco": {
11        "name": "Taco TC14",
12        "ingredientIds": ["FLT0", "CARN"]
13      },
14      "quantity": 2
15    }
16  ]
17 }
```

Body Cookies 1 Headers 12 Test Results 4/4 201 Created

{ } JSON Preview Visualize

```
24   "type": "PROTEIN",
25   "unitPrice": 2.50,
26   "available": true
27 }
28 ]
29 },
30 "quantity": 2,
31 "unitPriceAtPurchase": 3.75,
32 "subtotal": 7.50
33 }
34 ],
35 "total": 7.50,
36 "currency": "USD"
37 }
```

Con manipulación

Pruebas TacoCloud > TC-14 - Calcular precios y cantidades del lado servidor > quantity=2

POST http://localhost:8080/api/orders

Docs Params Authorization Headers 11 Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
8 "total": 0.01,
9 "currency": "MXN",
10 "items": [
11   {
12     "taco": {
13       "name": "Taco TC14",
14       "ingredientIds": ["FLT0", "CARN"]
15     },
16     "quantity": 2,
17     "unitPriceAtPurchase": 0.01,
18     "subtotal": 0.02
19   }
20 ]
21 }
```

Body Cookies 1 Headers 12 Test Results 4/4 201 Created

{ } JSON Preview Visualize

```
24   "type": "PROTEIN",
25   "unitPrice": 2.50,
26   "available": true
27 }
28 ]
29 },
30 "quantity": 2,
31 "unitPriceAtPurchase": 3.75,
32 "subtotal": 7.50
33 }
34 ],
35 "total": 7.50,
36 "currency": "USD"
37 }
```

Cuando se pone 0 en cantidad:

The screenshot shows a REST client interface with a POST request to `http://localhost:8080/api/orders`. The request body is a JSON object with `quantity: 0`. The response is a 400 Bad Request with a JSON body indicating a validation error.

```
POST http://localhost:8080/api/orders

{
  "total": 0.01,
  "currency": "MXN",
  "items": [
    {
      "taco": {
        "name": "Taco TC14",
        "ingredientIds": ["FLT0", "CARN"]
      },
      "quantity": 0,
      "unitPriceAtPurchase": 0.01,
      "subtotal": 0.02
    }
  ]
}
```

Body: Cookies 1 Headers 11 Test Results 400 Bad Request

```
{
  "type": "urn:tacocloud:problem:validation",
  "title": "Request validation failed",
  "status": 400,
  "detail": "One or more fields are invalid.",
  "instance": "/api/orders",
  "code": "VALIDATION_FAILED",
  "violations": [
    {
      "field": "items[0].quantity",
      "message": "Quantity must be at least 1"
    }
  ]
}
```

Cuando se supera el límite permitido

The screenshot shows a REST client interface with a POST request to `http://localhost:8080/api/orders`. The request body is a JSON object with `quantity: 11`. The response is a 422 Unprocessable Entity with a JSON body indicating a business rule violation.

```
POST http://localhost:8080/api/orders

{
  "total": 0.01,
  "currency": "MXN",
  "items": [
    {
      "taco": {
        "name": "Taco TC14",
        "ingredientIds": ["FLT0", "CARN"]
      },
      "quantity": 11,
      "unitPriceAtPurchase": 0.01,
      "subtotal": 0.02
    }
  ]
}
```

Body: Cookies 1 Headers 12 Test Results 422 Unprocessable Entity (WebDAV) (RFC 4918)

```
{
  "type": "urn:tacocloud:problem:business-rule",
  "title": "Business rule violation",
  "status": 422,
  "detail": "Item quantity cannot exceed 10.",
  "instance": "/api/orders",
  "code": "ITEM_QUANTITY_LIMIT_EXCEEDED",
  "violations": []
}
```

TC-15 – Motor de cupones con reglas y fechas de expiración

Se tiene que aplicar descuentos únicamente sobre precios calculados por TC-14 y ofrecer una cotización sin guardar órdenes, publicar eventos ni reservar inventario.

Para esto se modificaron:

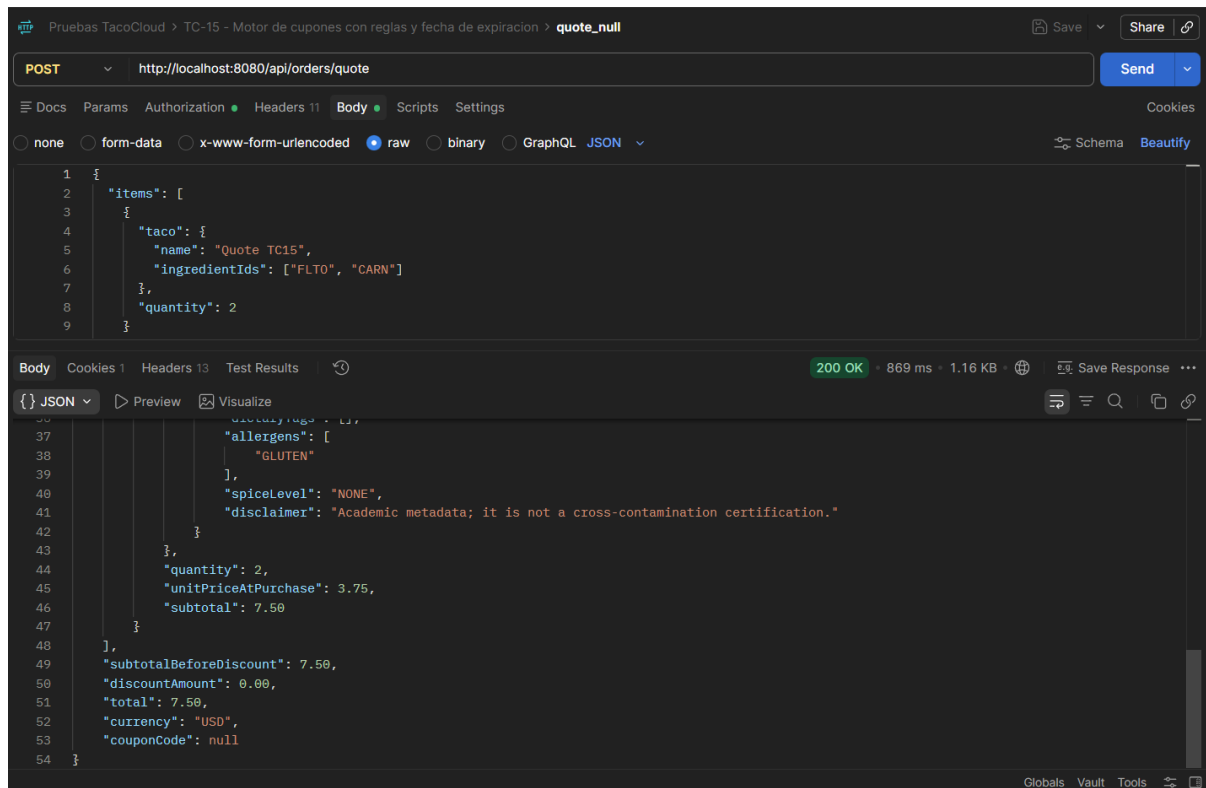
- OrderCreateRequest, OrderResponse, TacoOrder
- OrderMapper, OrderService, OrderApiController
- ApiErrorCodes
- application.yml
- OrderServiceTest, OrderMapperTest

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.coupon.CouponServiceTest
12:18:10.419 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.283 s - in tacos.coupon.CouponServiceTest
[INFO] Results:
[INFO]
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman

Sin cupón



Con cupón del 10%

The screenshot shows a REST client interface with the following details:

- Method:** POST
- URL:** http://localhost:8080/api/orders/quote
- Body (raw):**

```
1 {
2   "items": [
3     {
4       "taco": {
5         "name": "Quote TC15",
6         "ingredientIds": ["FLT0", "CARN"]
7       },
8       "quantity": 2
9     }
10  ],
11  "couponCode": "SAVE10"
12 }
```
- Response (JSON):**

```
39 {
40   "spiceLevel": "NONE",
41   "disclaimer": "Academic metadata; it is not a cross-contamination certification."
42 },
43 {
44   "quantity": 2,
45   "unitPriceAtPurchase": 3.75,
46   "subtotal": 7.50
47 },
48 ],
49 "subtotalBeforeDiscount": 7.50,
50 "discountAmount": 0.75,
51 "total": 6.75,
52 "currency": "USD",
53 "couponCode": "SAVE10"
54 }
```
- Status:** 200 OK
- Time:** 48 ms
- Size:** 1.09 KB

Cupón vencido

The screenshot shows a REST client interface with the following details:

- Method:** POST
- URL:** http://localhost:8080/api/orders/quote
- Body (raw):**

```
1 {
2   "items": [
3     {
4       "taco": {
5         "name": "Quote TC15",
6         "ingredientIds": ["FLT0", "CARN"]
7       },
8       "quantity": 2
9     }
10  ],
11  "couponCode": "OLD10"
12 }
```
- Response (JSON):**

```
1 {
2   "type": "urn:tacocloud:problem:business-rule",
3   "title": "Business rule violation",
4   "status": 422,
5   "detail": "The coupon cannot be applied to this order.",
6   "instance": "/api/orders/quote",
7   "code": "COUPON_NOT_APPLICABLE",
8   "violations": []
9 }
```
- Status:** 422 Unprocessable Entity (WebDAV) (RFC 4918)
- Time:** 39 ms
- Size:** 663 B

TC-16 – Reservar y liberar inventario sin vender aire

Este reto busca evitar sobreventa mediante actualizaciones Mongo atómicas, compensación, idempotencia interna y liberación al cancelar.

Para cumplir con este objetivo se modificaron:

- OrderService — reserva, compensación, confirmación y liberación.
- TacoOrder — clave interna de reserva no expuesta.
- ApiErrorCodes — INSUFFICIENT_STOCK.
- tacocloud-api/pom.xml — Mongo embebido sólo para tests.
- OrderServiceTest.

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.InventoryConcurrencyIntegrationTest
12:38:52.936 [main] DEBUG org.springframework.test.context.BootstrapUtils - Instantiating CacheAwareContextLoaderDelegate from class [org.springframework.test.context.cache.DefaultCacheAwareContextLoaderDelegate]
12:38:52.957 [main] DEBUG org.springframework.test.context.BootstrapUtils - Instantiating BootstrapContext using constructor [public org.springframework.test.context.support.DefaultBootstrapContext(java.lang.Class,org.springframework.test.context.CacheAwareContextLoaderDelegate)]
12:38:53.020 [main] DEBUG org.springframework.test.context.BootstrapUtils - Instantiating TestContextBootstrapper for test class [tacos.web.api.InventoryConcurrencyIntegrationTest] from class [org.springframework.boot.test.autoconfigure.data.mongo.DataMongoTestContextBootstrapper]
12:38:53.042 [main] INFO org.springframework.boot.test.autoconfigure.data.mongo.DataMongoTestContextBootstrapper - Neither @ContextConfiguration nor @Context

spring Boot-TacoCloudApplication<tacocloud_tc14> (Taco-Cloud-Practice) Live Share Java: Ready Spaces: 4 UTF-8 CRLF JSON with Comments Go Live Prettier

PS C:\Users\luisa\Documents\NTTDATA\Taco-Cloud-Practice\tacocloud> mvn test -Dtest=InventoryConcurrencyIntegrationTest -DfailIfNoTests=false
2026-09-29 12:39:01.076 INFO 4512 --- [ Thread-1] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-29T12:39:01.077-0600 I STORAGE [conn6] shutdown
: removing fs lock...
2026-09-29 12:39:01.077 INFO 4512 --- [ Thread-1] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-29T12:39:01.077-0600 I CONTROL [conn6] now exit
ing
2026-09-29 12:39:01.077 INFO 4512 --- [ Thread-1] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-29T12:39:01.077-0600 I CONTROL [conn6] shutting
down with code:0
2026-09-29 12:39:01.077 INFO 4512 --- [ Thread-1] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-29T12:39:01.077-0600 I CONTROL [initandlisten]
shutting down with code:0
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.OrderServiceTest
12:41:21.596 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 14, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.906 s - in tacos.web.api.OrderServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 14, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman

Se realizo una orden de dos tacos:

```
Pruebas TacoCloud > TC-16 - Reservar y liberar inventario sin vender aire > postOrder

POST http://localhost:8080/api/orders

Body
  none form-data x-www-form-urlencoded raw binary GraphQL JSON
  raw
  {
    "deliveryState": "ST",
    "deliveryZip": "00000",
    "paymentMethodId": "6abc07791a62d353a96bc946",
    "items": [
      {
        "taco": {
          "name": "Taco inventario",
          "ingredientIds": ["FLTO", "CARN"]
        },
        "quantity": 2
      }
    ]
  }

Body Cookies Headers Test Results
  JSON Preview Visualize
  {
    "disclaimer": "Academic metadata; it is not a cross-contamination certification.",
    },
    "quantity": 2,
    "unitPriceAtPurchase": 3.75,
    "subtotal": 7.50
  },
  "subtotalBeforeDiscount": 7.50,
  "discountAmount": 0.00,
  "total": 7.50,
  "currency": "USD",
  "couponCode": null
}
```

Se quita stock en el inventario:

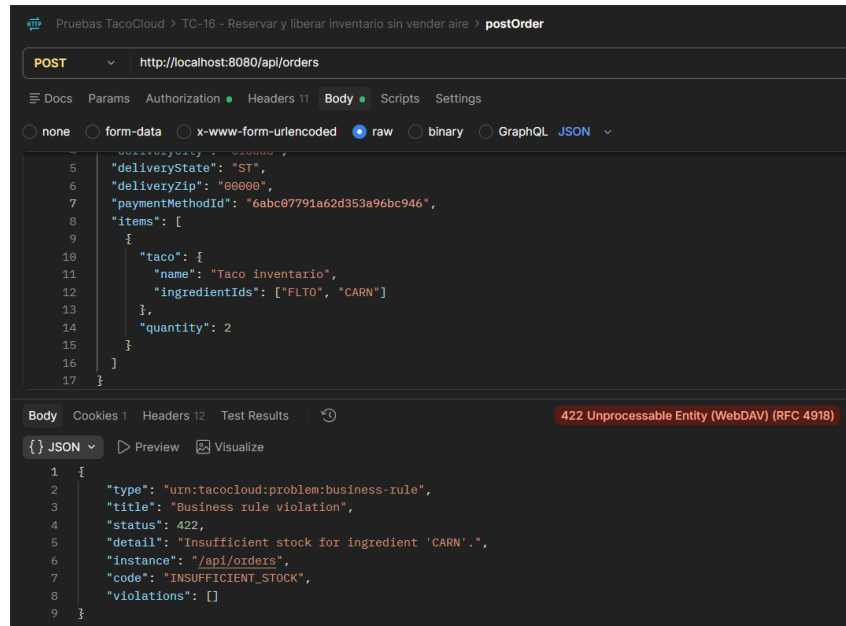
```
Pruebas TacoCloud > TC-16 - Reservar y liberar inventario sin vender aire > preparacionCaso

POST http://localhost:8080/api/admin/ingredients/CARN/stock-adjustments

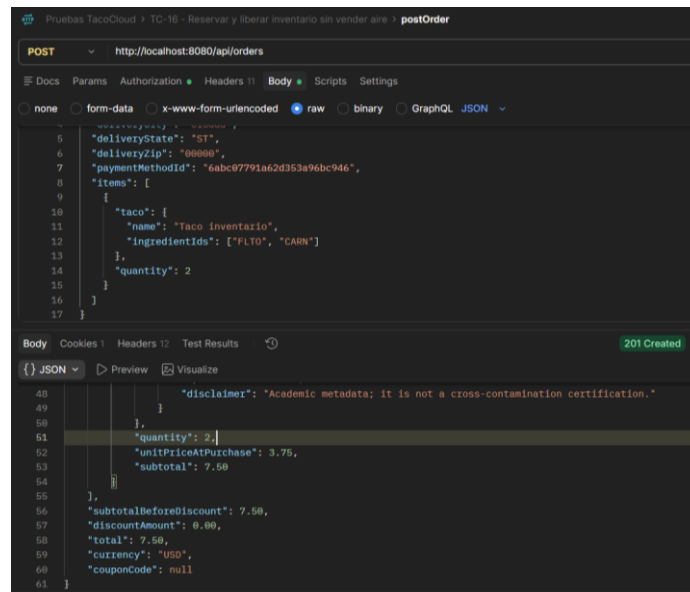
Body
  none form-data x-www-form-urlencoded raw binary GraphQL JSON
  raw
  {
    "adjustment": "-97"
  }

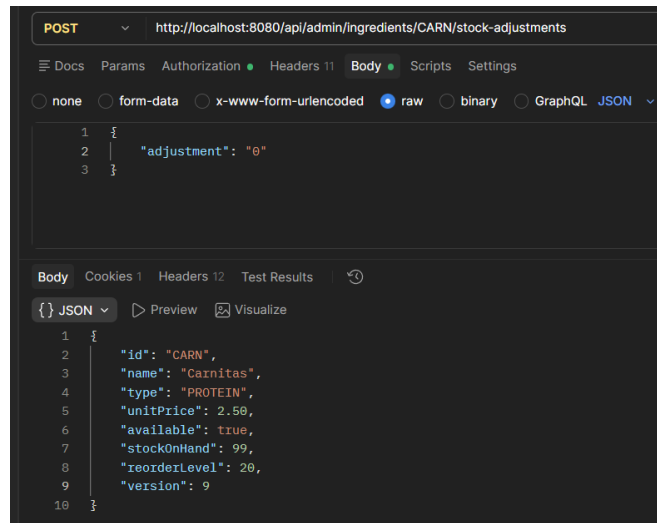
Body Cookies Headers Test Results
  JSON Preview Visualize
  {
    "id": "CARN",
    "name": "Carnitas",
    "type": "PROTEIN",
    "unitPrice": 2.50,
    "available": true,
    "stockOnHand": 1,
    "reorderLevel": 20,
    "version": 3
  }
```

Cuando pide la orden de dos tacos, stock insuficiente:

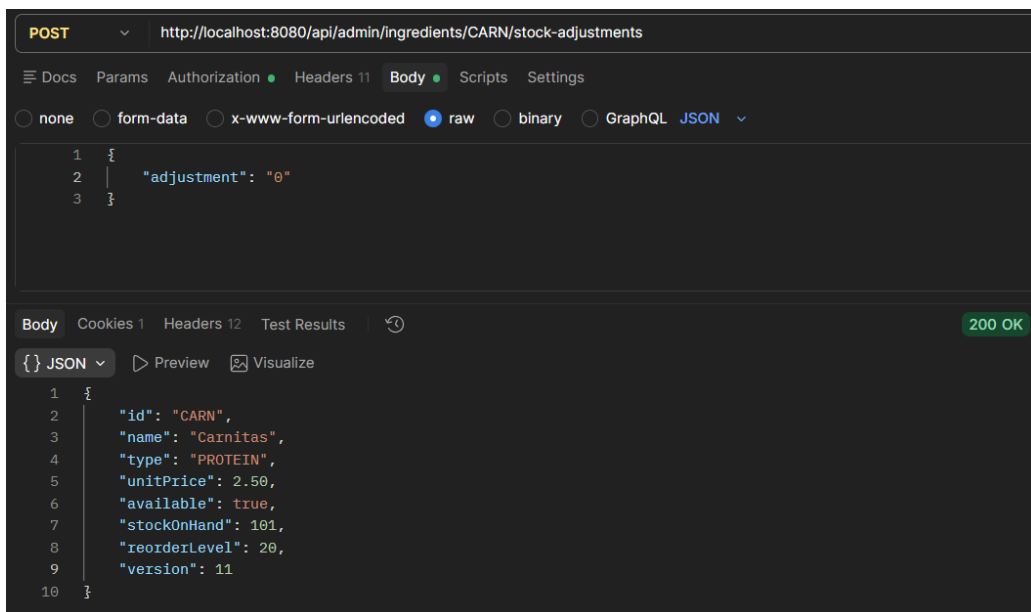
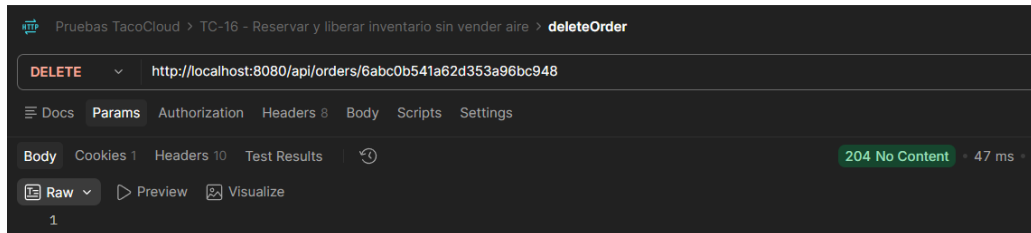


Cancelando una orden, con id 6abc0b541a62d353a96bc948





Al cancelarla, regresa el stock:



TC-17 – Etiquetas dietarias, alérgenos y nivel de picante

El objetivo por cumplir es agregar metadata tipada a ingredientes y derivar la clasificación del taco exclusivamente del lado servidor.

Para esto se agregaron diferentes políticas

- VEGAN, VEGETARIAN y GLUTEN_FREE: intersección; todos los ingredientes deben cumplir.
- allergens: unión exacta.
- spiceLevel: máxima severidad explícita, sin depender de ordinal().
- Se incluye disclaimer académico.
- Campos enviados por el cliente son ignorados.

Test

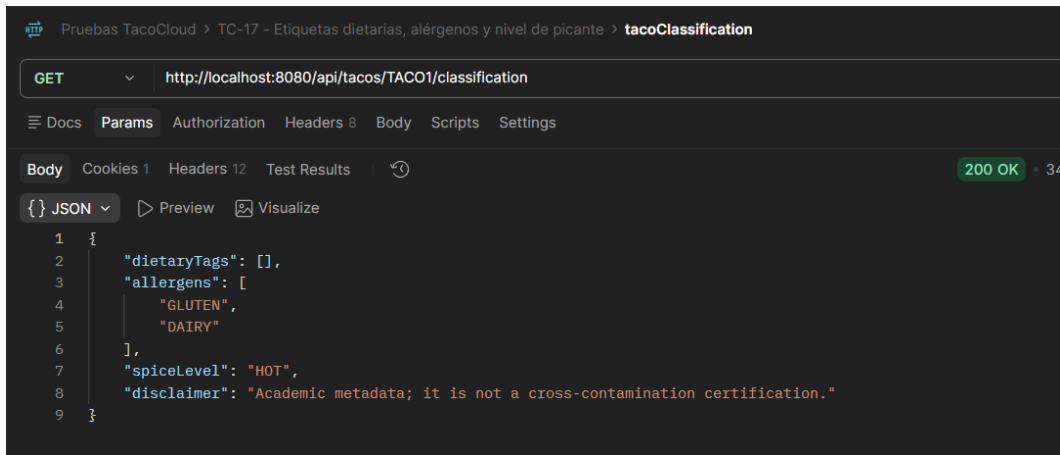
```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.TacoClassificationServiceTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.055 s - in tacos.web.api.TacoClassificationServiceTest
[INFO] Results:
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
```

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.api.dto.TacoCreateRequestTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.422 s - in tacos.api.dto.TacoCreateRequestTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
```

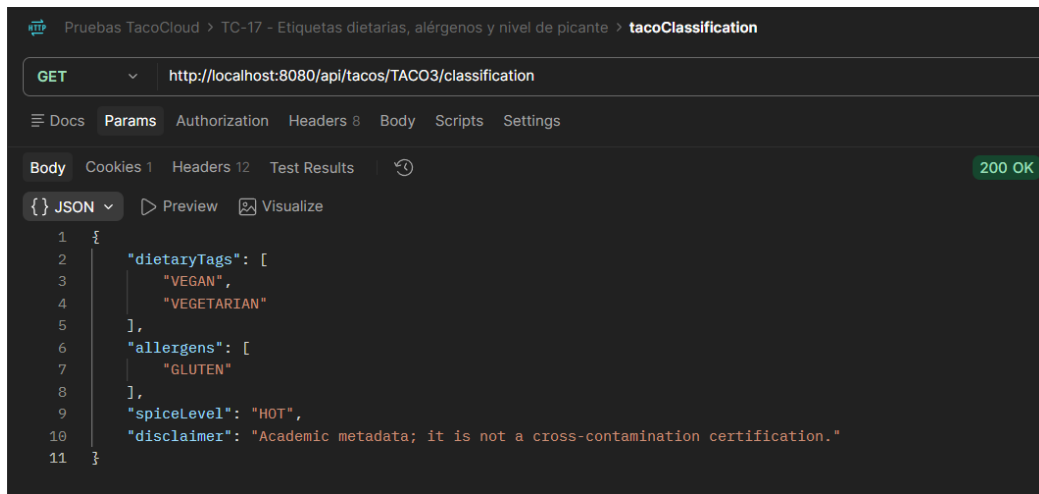
```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.TacoControllerTest
13:13:13.329 [main] DEBUG reactor.util.loggers - Using Slf4j logging framework
13:13:13.593 [main] DEBUG org.springframework.context.annotation.AnnotationConfigApplicationContext - Refreshing org.springframework.context.annotation.AnnotationConfigApplicationContext@3ecedf21
13:13:13.615 [main] DEBUG org.springframework.beans.factory.support.DefaultListableBeanFactory - Creating shared instance of singleton bean 'org.springframework.context.annotation.internalConfigurationAnnotationProcessor'
13:13:13.775 [main] DEBUG org.springframework.beans.factory.support.DefaultListableBeanFactory - Creating shared instance of singleton bean 'org.springframework.context.event.internalEventListenerProcessor'
Spring Boot-TacoCloudApplication<tacocloud_tcl4> (Taco-Cloud-Practice) Live Share Java: Ready Spaces: 4 UTF-8 CRLF JSON with Comments Go Live Prettier
PS C:\Users\Luisa\Documents\WTTDATA\Taco-Cloud-Practice\tacocloud> mvn test -Dtest=TacoControllerTest -DfailIfNoTests=false
13:13:16.410 [main] DEBUG com.jayway.jsonpath.internal.path.CompiledPath - Evaluating path: $[0]['id']
13:13:16.411 [main] DEBUG com.jayway.jsonpath.internal.path.CompiledPath - Evaluating path: $[0]['name']
13:13:16.412 [main] DEBUG com.jayway.jsonpath.internal.path.CompiledPath - Evaluating path: $[1]['id']
13:13:16.413 [main] DEBUG com.jayway.jsonpath.internal.path.CompiledPath - Evaluating path: $[1]['name']
13:13:16.414 [main] DEBUG com.jayway.jsonpath.internal.path.CompiledPath - Evaluating path: $[11]['id']
13:13:16.415 [main] DEBUG com.jayway.jsonpath.internal.path.CompiledPath - Evaluating path: $[11]['name']
13:13:16.416 [main] DEBUG com.jayway.jsonpath.internal.path.CompiledPath - Evaluating path: $[12]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 4.059 s - in tacos.web.api.TacoControllerTest
[INFO] Results:
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
```

Postman

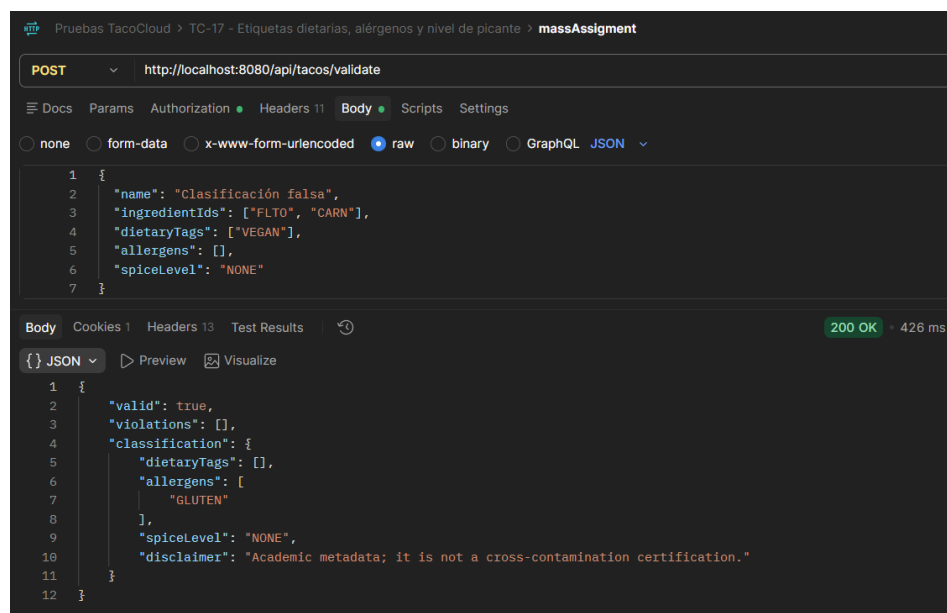
Clasificación normal, donde se observa los alérgenos y el nivel de picante:



Taco vegano:



Prueba de mass assignment:



TC-18 – Taco Physics: reglas compatibles de diseño

Se tiene como objetivo centralizar las reglas del diseño en componentes extensibles y reutilizarlas en validación, quote y creación.

Para esto se modificaron los archivos:

- Ingredient.Type — agrega BOWL.
- TacoCreateRequest — máximo 12.
- OrderService — mismas reglas para create y quote.
- ApiExceptionHandler — violations en ApiProblem.
- SecurityConfig — validate para USER/ADMIN.
- application.yml — fun rules por tipo, sin IDs mágicos.

Donde se implementaron:

- Exactamente una base WRAP o BOWL.
- Entre 2 y 12 ingredientes.
- Sin duplicados.
- Sólo ingredientes disponibles.
- HOT requiere VEGGIES.
- CHEESE requiere SAUCE.
- Todas las violaciones se agregan.
- TacoDesignValidator recibe List<TacoDesignRule>.
- Añadir una regla no requiere modificar el validador.

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.physics.TacoDesignValidatorTest
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.063 s - in tacos.physics.TacoDesignValidatorTest
[INFO] Results:
[INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0
```

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.OrderServiceTest
13:27:53.902 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 14, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.836 s - in tacos.web.api.OrderServiceTest
[INFO] Results:
[INFO] Tests run: 14, Failures: 0, Errors: 0, Skipped: 0
```

Postman

Ejemplo valido

The screenshot shows a REST client interface for a test named "postvalidate". The request is a POST to "http://localhost:8080/api/tacos/validate" with a JSON body. The response is a 200 OK with a JSON body indicating the taco is valid.

```
POST http://localhost:8080/api/tacos/validate

{
  "name": "Taco TC18",
  "ingredientIds": ["FLT0", "CARN"]
}
```

Body: Cookies 1 Headers 13 Test Results 200 OK

```
{
  "valid": true,
  "violations": [],
  "classification": {
    "dietaryTags": [],
    "allergens": [
      "GLUTEN"
    ],
    "spiceLevel": "NONE",
    "disclaimer": "Academic metadata; it is not a cross-contamination certification."
  }
}
```

Sin base

The screenshot shows a REST client interface for a test named "postvalidate". The request is a POST to "http://localhost:8080/api/tacos/validate" with a JSON body. The response is a 200 OK with a JSON body indicating the taco is invalid due to missing a base.

```
POST http://localhost:8080/api/tacos/validate

{
  "name": "Sin base",
  "ingredientIds": ["CARN", "LETC"]
}
```

Body: Cookies 1 Headers 12 Test Results 200 OK · 19 ms · 7

```
{
  "valid": false,
  "violations": [
    {
      "code": "INVALID_BASE_COUNT",
      "message": "A taco must contain exactly one base."
    }
  ],
  "classification": {
    "dietaryTags": [
      "GLUTEN_FREE"
    ],
    "allergens": [],
    "spiceLevel": "NONE",
    "disclaimer": "Academic metadata; it is not a cross-contamination certification."
  }
}
```

Dos bases

```
Pruebas TacoCloud > TC-18 - Taco Pysics: reglas compatibles de diseño > postvalidate

POST http://localhost:8080/api/tacos/validate

Body
  none form-data x-www-form-urlencoded raw binary GraphQL JSON
  1 {
  2   "name": "Dos bases",
  3   "ingredientIds": ["FLT0","COT0"]
  4 }

Body Cookies 1 Headers 12 Test Results 200 OK
  {} JSON Preview Visualize
  1 {
  2   "valid": false,
  3   "violations": [
  4     {
  5       "code": "INVALID_BASE_COUNT",
  6       "message": "A taco must contain exactly one base."
  7     }
  8   ],
  9   "classification": {
 10     "dietaryTags": [
 11       "VEGAN",
 12       "VEGETARIAN"
 13     ],
 14     "allergens": [
 15       "GLUTEN"
 16     ],
 17     "spiceLevel": "NONE",
 18     "disclaimer": "Academic metadata; it is not a cross-contamination certification."
 19   }
 20 }
```

Ingrediente no disponible

```
Pruebas TacoCloud > TC-18 - Taco Pysics: reglas compatibles de diseño > postvalidate

POST http://localhost:8080/api/tacos/validate

Body
  none form-data x-www-form-urlencoded raw binary GraphQL JSON
  1 {
  2   "name": "Dos bases",
  3   "ingredientIds": ["FLT0","CARN"]
  4 }

Body Cookies 1 Headers 12 Test Results 200 OK
  {} JSON Preview Visualize
  1 {
  2   "valid": false,
  3   "violations": [
  4     {
  5       "code": "INGREDIENT_NOT_AVAILABLE",
  6       "message": "Ingredient 'CARN' is not available."
  7     }
  8   ],
  9   "classification": {
 10     "dietaryTags": [],
 11     "allergens": [
 12       "GLUTEN"
 13     ],
 14     "spiceLevel": "NONE",
 15     "disclaimer": "Academic metadata; it is not a cross-contamination certification."
 16   }
 17 }
```

Orden sin base

Pruebas TacoCloud > TC-18 - Taco Physics: reglas compatibles de diseño > postQuote

POST http://localhost:8080/api/orders/quote

Docs Params Authorization Headers 10 Body Scripts Settings

none form-data x-www-form-urlencoded raw binary GraphQL JSON

```
4 {
5   "deliveryState": "ST",
6   "deliveryZip": "00000",
7   "paymentMethodId": "6abc07791a62d353a96bc946",
8   "items": [
9     {
10      "taco": {
11        "name": "Taco sin base",
12        "ingredientIds": ["LETC", "CARN"]
13      },
14      "quantity": 2
15    }
16  ]
17 }
```

Body Cookies 1 Headers 12 Test Results 422 Unprocessable Entity (WebDAV) (RFC 4918)

{ JSON Preview Visualize

```
1 {
2   "type": "urn:tacocloud:problem:business-rule",
3   "title": "Business rule violation",
4   "status": 422,
5   "detail": "The taco design violates one or more rules.",
6   "instance": "/api/orders/quote",
7   "code": "INVALID_TACO_DESIGN",
8   "violations": [
9     {
10      "field": "INVALID_BASE_COUNT",
11      "message": "A taco must contain exactly one base."
12    }
13  ]
14 }
```

Laboratorio 4 – Funciones que sí dan ganas de usar

TC-19 – Buscar, filtrar, ordenar y paginar tacos

Se realizaron cambios o modificaciones en los archivos

- TacoCatalogService.java
- TacoController.java
- PagedResponse.java
- TacoSearchCriteria.java
- TacoMapper.java
- TacoResponse.java
- RecentTacosService.ts
- recents.component.ts

Test

```
[INFO] -----  
[INFO] T E S T S  
[INFO] -----  
[INFO] Running tacos.web.api.Tc19CatalogValidationTest  
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.535 s - in tacos.web.api.Tc19CatalogValidationTest  
[INFO] Results:  
[INFO]  
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0  
[INFO]
```

Postman

Catálogo filtrado y paginado:

```
Pruebas TacoCloud > TC-19 - Buscar, filtrar, ordenar y paginar tacos > getTacos

GET http://localhost:8080/api/tacos?name=Carn&ingredientId=FLTO&page=0&size=10&sort=name&direction=asc

Body Cookies 1 Headers 12 Test Results 1/1 200 OK

{} JSON Preview Visualize

1  {
2    "items": [
3      {
4        "id": "TAC01",
5        "createdAt": "2026-09-29T21:35:48.148+00:00",
6        "name": "Carnivore",
7        "ingredients": [
8          {
9            "id": "FLTO",
10           "name": "Flour Tortilla",
11           "type": "WRAP",
12           "unitPrice": 1.25,
13           "available": true,
14           "dietaryTags": [
15             "VEGAN",
16             "VEGETARIAN"
17           ],
18           "allergens": [
19             "GLUTEN"
20           ],
21           "spiceLevel": "NONE"
22         },
23         {
24           "id": "GRBF",
25           "name": "Ground Beef",
26           "type": "PROTEIN",
27           "unitPrice": 2.35,
28           "available": true,
29           "dietaryTags": [
30             "GLUTEN_FREE"
```

```
Pruebas TacoCloud > TC-19 - Buscar, filtrar, ordenar y paginar tacos > getTacos

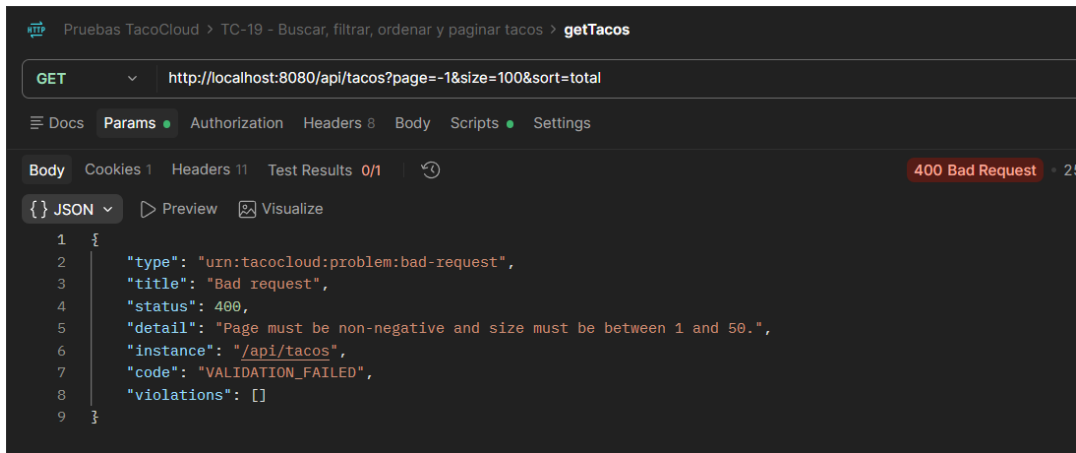
GET http://localhost:8080/api/tacos?name=Carn&ingredientId=FLTO&page=0&size=10&sort=name&direction=asc

Body Cookies 1 Headers 12 Test Results 1/1 200 OK

{} JSON Preview Visualize

79     "type": "CHEESE",
80     "unitPrice": 1.20,
81     "available": true,
82     "dietaryTags": [
83       "VEGETARIAN",
84       "GLUTEN_FREE"
85     ],
86     "allergens": [
87       "DAIRY"
88     ],
89     "spiceLevel": "NONE"
90   },
91   ],
92   "classification": {
93     "dietaryTags": [],
94     "allergens": [
95       "GLUTEN",
96       "DAIRY"
97     ],
98     "spiceLevel": "HOT",
99     "disclaimer": "Academic metadata; it is not a cross-contamination certification."
100  },
101  },
102  ],
103  "page": 0,
104  "size": 10,
105  "totalElements": 1,
106  "totalPages": 1,
107  "hasNext": false
108 }
```

Validación de paginación:



TC-20 – Taco del día determinista y comprobable

Archivos modificados:

- TacoOfTheDayService.java
- TacoOfTheDayResponse.java
- TacoController.java
- TimeConfiguration.java
- application.yml

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.Tc20TacoOfTheDayServiceTest
15:49:25.567 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.239 s - in tacos.web.api.Tc20TacoOfTheDayServiceTest
[INFO] Results:
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman

Obteniendo taco del día:

Pruebas TacoCloud > TC-20 - Taco del día determinista y comprobable > **getTacos**

GET **http://localhost:8080/api/tacos/today**

Docs Params Authorization Headers 8 Body Scripts Settings

Body Cookies 1 Headers 12 Test Results 200 OK

{ } JSON Preview Visualize

```
1 {
2   "taco": {
3     "id": "TAC02",
4     "createdAt": "2026-09-29T21:52:46.566+00:00",
5     "name": "Bovine Bounty",
6     "ingredients": [
7       {
8         "id": "COTO",
9         "name": "Corn Tortilla",
10        "type": "WRAP",
11        "unitPrice": 1.10,
12        "available": true,
13        "dietaryTags": [
14          "VEGAN",
15          "VEGETARIAN",
16          "GLUTEN_FREE"
17        ],
18        "allergens": [],
19        "spiceLevel": "NONE"
20      },
21      {
22        "id": "GRBF",
23        "name": "Ground Beef",
24        "type": "PROTEIN",
25        "unitPrice": 2.35,
26        "available": true,
27        "dietaryTags": [
28          "GLUTEN_FREE"
29        ],
30        "allergens": [],
```

Pruebas TacoCloud > TC-20 - Taco del día determinista y comprobable > **getTacos**

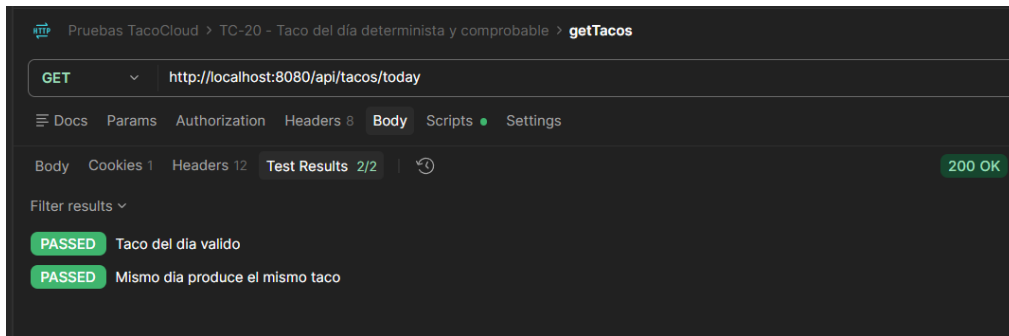
GET **http://localhost:8080/api/tacos/today**

Docs Params Authorization Headers 8 Body Scripts Settings

Body Cookies 1 Headers 12 Test Results 200 OK

{ } JSON Preview Visualize

```
63 {
64   "id": "SRCR",
65   "name": "Sour Cream",
66   "type": "SAUCE",
67   "unitPrice": 0.95,
68   "available": true,
69   "dietaryTags": [
70     "VEGETARIAN",
71     "GLUTEN_FREE"
72   ],
73   "allergens": [
74     "DAIRY"
75   ],
76   "spiceLevel": "NONE"
77 },
78 ],
79 "classification": {
80   "dietaryTags": [
81     "GLUTEN_FREE"
82   ],
83   "allergens": [
84     "DAIRY"
85   ],
86   "spiceLevel": "NONE",
87   "disclaimer": "Academic metadata; it is not a cross-contamination certification."
88 },
89 },
90 "date": "2026-09-29",
91 "reason": "Deterministic daily selection among valid tacos with available ingredients."
92 }
```



TC-21 – Favoritos por usuario sin confiar en userId del cliente

Se agregó la funcionalidad de favoritos mediante tres operaciones:

- PUT /api/users/me/favorites/{tacold}
- DELETE /api/users/me/favorites/{tacold}
- GET /api/users/me/favorites?page=0&size=20

Se realizaron cambios en los archivos:

- Favorite.java
- FavoriteService.java
- FavoriteController.java
- FavoriteResponse.java
- FavoriteService.ts
- recents.component.html
- SecurityConfig.java

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.Tc21FavoriteServiceTest
16:06:36.533 [main] DEBUG reactor.util.loggers - Using Slf4j logging framework
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.952 s - in tacos.web.api.Tc21FavoriteServiceTest
[INFO] Results:
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman

Agregando favorito:

```
Pruebas TacoCloud > TC-21 - Favoritos por usuario sin confiar en userId del cliente > putFavorite

PUT http://localhost:8080/api/users/me/favorites/TACO1

Body 200 OK - 720 ms

{
  "tacoId": "TACO1",
  "createdAt": "2026-09-29T22:47:45.051Z",
  "taco": {
    "id": "TACO1",
    "createdAt": "2026-09-29T22:46:37.736+00:00",
    "name": "Carnivore",
    "ingredients": [
      {
        "id": "FLTO",
        "name": "Flour Tortilla",
        "type": "WRAP",
        "unitPrice": 1.25,
        "available": true,
        "dietaryTags": [
          "VEGAN",
          "VEGETARIAN"
        ],
        "allergens": [
          "GLUTEN"
        ],
        "spiceLevel": "NONE"
      },
      {
        "id": "GRBF",
        "name": "Ground Beef",
        "type": "PROTEIN",
        "unitPrice": 2.35,
        "available": true,
        "dietaryTags": [

```

Consultar favoritos:

```
Pruebas TacoCloud > TC-21 - Favoritos por usuario sin confiar en userId del cliente > getFavorites

GET http://localhost:8080/api/users/me/favorites?page=0&size=20

Body 200 OK - 53 ms - 1.

{
  "unitPrice": 1.25,
  "available": true,
  "dietaryTags": [
    "VEGETARIAN",
    "GLUTEN_FREE"
  ],
  "allergens": [
    "DAIRY"
  ],
  "spiceLevel": "NONE"
},
{
  "classification": {
    "dietaryTags": [],
    "allergens": [
      "GLUTEN",
      "DAIRY"
    ],
    "spiceLevel": "HOT",
    "disclaimer": "Academic metadata; it is not a cross-contamination certification."
  }
},
{
  "page": 0,
  "size": 20,
  "totalElements": 1,
  "totalPages": 1,
  "hasNext": false
}
```

Validando que otro usuario no pueda ver los favoritos de otro

Pruebas TacoCloud > TC-21 - Favoritos por usuario sin confiar en userId del cliente > getFavorites

GEThttp://localhost:8080/api/users/me/favorites?page=0&size=20

DocsParamsAuthorizationHeaders9BodyScriptsSettings

Auth type

Basic Auth

The authorization header will be automatically generated when you send the request. Learn more about [authorization](#)

Basic Auth

Usernameuser-a

Password

BodyCookies1Headers12Test Results200 OK99 ms

JSON

PreviewVisualize

```
100  ],
101      "spiceLevel": "HOT",
102      "disclaimer": "Academic metadata; it is not a cross-contamination certification."
103  }
104  }
105  ],
106  "page": 0,
107  "size": 20,
108  "totalElements": 1,
109  "totalPages": 1,
110  "hasNext": false
111  }
112  }
```

Pruebas TacoCloud > TC-21 - Favoritos por usuario sin confiar en userId del cliente > getFavorites

GEThttp://localhost:8080/api/users/me/favorites?page=0&size=20

DocsParamsAuthorizationHeaders9BodyScriptsSettings

Auth type

Basic Auth

The authorization header will be automatically generated when you send the request. Learn more about [authorization](#)

Basic Auth

Usernameuser-b

Password

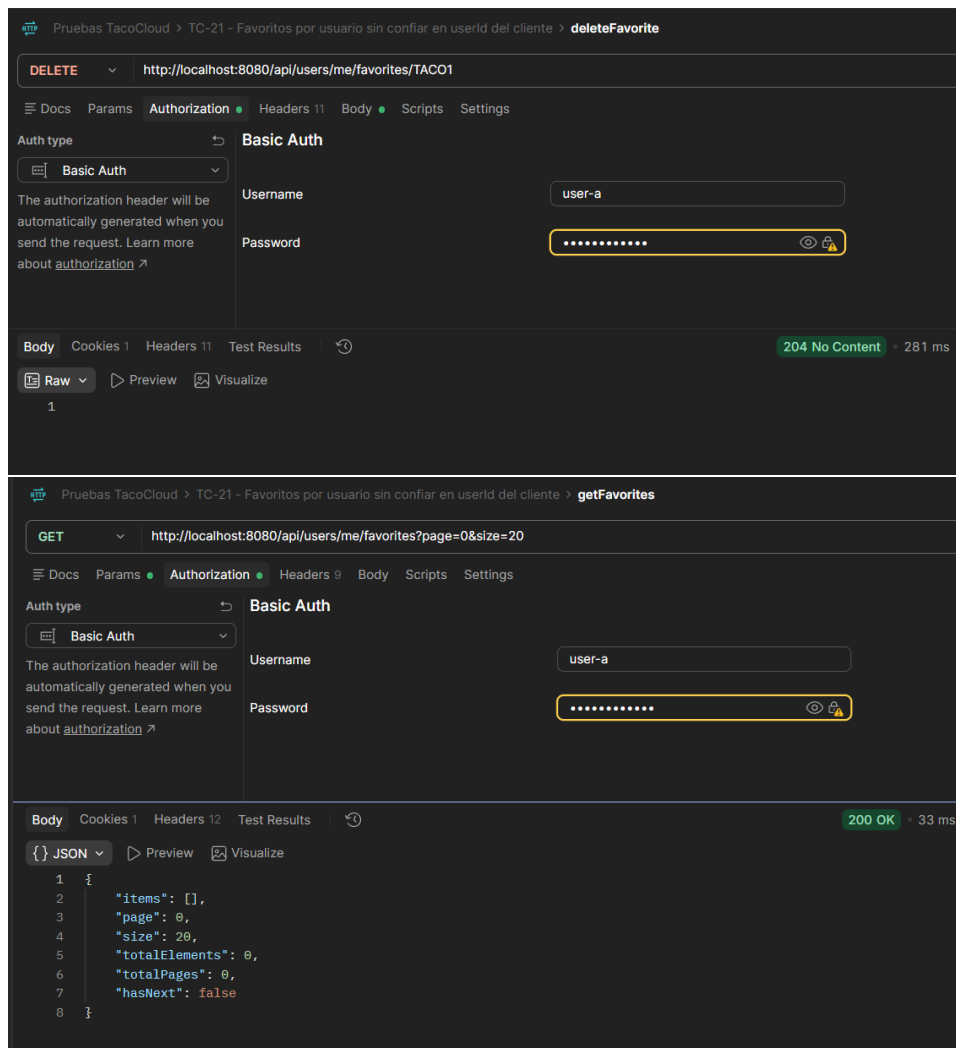
BodyCookies1Headers12Test Results200 OK110 ms

JSON

PreviewVisualize

```
1  {
2      "items": [],
3      "page": 0,
4      "size": 20,
5      "totalElements": 0,
6      "totalPages": 0,
7      "hasNext": false
8  }
```

Eliminando favorito:



TC-22 – Calificaciones y ranking de tacos

Se incorporó un sistema de calificaciones mediante:

- PUT /api/tacos/{tacold}/rating
- GET /api/tacos/top?limit=10

Cada usuario puede asignar una puntuación entre 1 y 5. La relación userId + tacold posee un índice único, por lo que un segundo PUT realizado por el mismo usuario no crea otro voto, sino que actualiza su calificación anterior.

Se crearon/realizaron cambios en los siguientes archivos:

- TacoRating.java
- TacoRatingService.java
- TacoRatingController.java
- RatingRequest.java

- RatingSummaryResponse.java
- application.yml

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.Tc22RatingValidationTest
17:55:41.496 [main] DEBUG org.jboss.logging - Logging Provider: org.jboss.logging.Log4j2LoggerProvider
17:55:41.508 [main] INFO org.hibernate.validator.internal.util.Version - HV000001: Hibernate Validator 6.2.0.Final
17:55:41.531 [main] DEBUG org.hibernate.validator.internal.xml.config.ValidationXmlParser - Trying to load META-INF/validation.xml for XML based Validator configuration.
17:55:41.535 [main] DEBUG org.hibernate.validator.internal.xml.config.ResourceLoaderHelper - Trying to load META-INF/validation.xml via TCCL
17:55:41.537 [main] DEBUG org.hibernate.validator.internal.xml.config.ResourceLoaderHelper - Trying to load META-INF/validation.xml via Hibernate Validator's class loader
17:55:41.538 [main] DEBUG org.hibernate.validator.internal.xml.config.ValidationXmlParser - No META-INF/validation.xml found. Using annotation based configuration only.
17:55:41.595 [main] DEBUG org.hibernate.validator.internal.engine.resolver.TraversableResolvers - Cannot find javax.persistence.Persistence on classpath. Assuming non JPA 2 environment. All properties will per default be traversable.
17:55:41.710 [main] DEBUG org.hibernate.validator.messageinterpolation.ResourceBundleMessageInterpolator - Loaded expression factory via original TCCL

spring Boot-TacoCloudApplication<tacocloud_t41> (Taco-Cloud-Practice) Live Share Java: Ready Ln 58, Col 27 Spaces: 4 UTF-8 LF Java Go Live Prettier

PS C:\Users\luisa\Documents\WTIDATA\Taco-Cloud-Practice\tacocloud> mvn test -Dtest=Tc22RatingValidationTest -DfailIfNoTests=false
1.ExecutableParameterNameProvider as ValidatorFactory-scoped parameter name provider.
17:55:44.134 [main] DEBUG org.hibernate.validator.internal.engine.ValidatorFactoryConfigurationHelper - HV000234: Using org.hibernate.validator.internal.engine.DefaultClockProvider as ValidatorFactory-scoped clock provider.
17:55:44.134 [main] DEBUG org.hibernate.validator.internal.engine.ValidatorFactoryConfigurationHelper - HV000234: Using org.hibernate.validator.internal.engine.scripting.DefaultScriptEvaluatorFactory as ValidatorFactory-scoped script evaluator factory.
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.031 s - in tacos.web.api.Tc22RatingValidationTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman

Creando voto:

Pruebas TacoCloud > TC-22 - Calificaciones y ranking de tacos > putRating

PUT http://localhost:8080/api/tacos/TACO1/rating

Body

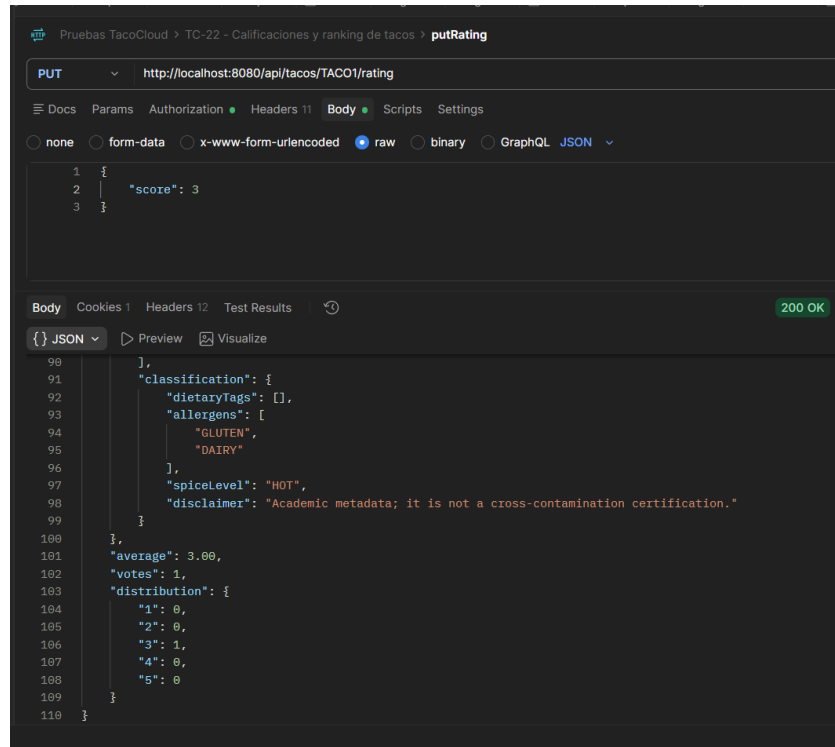
```
1 {
2   "score": 5
3 }
```

Body

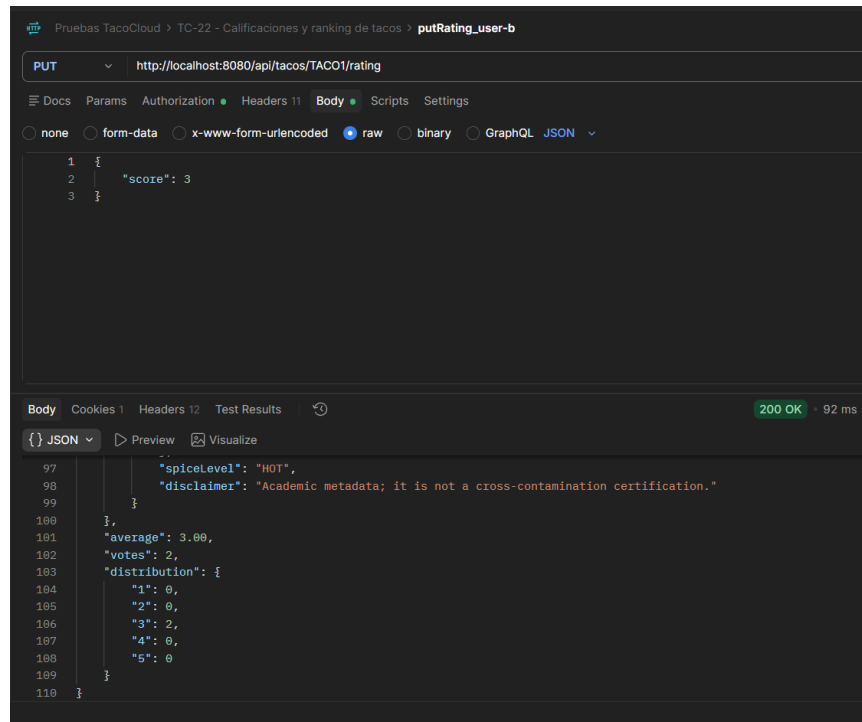
```
90 ],
91   "classification": {
92     "dietaryTags": [],
93     "allergens": [
94       "GLUTEN",
95       "DAIRY"
96     ],
97     "spiceLevel": "HOT",
98     "disclaimer": "Academic metadata; it is not a cross-contamination certification."
99   },
100   "average": 5.00,
101   "votes": 1,
102   "distribution": {
103     "1": 0,
104     "2": 0,
105     "3": 0,
106     "4": 0,
107     "5": 1
108   }
109 }
110 }
```

200 OK - 805 ms

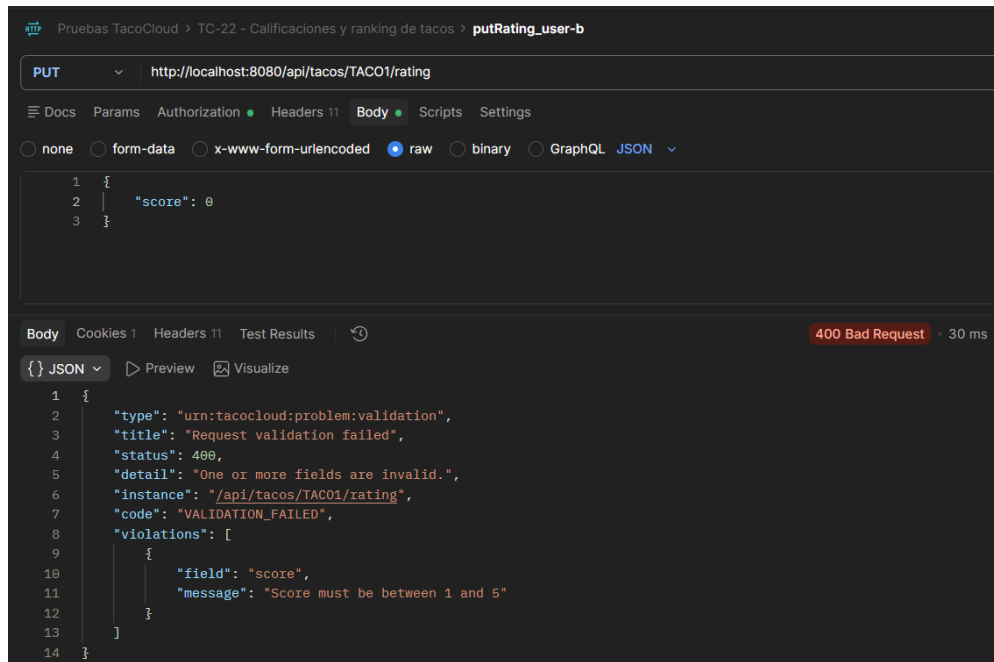
Actualizando voto:



Creando voto con otro usuario:



Puntuación inválida:



TC-23 – Historial paginado y privado de órdenes

Se reemplazó para usuarios normales el acceso global a las órdenes por rutas asociadas explícitamente al usuario autenticado:

- GET /api/users/me/orders?page=0&size=10
- GET /api/users/me/orders/{orderId}

También se crearon rutas administrativas independientes:

- GET /api/admin/orders?page=0&size=10
- GET /api/admin/orders/{orderId}

Archivos creados/modificados:

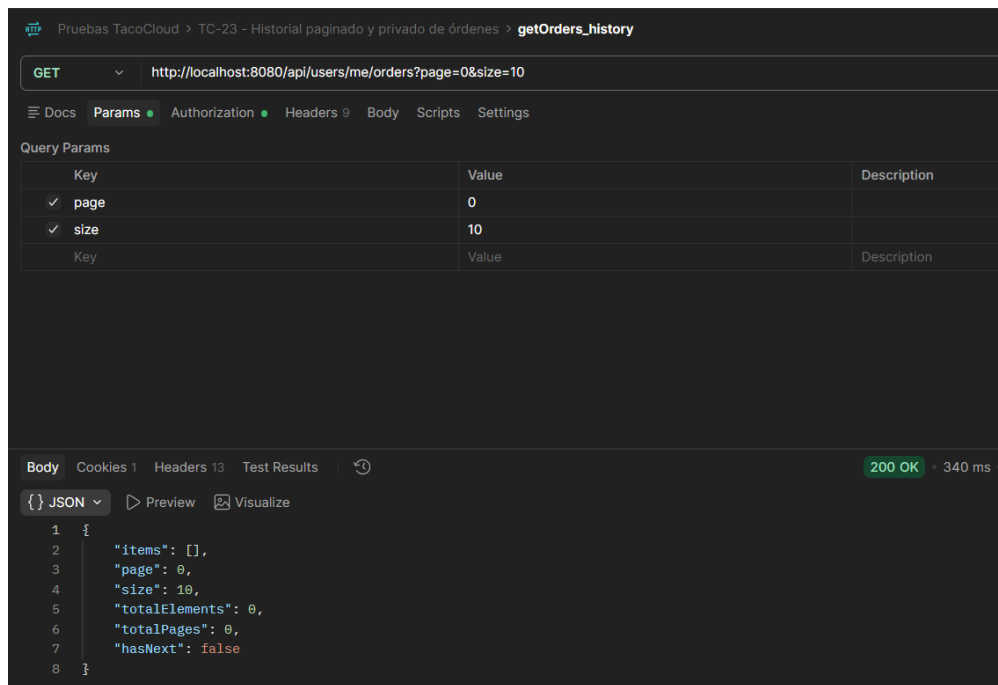
- OrderHistoryService.java
- OrderHistoryController.java
- AdminOrderHistoryController.java
- OrderSummaryResponse.java
- OrderRepository.java

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.Tc23OrderHistoryServiceTest
18:14:57.978 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.311 s - in tacos.web.api.Tc23OrderHistoryServiceTest
[INFO] Results:
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman

Viendo historial:



Pruebas TacoCloud > TC-23 - Historial paginado y privado de órdenes > **getOrders_history**

GET ▼ <http://localhost:8080/api/users/me/orders?page=0&size=10>

Docs Params **Authorization** Headers 9 Body Scripts Settings

Query Params

Key	Value	Description
✓ page	0	
✓ size	10	
Key	Value	Description

Body Cookies 1 Headers 13 Test Results ↺ **200 OK** - 340 ms

JSON ▼ ▶ Preview 🖨 Visualize

```
1 {
2   "items": [],
3   "page": 0,
4   "size": 10,
5   "totalElements": 0,
6   "totalPages": 0,
7   "hasNext": false
8 }
```

Visualizando una orden:

Primero se creó la orden

Pruebas TacoCloud > TC-16 - Reservar y liberar inventario sin vender aire > **postOrder**

POST **http://localhost:8080/api/orders**

Docs Params Authorization Headers 11 **Body** Scripts Settings

Body Cookies 1 Headers 12 Test Results **201 Created** 131 ms

{ } JSON Preview Visualize

```
1 {
2   "id": "6abc5805864ac13474524ffa",
3   "placedAt": "2026-09-30T00:29:57.407+00:00",
4   "deliveryName": "Usuario TC16",
5   "deliveryStreet": "Calle Laboratorio 16",
6   "deliveryCity": "Ciudad",
7   "deliveryState": "ST",
8   "deliveryZip": "00000",
9   "items": [
10    {
11      "taco": {
12        "id": null,
13        "createdAt": "2026-09-30T00:29:57.405+00:00",
14        "name": "Taco inventario",
15        "ingredients": [
16          {
17            "id": "FLT0",
18            "name": "Flour Tortilla",
19            "type": "WRAP",
20            "unitPrice": 1.25,
21            "available": true,
22            "dietaryTags": [
23              "VEGAN",
24              "VEGETARIAN"
25            ],
26            "allergens": [
27              "GLUTEN"
28            ],
29            "spiceLevel": "NONE"
30          }
31        ]
32      }
33    }
34  ]
35 }
```

Visualizando la orden

Pruebas TacoCloud > TC-23 - Historial paginado y privado de órdenes > **getOrders_history**

GET **http://localhost:8080/api/users/me/orders/6abc5805864ac13474524ffa**

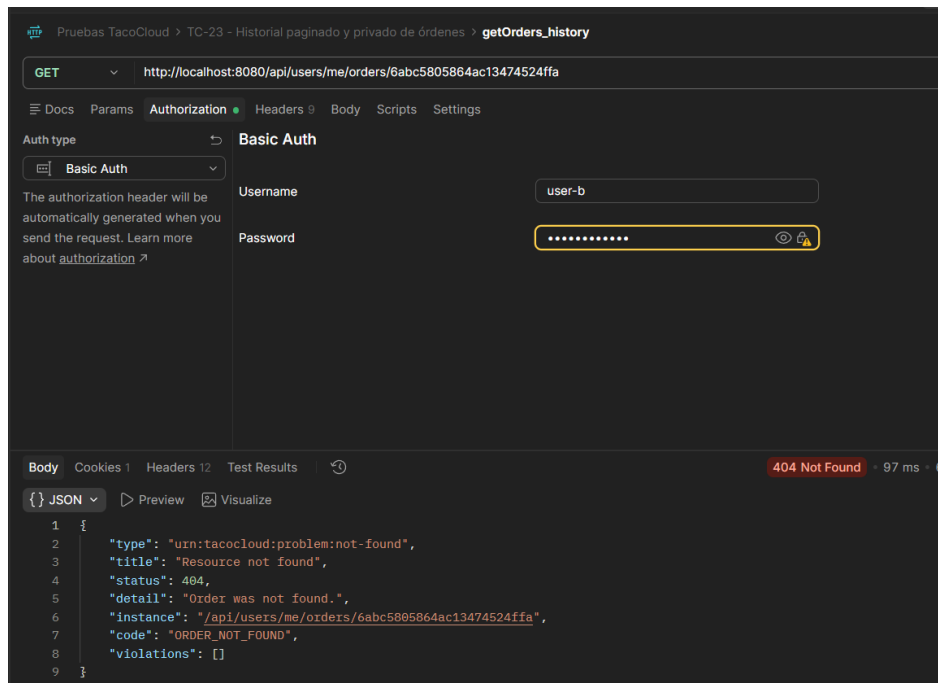
Docs Params Authorization Headers 9 **Body** Scripts Settings

Body Cookies 1 Headers 12 Test Results **200 OK** 42 ms

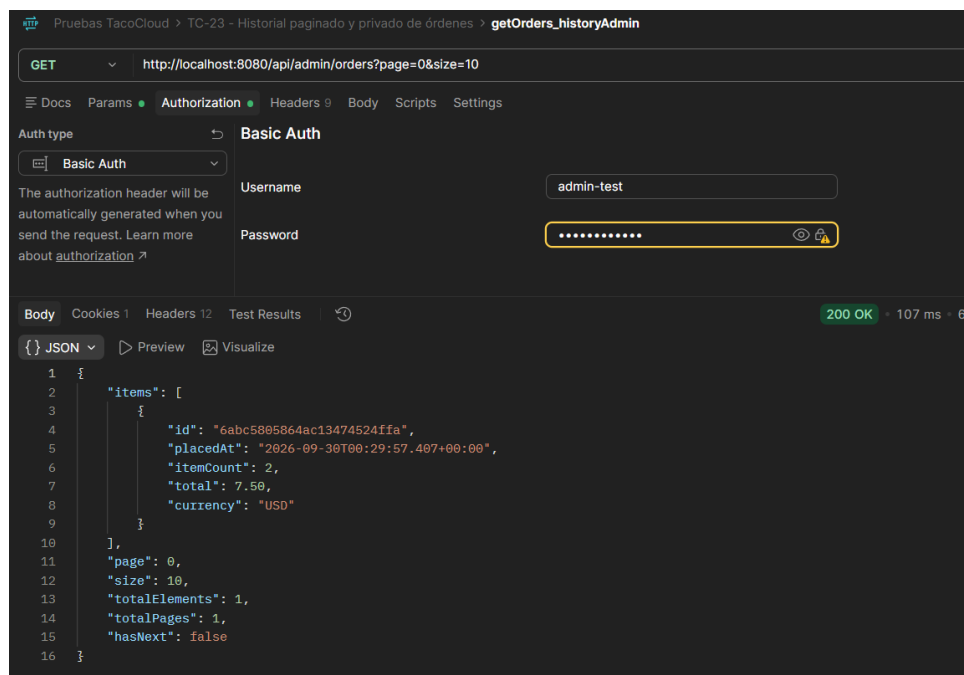
{ } JSON Preview Visualize

```
1 {
2   "id": "6abc5805864ac13474524ffa",
3   "placedAt": "2026-09-30T00:29:57.407+00:00",
4   "deliveryName": "Usuario TC16",
5   "deliveryStreet": "Calle Laboratorio 16",
6   "deliveryCity": "Ciudad",
7   "deliveryState": "ST",
8   "deliveryZip": "00000",
9   "items": [
10    {
11      "taco": {
12        "id": null,
13        "createdAt": "2026-09-30T00:29:57.405+00:00",
14        "name": "Taco inventario",
15        "ingredients": [
16          {
17            "id": "FLT0",
18            "name": "Flour Tortilla",
19            "type": "WRAP",
20            "unitPrice": 1.25,
21            "available": true,
22            "dietaryTags": [
23              "VEGAN",
24              "VEGETARIAN"
25            ],
26            "allergens": [
27              "GLUTEN"
28            ],
29            "spiceLevel": "NONE"
30          }
31        ]
32      }
33    }
34  ]
35 }
```

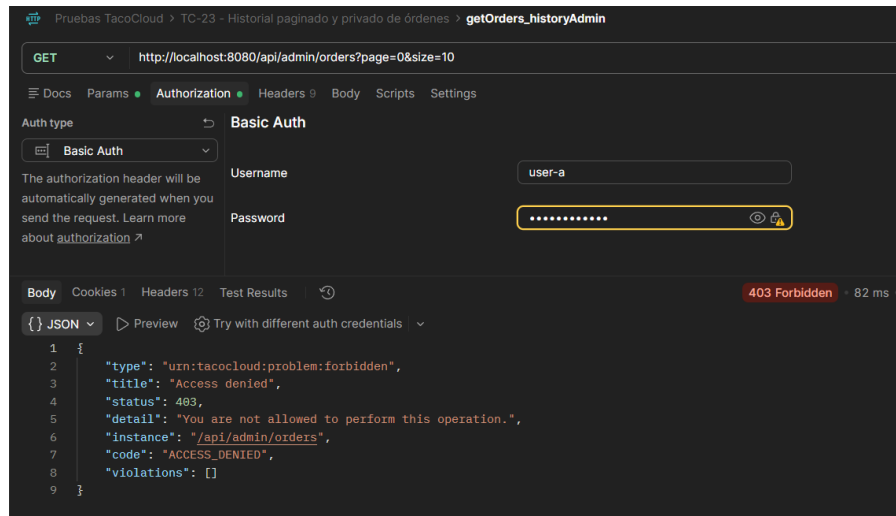
Buscando esa orden ajena con otro usuario:



Obteniendo las órdenes como administrador:



Si un usuario intenta hacer lo mismo:



TC-24 – Reordenar una compra anterior con reglas actuales

Se implementó el endpoint:

- POST /api/orders/{orderId}/reorder

Archivos creados/modificados:

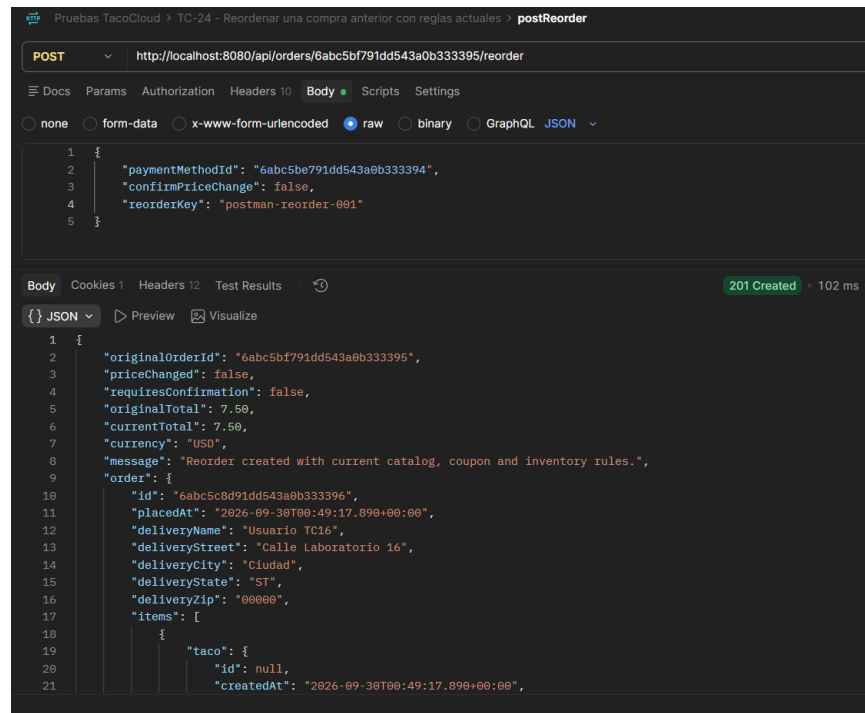
- ReorderService.java
- ReorderController.java
- ReorderOperation.java
- ReorderRequest.java
- ReorderResponse.java
- OrderService.java

Test

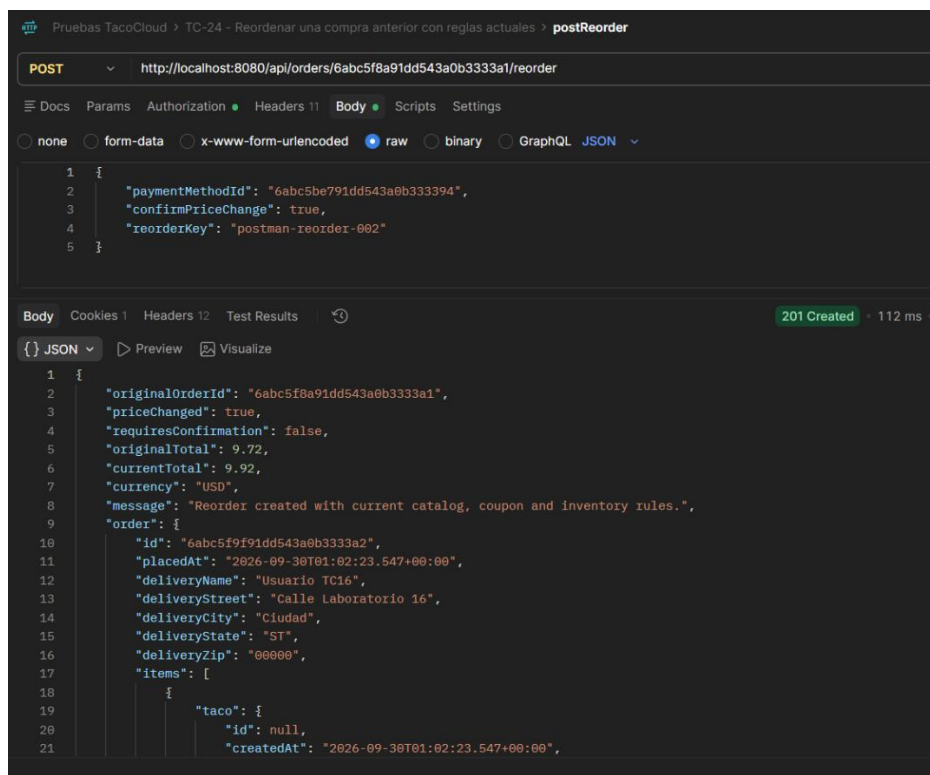
```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.Tc24ReorderValidationTest
18:39:10.863 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
18:39:11.212 [main] DEBUG org.jboss.logging - Logging Provider: org.jboss.logging.Log4j2LoggerProvider
18:39:11.214 [main] INFO org.hibernate.validator.internal.util.Version - HV000001: Hibernate Validator 6.2.0.Final
18:39:11.225 [main] DEBUG org.hibernate.validator.internal.xml.config.ValidationXmlParser - Trying to load META-INF/validation.xml for XML based Validator c
onfiguration.
18:39:11.227 [main] DEBUG org.hibernate.validator.internal.xml.config.ResourceLoaderHelper - Trying to load META-INF/validation.xml via TCCL
18:39:11.228 [main] DEBUG org.hibernate.validator.internal.xml.config.ResourceLoaderHelper - Trying to load META-INF/validation.xml via Hibernate Validator"
pring Boot-TacoCloudApplication<tacocloud_tc14> (Taco-Cloud-Practice) Live Share Java: Ready Ln 58, Col 27 Spaces: 4 UTF-8 LF Java Go Live Prettier
1. ExecutableParameterNameProvider as ValidatorFactory-scoped parameter name provider.
18:39:11.617 [main] DEBUG org.hibernate.validator.internal.engine.ValidatorFactoryConfigurationHelper - HV000234: Using org.hibernate.validator.internal.eng
ine.DefaultClockProvider as ValidatorFactory-scoped clock provider.
18:39:11.617 [main] DEBUG org.hibernate.validator.internal.engine.ValidatorFactoryConfigurationHelper - HV000234: Using org.hibernate.validator.internal.eng
ine.scripting.DefaultScriptEvaluatorFactory as ValidatorFactory-scoped script evaluator factory.
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.044 s - in tacos.web.api.Tc24ReorderValidationTest
[INFO] Results:
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman

Reorden normal:



Reorden con precio cambiado:



Repitiendo un reorden:

The screenshot shows a Postman test run for the endpoint `http://localhost:8080/api/orders/6abc5f2291dd543a0b33339f/reorder` using the `POST` method. The request body is a JSON object with the following fields:

```
{
  "paymentMethodId": "6abc5be791dd543a0b333394",
  "confirmPriceChange": true,
  "reorderKey": "postman-reorder-002"
}
```

The response is a `201 Created` status with a response time of 21 ms. The response body is a JSON object:

```
{
  "originalOrderId": "6abc5f2291dd543a0b33339f",
  "priceChanged": false,
  "requiresConfirmation": false,
  "originalTotal": 9.72,
  "currentTotal": 9.72,
  "currency": "USD",
  "message": "Existing idempotent reorder returned.",
  "order": {
    "id": "6abc5f3191dd543a0b3333a0",
    "placedAt": "2026-09-30T01:00:33.268+00:00",
    "deliveryName": "Usuario TC16",
    "deliveryStreet": "Calle Laboratorio 16",
    "deliveryCity": "Ciudad",
    "deliveryState": "ST",
    "deliveryZip": "00000",
    "items": [
      {
        "taco": {
          "id": null,
          "createdAt": "2026-09-30T01:00:33.268+00:00",

```

Probar orden con otro usuario:

The screenshot shows a Postman test run for the endpoint `http://localhost:8080/api/orders/6abc5f8a91dd543a0b3333a1/reorder` using the `POST` method. The `Authorization` tab is selected, showing `Basic Auth` with the username `user-b` and password `{{adminPass}}`. The response is a `404 Not Found` status with a response time of 97 ms. The response body is a JSON object:

```
{
  "type": "urn:tacocloud:problem:not-found",
  "title": "Resource not found",
  "status": 404,
  "detail": "Order was not found.",
  "instance": "/api/orders/6abc5f8a91dd543a0b3333a1/reorder",
  "code": "ORDER_NOT_FOUND",
  "violations": []
}
```

Laboratorio 5 – Cocina y mensajería confiable

TC-25 – Flujo de estados de una orden

Se establecieron dos nuevos endpoints:

- PATCH /api/orders/{orderId}/status
- POST /api/orders/{orderId}/cancel

Cambios/modificaciones de archivos:

- OrderStatus.java
- OrderStatusHistory.java
- TacoOrder.java
- OrderWorkflowService.java
- OrderWorkflowController.java
- OrderStatusRequest.java
- OrderStatusHistoryResponse.java

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.OrderWorkflowServiceTest
20:17:53.361 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.22 s - in tacos.web.api.OrderWorkflowServiceTest
[INFO] Results:
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] -----< sia:tacocloud-ui >-----
[INFO] Building tacocloud-ui 0.0.17-SNAPSHOT [7/17]
[INFO] from tacocloud-ui\pom.xml

Spring Boot-TacoCloudApplication<tacocloud_tc14> (Taco-Cloud-Practice) Live Share Java: Warning Ln 58, Col 27 Spaces: 4 UTF-8 LF
```

Postman

Crear orden:

Pruebas TacoCloud > TC-25 - Flujos de estados de una orden > **postOrder**

POST http://localhost:8080/api/orders

Docs Params Authorization Headers 11 **Body** Scripts Settings

none form-data x-www-form-urlencoded **raw** binary GraphQL JSON

```
1 {
2   "deliveryName": "Usuario Lab 5",
3   "deliveryStreet": "Calle Laboratorio 5",
4   "deliveryCity": "Ciudad",
5   "deliveryState": "ST",
6   "deliveryZip": "00000",
7   "paymentMethodId": "6abc7cd2df7c627f70497185",
8   "items": [
9     {
10      "taco": {
11        "name": "Taco Lab 5",
12        "ingredientIds": ["FLT0", "CARN"]
13      }
14    }
15  ]
16 }
```

Body Cookies 1 Headers 12 Test Results **201 Created** 530 ms

{ } JSON Preview Visualize

```
62 "couponCode": null,
63 "status": "CREATED",
64 "statusHistory": [
65   {
66     "from": null,
67     "to": "CREATED",
68     "actor": "user-a",
69     "occurredAt": "2026-09-30T03:07:13.093+00:00",
70     "origin": "API",
71     "reason": "Order created"
72   }
73 ],
74 "stationId": null
75 }
```

Transición válida:

Pruebas TacoCloud > TC-25 - Flujos de estados de una orden > **patchOrder**

PATCH http://localhost:8080/api/orders/6abc7ce1df7c627f70497186/status

Docs Params Authorization Headers 11 **Body** Scripts Settings

none form-data x-www-form-urlencoded **raw** binary GraphQL JSON

```
1 {
2   "status": "PREPARING",
3   "reason": "Inicio de preparación"
4 }
```

Body Cookies 1 Headers 12 Test Results **200 OK** 43 ms

{ } JSON Preview Visualize

```
78 "origin": "KITCHEN_CLAIM",
79 "reason": "Claimed from FIFO queue"
80 },
81 {
82   "from": "ACCEPTED",
83   "to": "PREPARING",
84   "actor": "kitchen-test",
85   "occurredAt": "2026-09-30T03:29:30.966+00:00",
86   "origin": "KITCHEN",
87   "reason": "Inicio de preparación"
88 }
89 ],
90 "stationId": "kitchen-test"
91 }
```

Transición inválida:

Pruebas TacoCloud > TC-25 - Flujos de estados de una orden > **patchOrder**

PATCH `http://localhost:8080/api/orders/6abc7ce1df7c627f70497186/status`

Docs Params Authorization Headers 11 **Body** Scripts Settings

☐ none ☐ form-data ☐ x-www-form-urlencoded ☒ raw ☐ binary ☐ GraphQL **JSON**

```
1 {
2   "status": "PREPARING",
3   "reason": "Inicio de preparación"
4 }
```

Body Cookies 1 Headers 12 Test Results **409 Conflict** · 177 ms

{ } JSON Preview Visualize

```
1 {
2   "type": "urn:tacocloud:problem:conflict",
3   "title": "Conflict",
4   "status": 409,
5   "detail": "The requested order status transition is not allowed.",
6   "instance": "/api/orders/6abc7ce1df7c627f70497186/status",
7   "code": "INVALID_ORDER_STATUS_TRANSITION",
8   "violations": []
9 }
```

Cancelar orden:

Pruebas TacoCloud > TC-25 - Flujos de estados de una orden > **postOrder_cancel**

POST `http://localhost:8080/api/orders/6abc8017df7c627f70497188/cancel`

Docs **Params** Authorization Headers 11 **Body** Scripts Settings

Body Cookies 1 Headers 12 Test Results **200 OK** · 75 ms

{ } JSON Preview Visualize

```
54   "unitPriceAtPurchase": 3.75,
55   "subtotal": 7.50
56 },
57 ],
58 "subtotalBeforeDiscount": 7.50,
59 "discountAmount": 0.00,
60 "total": 7.50,
61 "currency": "USD",
62 "couponCode": null,
63 "status": "CANCELLED",
64 "statusHistory": [
65   {
66     "from": null,
67     "to": "CREATED",
68     "actor": "user-a",
69     "occurredAt": "2026-09-30T03:20:55.360+00:00",
70     "origin": "API",
71     "reason": "Order created"
72   },
73   {
74     "from": "CREATED",
75     "to": "CANCELLED",
76     "actor": "user-a",
77     "occurredAt": "2026-09-30T03:33:35.540+00:00",
78     "origin": "CUSTOMER",
79     "reason": "Cancelled by customer"
80   }
81 ],
82 "stationId": null
83 }
```

TC-26 – Cola de cocina, claim atómico y tiempo estimado

Se implementó una cola de cocina formada únicamente por órdenes en estado CREATED, ordenadas mediante FIFO estable por placedAt e identificador.

Cambios/modificaciones de archivos:

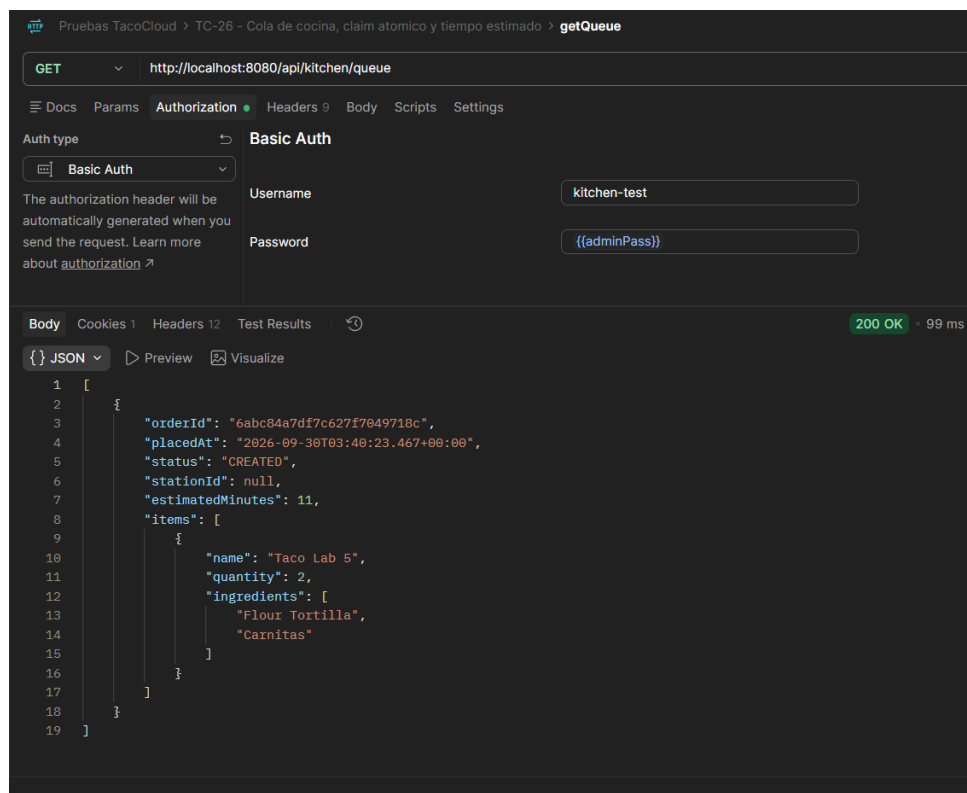
- KitchenQueueService.java
- KitchenQueueController.java
- KitchenOrderResponse.java
- KitchenOrderItemResponse.java
- kitchen.html

Test

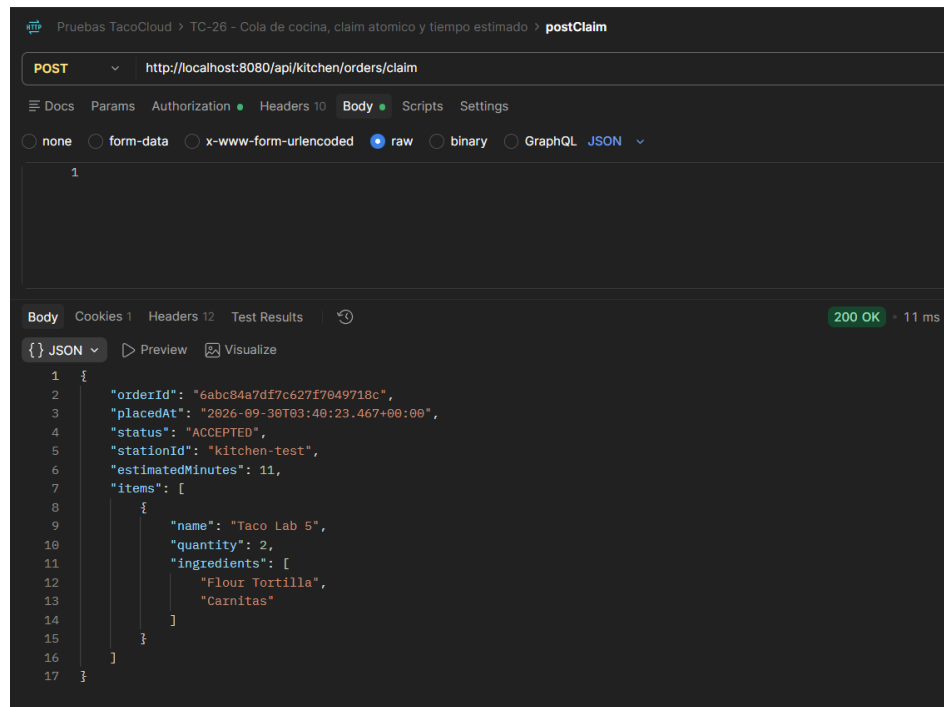
```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.KitchenQueueServiceTest
22:59:01.408 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.823 s - in tacos.web.api.KitchenQueueServiceTest
[INFO] Results:
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
```

Postman

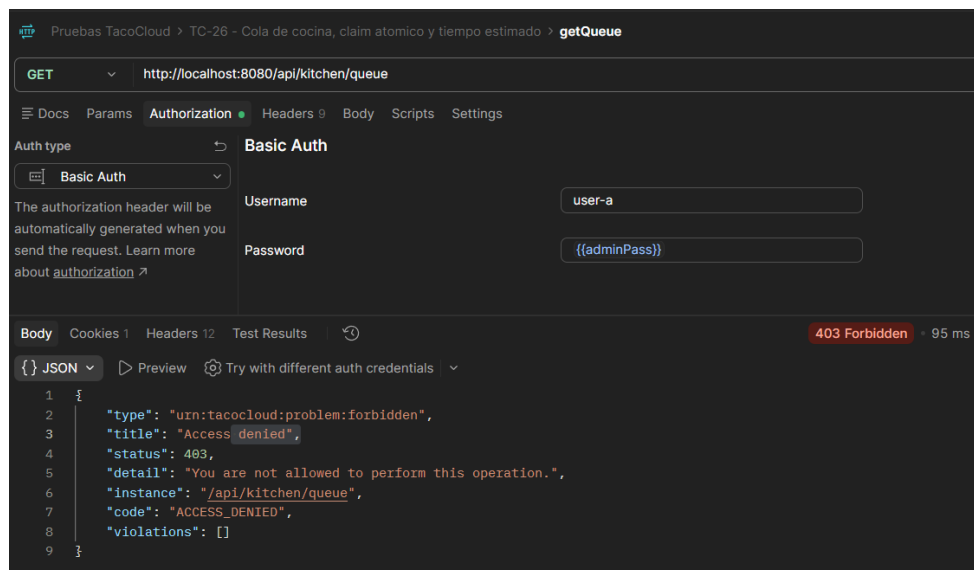
Ver en espera:



Aceptando el pedido:



Obteniendo la cola de espera como usuario:



TC-27 – Contrato único de eventos de orden

Se creó el nuevo módulo Maven `tacocloud-messaging-contract`, independiente de Spring, MongoDB y los brokers específicos. Este módulo centraliza el contrato compartido de mensajería mediante `OrderEvent`, `OrderEventType`, `OrderEventPayload` y `OrderMessagingService`.

Se crearon/modificaron los archivos en tacocloud-messaging-contract/

- OrderEvent.java
- OrderEventItem.java
- OrderEventPayload.java
- OrderEventType.java
- OrderMessagingService.java

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.messaging.OrderEventContractTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.762 s - in tacos.messaging.OrderEventContractTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
```

TC-28 – Elegir bróker en runtime, no editando el POM

Se eliminó la necesidad de editar el POM para cambiar el transporte de mensajería. La implementación utiliza ahora la propiedad `tacocloud.messaging.transport` para seleccionar entre `noop`, `jms`, `rabbit` y `kafka`.

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.Tc28MessagingTransportConfigurationTest
23:03:50.643 [main] DEBUG org.springframework.context.annotation.AnnotationConfigApplicationContext - Refreshing org.springframework.context.annotation.AnnotationConfigApplicationContext@e29591c
23:03:50.680 [main] DEBUG org.springframework.beans.factory.support.DefaultListableBeanFactory - Creating shared instance of singleton bean 'org.springframework.context.annotation.internalConfigurationAnnotationProcessor'
23:03:51.000 [main] DEBUG org.springframework.beans.factory.support.DefaultListableBeanFactory - Creating shared instance of singleton bean 'org.springframework.context.event.internalEventListenerProcessor'
23:03:51.008 [main] DEBUG org.springframework.beans.factory.support.DefaultListableBeanFactory - Creating shared instance of singleton bean 'org.springframework.context.event.internalEventListenerFactory'
23:03:51.011 [main] DEBUG org.springframework.beans.factory.support.DefaultListableBeanFactory - Creating shared instance of singleton bean 'org.springframework.context.annotation.internalAutowiredAnnotationProcessor'
23:03:51.018 [main] DEBUG org.springframework.beans.factory.support.DefaultListableBeanFactory - Creating shared instance of singleton bean 'org.springframework
PS C:\Users\luisa\Documents\WITDATA\Taco-Cloud-Practice\tacocloud> mvn test -Dtest=Tc28MessagingTransportConfigurationTest -DfailIfNoTests=false
23:03:53.517 [main] DEBUG org.springframework.beans.factory.support.DefaultListableBeanFactory - Creating shared instance of singleton bean 'jmsOrderMessagingService'
23:03:53.659 [main] DEBUG org.springframework.core.env.PropertySourcesPropertyResolver - Found key 'tacocloud.messaging.destination' in PropertySource 'configurationProperties' with value of type String
23:03:53.659 [main] DEBUG org.springframework.beans.factory.support.DefaultListableBeanFactory - Autowiring by type from bean name 'jmsOrderMessagingService' via constructor to bean named 'jmsTemplate'
23:03:53.661 [main] DEBUG org.springframework.context.annotation.AnnotationConfigApplicationContext - Closing org.springframework.context.annotation.AnnotationConfigApplicationContext@e11ecfa, started on Tue Sep 29 23:03:53 CST 2026
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 3.612 s - in tacos.Tc28MessagingTransportConfigurationTest
[INFO] Results:
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
```

TC-29 – Outbox transaccional para no perder órdenes

Se implementó el patrón Transaccional Outbox para eliminar la ventana de fallo existente entre guardar una orden y publicar su evento. La creación de una orden y el registro de su OutboxEvent forman ahora parte de la misma transacción MongoDB

Archivos creados/modificados:

- OutboxEvent.java
- OutboxStatus.java
- OutboxEventRepository.java
- OrderOutboxService.java
- OutboxPublisher.java
- OutboxScheduler.java
- MongoTransactionConfiguration.java
- OrderService.java

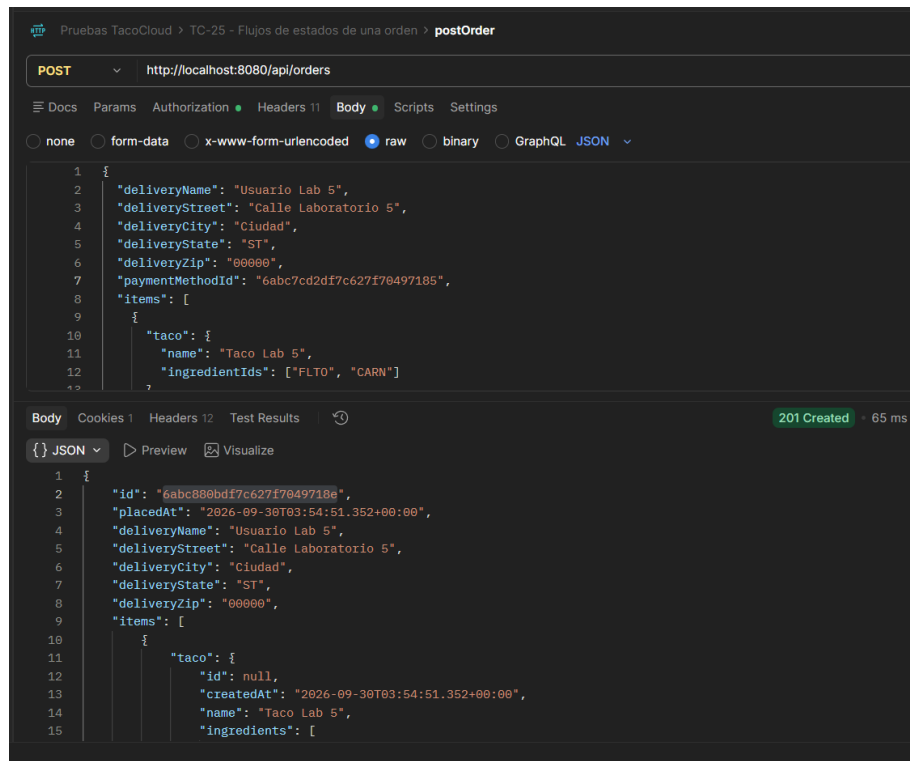
Test

```
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 30.626 s - in tacos.web.api.Tc29MongoOutboxTransactionTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

```
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.web.api.OutboxPublisherTest
23:08:59.113 [main] DEBUG reactor.util.Loggers - Using Slf4j logging framework
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.471 s - in tacos.web.api.OutboxPublisherTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman

Se crea una orden:



Estado en outbox_events:

```

_id: ObjectId('6abc880bdf7c627f7049718f')
eventId: "57a38d0f-3c4e-477d-b51e-116725afcd87"
eventType: "CREATED"
eventVersion: 1
aggregateId: "6abc880bdf7c627f7049718e"
correlationId: "2c6e7974-af40-4294-ac7f-c37eac8e665"
payload: "{\"eventId\":\"57a38d0f-3c4e-477d-b51e-116725afcd87\",\"type\":\"CREATED\",\"version\":1,\"occurredAt\":\"2026-09-30T03:54:51.382+00:00\",...\"status\":\"PUBLISHED\"}
attempts: 0
createdAt: ISODate('2026-09-30T03:54:51.382+00:00') 30 de septiembre de 2026 a las 3:54:51 a.m. UTC
updatedAt: ISODate('2026-09-30T03:54:52.044+00:00') 30 de septiembre de 2026 a las 3:54:52 a.m. UTC
nextAttemptAt: ISODate('2026-09-30T03:54:51.382+00:00') 30 de septiembre de 2026 a las 3:54:51 a.m. UTC
_class: "tacos.OutboxEvent"
publishedAt: ISODate('2026-09-30T03:54:52.044+00:00') 30 de septiembre de 2026 a las 3:54:52 a.m. UTC

```

TC-30 – Consumidor idempotente, retry limitado y DLQ

Se implementó un consumidor idempotente para los eventos de orden. La deduplicación utiliza eventId como identificador único, no orderId, y los eventos procesados quedan registrados mediante ProcessedEvent.

Archivos creados/modificados:

- KitchenEventProcessor.java
- ProcessedEvent.java
- ProcessedEventRepository.java
- KitchenOrderView.java
- Rabbit OrderListener

- Rabbit MessagingConfig
- KitchenMongoTransactionConfiguration.java

Test

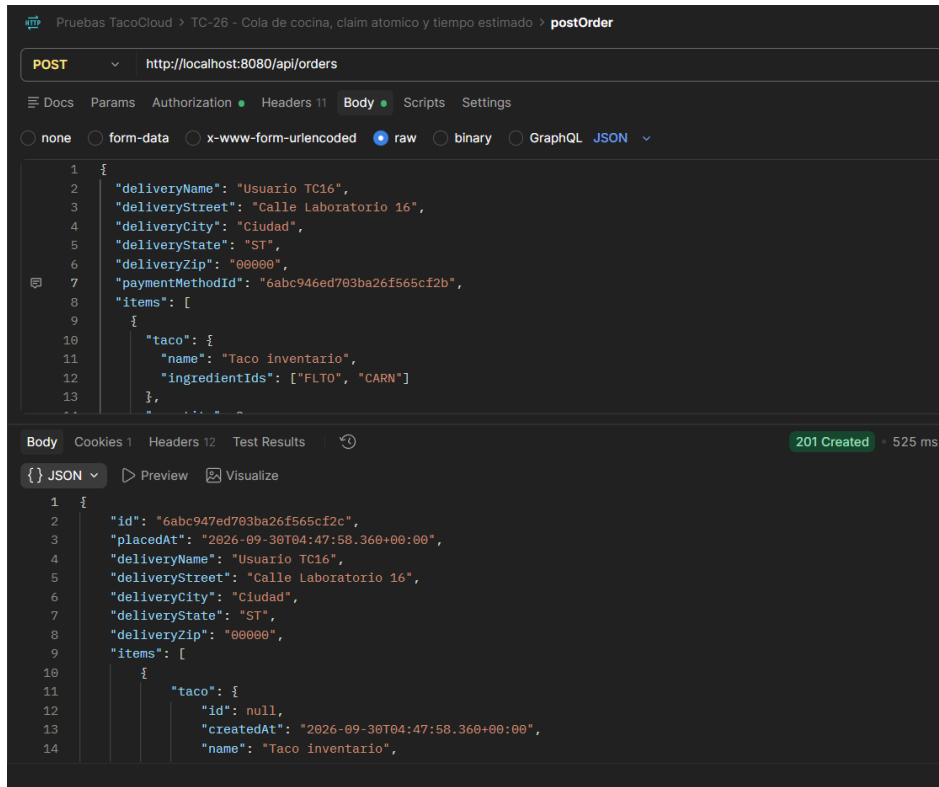
```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.kitchen.KitchenEventProcessorTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.918 s - in tacos.kitchen.KitchenEventProcessorTest
[INFO] Results:
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.kitchen.RabbitOrderListenerTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.824 s - in tacos.kitchen.RabbitOrderListenerTest
[INFO] Results:
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

```
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 20.605 s - in tacos.kitchen.Tc30RabbitContractE2eTest
[INFO] Results:
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman

Se crea una orden:



Se puede ver en las colas de RabbitMQ:

RabbitMQ 3.11.28 Erlang 25.3.2.9

Refreshed 2026-09-29 22:52:29 Refresh every 5 seconds

Virtual host All

Cluster rabbit@961a2dc41de4 User guest Log out

Overview Connections Channels Exchanges **Queues** Admin

Queues

▼ All queues (3)

Pagination

Page 1 of 1 - Filter: Regex

Displaying 3 items, page size up to: 100

Overview					Messages			Message rates		
Virtual host	Name	Type	Features	State	Ready	Unacked	Total	incoming	deliver	get / ack
/	tacocloud.order.dlq	classic	D	idle	0	0	0			
/	tacocloud.order.queue	classic	D DLX	idle	0	0	0	0.00/s	0.00/s	0.00/s
/	tacocloud.order.queue.retry	classic	D TTL DLX DLK	idle	0	0	0			

➤ Add a new queue

HTTP API Documentation Tutorials New releases Commercial edition Commercial support Discussions Discord Slack Plugins GitHub

Overview Connections Channels Exchanges Queues Admin



Message rates last minute



Publish	0.00/s	Consumer	0.00/s	Get (auto ack)	0.00/s
Deliver (manual ack)	0.00/s	Redelivered	0.00/s	Get (empty)	0.00/s
Deliver (auto ack)	0.00/s	Get (manual ack)	0.00/s		

Details

Features	x-dead-letter-exchange: tacocloud.order.dlx	State	idle
	durable: true	Consumers	1
	queue storage version: 1	Consumer capacity	100%
Policy		Messages	0
Operator policy		Message body bytes	0 B
Effective policy definition		Process memory	11 KiB

Consumers (1)

Channel	Consumer tag	Ack required	Exclusive	Prefetch count	Active	Activity status	Arguments
172.17.0.1:52944 (1)	amq.ctag-CKGeId-ac8zZeCOYo1PLMA	*	*	250	*	up	

Bindings (1)

Laboratorio 6 – Operación y calidad

TC-31 – Correlation ID de HTTP a evento y logs

Se implementó el manejo de X-Correlation-Id desde la entrada HTTP hasta el procesamiento en cocina. El identificador se genera cuando no es proporcionado, se válida para evitar valores maliciosos, se devuelve en la respuesta y se propaga por el flujo de órdenes, outbox, eventos, RabbitMQ, retry, DLQ y Kitchen. También se añadió soporte de MDC y limpieza del contexto al finalizar cada petición.

Archivos creados/modificados:

- CorrelationIds.java
- CorrelationIdFilter.java
- OrderService.java
- OrderWorkflowService.java
- OutboxPublisher.java
- RabbitOrderMessagingService.java
- OrderListener.java

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.Tc31CorrelationIdFilterTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.233 s - in tacos.Tc31CorrelationIdFilterTest
[INFO] Results:
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0
[INFO] -----
[INFO] -----
```

Postman

Conservar un correlation ID:

Pruebas TacoCloud > TC-31 - Correlation ID de HTTP a evento y logs > **getCorrelation** Save Share

GET ▼ http://localhost:8080/api/v1/ingredients Send ▼

Docs Params Authorization **Headers 9** Body Scripts Settings Cookies

Headers ↔ 8 hidden

Key	Value	Description	Bulk Edit	Presets	...
✓ X-Correlation-Id	postman.CORR-31				
Key	Value	Description			

Body Cookies 1 **Headers 13** Test Results ↺ 200 OK · 32 ms · 2.06 KB 🌐 📄 Save Response ...

Key	Value
X-Correlation-Id	postman.CORR-31
Vary	Origin, Access-Control-Request-Method, Access-Control-Request-Headers
X-Content-Type-Options	nosniff
X-Xss-Protection	1; mode=block
Cache-Control	no-cache, no-store, max-age=0, must-revalidate
Pragma	no-cache
Expires	0
X-Frame-Options	SAMEORIGIN
Content-Type	application/json
Transfer-Encoding	chunked
Date	Wed, 30 Sep 2026 08:53:43 GMT
Keep-Alive	timeout=60
Connection	keep-alive

Globals Vault Tools ⚙️ 📄

Sin correlation ID:

Pruebas TacoCloud > TC-31 - Correlation ID de HTTP a evento y logs > **getCorrelation** Save Share

GET ▼ http://localhost:8080/api/v1/ingredients Send ▼

Docs Params Authorization **Headers 9** Body Scripts Settings Cookies

Headers ↔ 8 hidden

Key	Value	Description	Bulk Edit	Presets	...
X-Correlation-Id	postman.CORR-31				
Key	Value	Description			

Body Cookies 1 **Headers 13** Test Results ↺ 200 OK · 16 ms · 2.08 KB 🌐 📄 Save Response ...

Key	Value
X-Correlation-Id	2653b6be-cfc1-40c4-8a18-f14d34e3f730
Vary	Origin, Access-Control-Request-Method, Access-Control-Request-Headers
X-Content-Type-Options	nosniff
X-Xss-Protection	1; mode=block
Cache-Control	no-cache, no-store, max-age=0, must-revalidate
Pragma	no-cache
Expires	0
X-Frame-Options	SAMEORIGIN
Content-Type	application/json
Transfer-Encoding	chunked
Date	Wed, 30 Sep 2026 08:54:40 GMT
Keep-Alive	timeout=60
Connection	keep-alive

Globals Vault Tools ⚙️ 📄

TC-32 – Métricas y salud que explican el negocio

Se amplió la observabilidad del sistema mediante métricas de negocio, timers y un gauge del backlog del outbox. También se agregó un indicador de salud específico para el outbox, probes de liveness/readiness y configuración de Actuator con acceso restringido. Las métricas cubren órdenes, inventario, outbox y procesamiento de eventos en Kitchen.

Archivos creados/modificados:

- BusinessMetrics.java
- OutboxHealthIndicator.java
- application.yml
- OrderService.java
- OrderWorkflowService.java
- OrderListener.java
- SecurityConfig.java

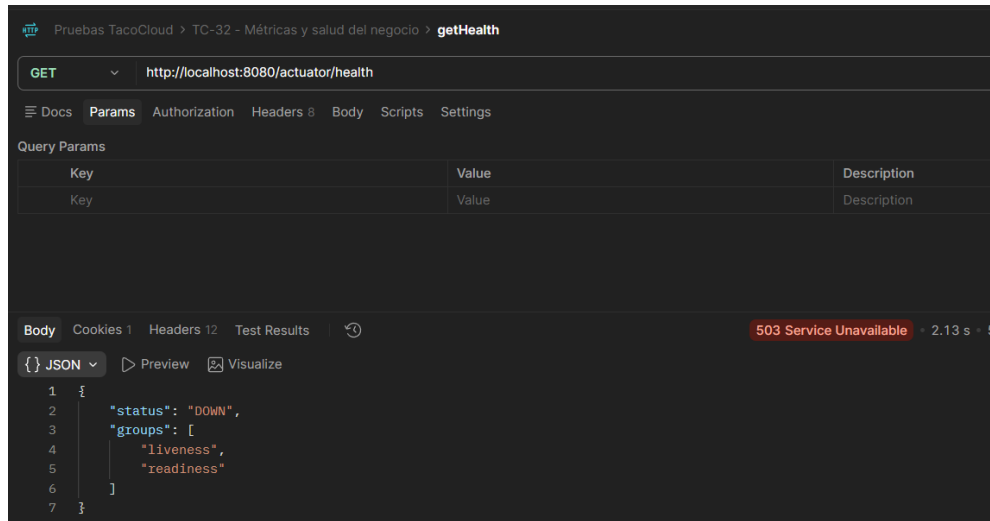
Test

```
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.Tc11AuthorizationIntegrationTest
02:33:23.022 [main] DEBUG org.springframework.test.context.BootstrapUtils - Instantiating CacheAwareContextLoaderDelegate from class [org.springframework.test.context.cache.DefaultCacheAwareContextLoaderDelegate]
02:33:23.061 [main] DEBUG org.springframework.test.context.BootstrapUtils - Instantiating BootstrapContext using constructor [public org.springframework.test.context.support.DefaultBootstrapContext(java.lang.Class,org.springframework.test.context.CacheAwareContextLoaderDelegate)]
02:33:23.161 [main] DEBUG org.springframework.test.context.BootstrapUtils - Instantiating TestContextBootstrapper for test class [tacos.Tc11AuthorizationIntegrationTest] from class [org.springframework.boot.test.context.SpringBootTestContextBootstrapper]
02:33:23.196 [main] INFO org.springframework.boot.test.context.SpringBootTestContextBootstrapper - Neither @ContextConfiguration nor @ContextHierarchy found for test class [tacos.Tc11AuthorizationIntegrationTest], using SpringBootContextLoader
02:33:23.208 [main] DEBUG org.springframework.test.context.support.AbstractContextLoader - Did not detect default resource location for test class [tacos.Tc11AuthorizationIntegrationTest]: class path resource [tacos/Tc11AuthorizationIntegrationTest-context.xml] does not exist

2026-09-30 02:33:55.081 INFO [corr:none] 29220 --- [ Thread-2] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-30T02:33:55.081-0600 I NETWORK [co
nn15] closing listening socket: 656
2026-09-30 02:33:55.081 INFO [corr:none] 29220 --- [ Thread-2] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-30T02:33:55.081-0600 I NETWORK [co
nn15] shutdown: going to flush diaglog...
2026-09-30 02:33:55.081 INFO [corr:none] 29220 --- [ Thread-2] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-30T02:33:55.081-0600 I FTDC [co
nn15] Shutting down full-time diagnostic data capture
2026-09-30 02:33:55.082 INFO [corr:none] 29220 --- [ Thread-2] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-30T02:33:55.082-0600 I STORAGE [co
nn15] WiredTigerKVEngine shutting down
2026-09-30 02:33:55.103 INFO [corr:none] 29220 --- [ Thread-2] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-30T02:33:55.103-0600 I STORAGE [co
nn15] shutdown: removing fs lock...
2026-09-30 02:33:55.103 INFO [corr:none] 29220 --- [ Thread-2] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-30T02:33:55.104-0600 I CONTROL [co
nn15] now exiting
2026-09-30 02:33:55.103 INFO [corr:none] 29220 --- [ Thread-2] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-30T02:33:55.104-0600 I CONTROL [co
nn15] shutting down with code:0
2026-09-30 02:33:55.103 INFO [corr:none] 29220 --- [ Thread-2] o.s.b.a.mongo.embedded.EmbeddedMongo : 2026-09-30T02:33:55.104-0600 I CONTROL [in
itandlisten] shutting down with code:0
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 22, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

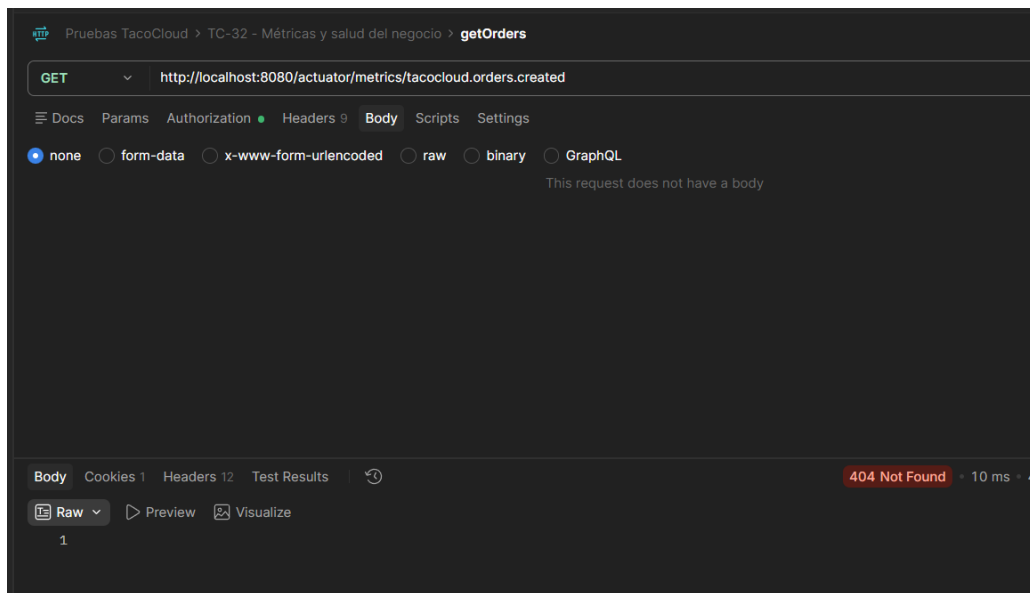
Postman

Obteniendo la salud de la aplicación:

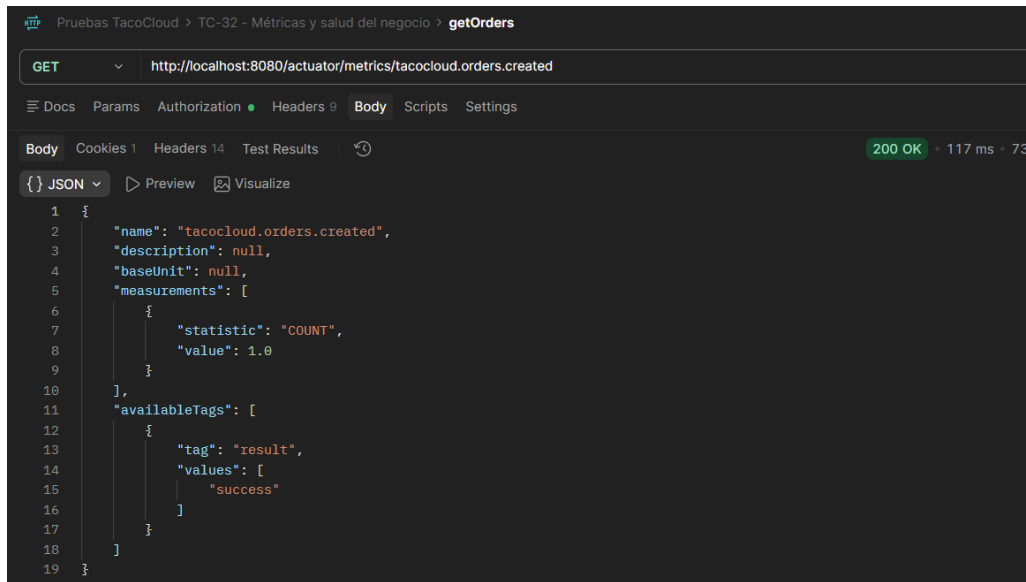


Métrica de errores:

Obteniendo ordenes (vacío)



Cuando ya se tiene al menos una orden



TC-33 – Reemplazar Notes por anuncios operativos seguros

Se sustituyó el antiguo endpoint de notas en memoria por un sistema de anuncios operativos persistentes en MongoDB. Se incorporaron identificadores estables, severidad, expiración, validación de contenido, límite de anuncios activos y control de concurrencia. La administración de anuncios quedó restringida a usuarios con rol ADMIN y la información interna del autor no se expone en las respuestas.

Archivos creados/modificados:

- OpsAnnouncement.java
- AnnouncementSeverity.java
- OpsAnnouncementRepository.java
- OpsAnnouncementService.java
- OpsAnnouncementController.java
- AnnouncementRequest.java
- AnnouncementResponse.java
- NotesEndpoint.java

Test

```

alive for 3 MINUTES
02:36:01.988 [main] DEBUG com.github.dockerjava.zerodep.shaded.org.apache.hc.client5.http.impl.classic.InternalHttpClient - ep-00000015: releasing valid endpoint
02:36:01.988 [main] DEBUG com.github.dockerjava.zerodep.shaded.org.apache.hc.client5.http.impl.io.PoolingHttpClientConnectionManager - ep-00000015: releasing endpoint
02:36:01.988 [main] DEBUG com.github.dockerjava.zerodep.shaded.org.apache.hc.client5.http.impl.io.PoolingHttpClientConnectionManager - ep-00000015: connection http-outgoing-2 can be kept alive for 3 MINUTES
02:36:01.988 [main] DEBUG com.github.dockerjava.zerodep.shaded.org.apache.hc.client5.http.impl.io.PoolingHttpClientConnectionManager - ep-00000015: connection released [route: {}->npipe://localhost:2375][total available: 1; route allocated: 1 of 2147483647; total allocated: 1 of 2147483647]
02:36:01.988 [main] DEBUG org.testcontainers.utility.ResourceReaper - Removed container and associated volume(s): mongo:4.4.29
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 5.028 s - in tacos.web.api.Tc33MongoAnnouncementIntegrationTest
[INFO] Running tacos.web.api.Tc33OpsAnnouncementServiceTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.1 s - in tacos.web.api.Tc33OpsAnnouncementServiceTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0
[INFO]

```

Postman

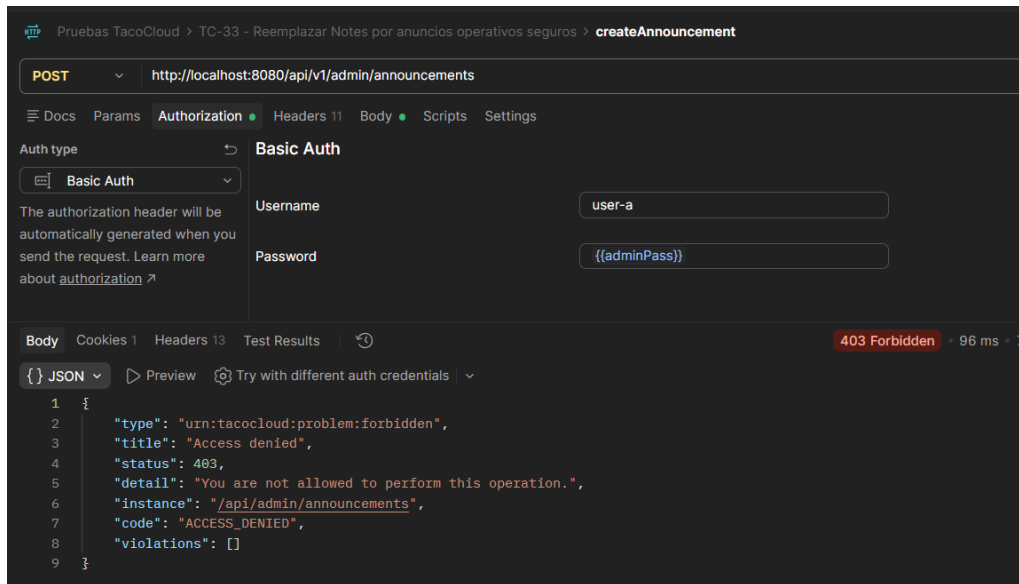
Creando un anuncio:

The screenshot shows a Postman interface for a REST client. The request is a POST to `http://localhost:8080/api/v1/admin/announcements`. The body is raw JSON: `{ "text": "Mantenimiento programado", "severity": "WARNING", "expiresAt": "2030-12-31T23:59:59Z" }`. The response is a 201 Created status with a response time of 79 ms. The response body is JSON: `{ "id": "6abcd0d087a4fa124a7cce7b", "text": "Mantenimiento programado", "severity": "WARNING", "createdAt": "2026-09-30T09:05:20.515Z", "expiresAt": "2030-12-31T23:59:59Z" }`.

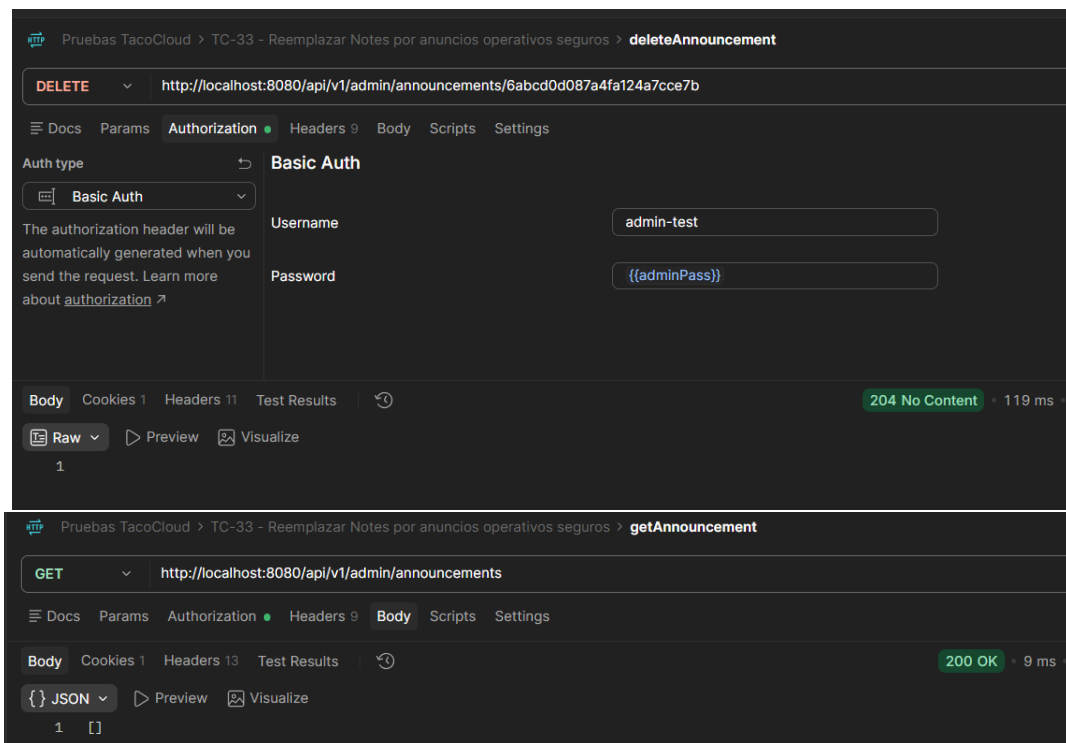
Obteniendo los anuncios:

The screenshot shows a Postman interface for a REST client. The request is a GET to `http://localhost:8080/api/v1/admin/announcements`. The response is a 200 OK status with a response time of 46 ms. The response body is JSON: `[ { "id": "6abcd0d087a4fa124a7cce7b", "text": "Mantenimiento programado", "severity": "WARNING", "createdAt": "2026-09-30T09:05:20.515Z", "expiresAt": "2030-12-31T23:59:59Z" } ]`.

Comprobando seguridad al evitar que los usuarios escriban anuncios:



Anuncio eliminado:



TC-34 – Idempotency Key en creación de órdenes

Se implementó idempotencia para la creación de órdenes mediante el header `Idempotency-Key`. Las solicitudes se identifican por usuario y clave, utilizando un hash SHA-256 calculado sobre una representación canónica del request. Una repetición de la misma solicitud recupera la misma orden, mientras que reutilizar la clave con un contenido distinto genera un conflicto 409. También

se añadió control de concurrencia, estados de procesamiento y coordinación con el flujo transaccional y outbox existente.

Archivos creados/modificados:

- IdempotencyRecord.java
- IdempotencyStatus.java
- IdempotencyRecordRepository.java
- CanonicalOrderHasher.java
- IdempotentOrderService.java
- OrderApiController.java
- OrderService.java
- OrderWorkflowService.java

Test

```
02:38:17.937 [main] DEBUG com.github.dockerjava.zerodep.shaded.org.apache.hc.client5.http.headers - http-outgoing-3 << Server: Docker/29.7.2 (linux)
02:38:17.938 [main] DEBUG com.github.dockerjava.zerodep.shaded.org.apache.hc.client5.http.impl.classic.MainClientExec - ex-00000025: connection can be kept alive for 3 MINUTES
02:38:17.938 [main] DEBUG com.github.dockerjava.zerodep.shaded.org.apache.hc.client5.http.impl.classic.InternalHttpClient - ep-00000024: releasing valid endpoint
02:38:17.938 [main] DEBUG com.github.dockerjava.zerodep.shaded.org.apache.hc.client5.http.impl.io.PoolingHttpClientConnectionManager - ep-00000024: releasing endpoint
02:38:17.938 [main] DEBUG com.github.dockerjava.zerodep.shaded.org.apache.hc.client5.http.impl.io.PoolingHttpClientConnectionManager - ep-00000024: connection http-outgoing-3 can be kept alive for 3 MINUTES
02:38:17.938 [main] DEBUG com.github.dockerjava.zerodep.shaded.org.apache.hc.client5.http.impl.io.PoolingHttpClientConnectionManager - ep-00000024: connection released [route: {}->pipe://localhost:2375][total available: 1; route allocated: 1 of 2147483647; total allocated: 1 of 2147483647]
02:38:17.938 [main] DEBUG org.testcontainers.utility.ResourceReaper - Removed container and associated volume(s): mongo:4.4.29
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.716 s - in tacos.web.api.Tc34MongoIdempotencyConcurrencyTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman

Creando una orden:

Pruebas TacoCloud > TC-34 - Idempotency Key en creación de órdenes > **postOrder**

POST ▼ <http://localhost:8080/api/v1/orders>

Docs Params Authorization Headers 13 Body Scripts Settings

Headers 11 hidden

Key	Value	Description
✓ Idempotency-Key	postman-order-0001	
✓ X-Correlation-Id	postman.ORDER-34	
Key	Value	Description

Body Cookies 1 Headers 13 Test Results 201 Created · 102 ms

{ } JSON ▼ Preview Visualize

```
1 {
2   "id": "6abcd33787a4fa124a7cce83",
3   "placedAt": "2026-09-30T09:15:35.299+00:00",
4   "deliveryName": "Usuario TC34",
5   "deliveryStreet": "Calle Laboratorio 6",
6   "deliveryCity": "Ciudad",
7   "deliveryState": "ST",
8   "deliveryZip": "00000",
9   "items": [
10    {
11      "taco": {
12        "id": null,
13        "createdAt": "2026-09-30T09:15:35.299+00:00",
14        "name": "Taco TC34",
15        "ingredients": [
16          {
17            "id": "FLT0",
18            "name": "Flour Tortilla",
19            "type": "WRAP",
20            "unitPrice": 1.25,
21            "available": true,
```

Si se Vuelve a realizar una orden, el ID no cambia:

Pruebas TacoCloud > TC-34 - Idempotency Key en creación de órdenes > **postOrder**

POST ▼ <http://localhost:8080/api/v1/orders>

Docs Params Authorization Headers 13 Body Scripts Settings

Headers 11 hidden

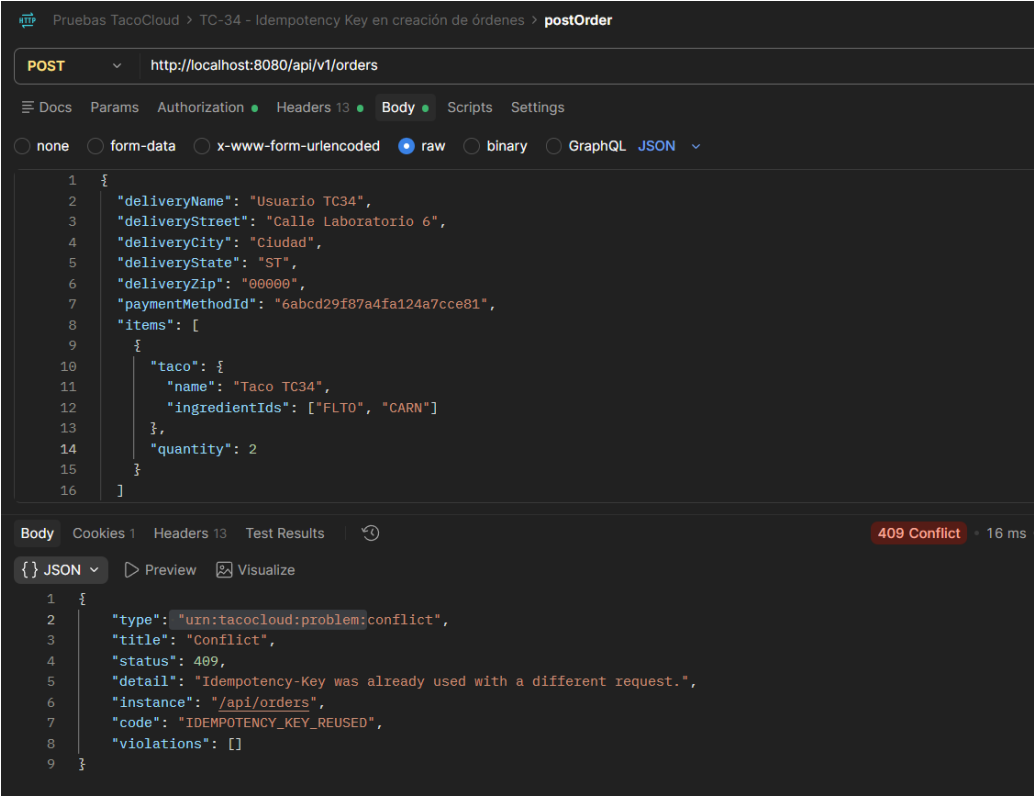
Key	Value	Description
✓ Idempotency-Key	postman-order-0001	
✓ X-Correlation-Id	postman.ORDER-34	
Key	Value	Description

Body Cookies 1 Headers 13 Test Results 201 Created · 19 ms

{ } JSON ▼ Preview Visualize

```
1 {
2   "id": "6abcd33787a4fa124a7cce83",
3   "placedAt": "2026-09-30T09:15:35.299+00:00",
4   "deliveryName": "Usuario TC34",
5   "deliveryStreet": "Calle Laboratorio 6",
6   "deliveryCity": "Ciudad",
7   "deliveryState": "ST",
8   "deliveryZip": "00000",
9   "items": [
10    {
11      "taco": {
12        "id": null,
13        "createdAt": "2026-09-30T09:15:35.299+00:00",
14        "name": "Taco TC34",
15        "ingredients": [
16          {
17            "id": "FLT0",
18            "name": "Flour Tortilla",
19            "type": "WRAP",
20            "unitPrice": 1.25,
21            "available": true,
```

Si se realiza un cambio en el contenido:



The screenshot shows a REST client interface with a POST request to `http://localhost:8080/api/v1/orders`. The request body is a JSON object with delivery details and a list of items. The response is a 409 Conflict status with a detailed error message.

```
1 {
2   "deliveryName": "Usuario TC34",
3   "deliveryStreet": "Calle Laboratorio 6",
4   "deliveryCity": "Ciudad",
5   "deliveryState": "ST",
6   "deliveryZip": "00000",
7   "paymentMethodId": "6abcd29f87a4fa124a7cce81",
8   "items": [
9     {
10      "taco": {
11        "name": "Taco TC34",
12        "ingredientIds": ["FLT0", "CARN"]
13      },
14      "quantity": 2
15    }
16  ]
17 }
```

```
1 {
2   "type": "urn:tacocloud:problem:conflict",
3   "title": "Conflict",
4   "status": 409,
5   "detail": "Idempotency-Key was already used with a different request.",
6   "instance": "/api/orders",
7   "code": "IDEMPOTENCY_KEY_REUSED",
8   "violations": []
9 }
```

TC-35 – Versionar la API y publicar contrato OpenApi

Se introdujo la versión pública `/api/v1/**` conservando temporalmente la compatibilidad con `/api/**`. Las rutas antiguas son marcadas mediante headers de deprecación y se añadió un contrato OpenAPI 3.0.3 que documenta los endpoints disponibles, autenticación, DTOs, errores, paginación y los headers `X-Correlation-Id` e `Idempotency-Key`. El contrato evita exponer entidades Mongo y datos sensibles.

Archivos creados/modificados:

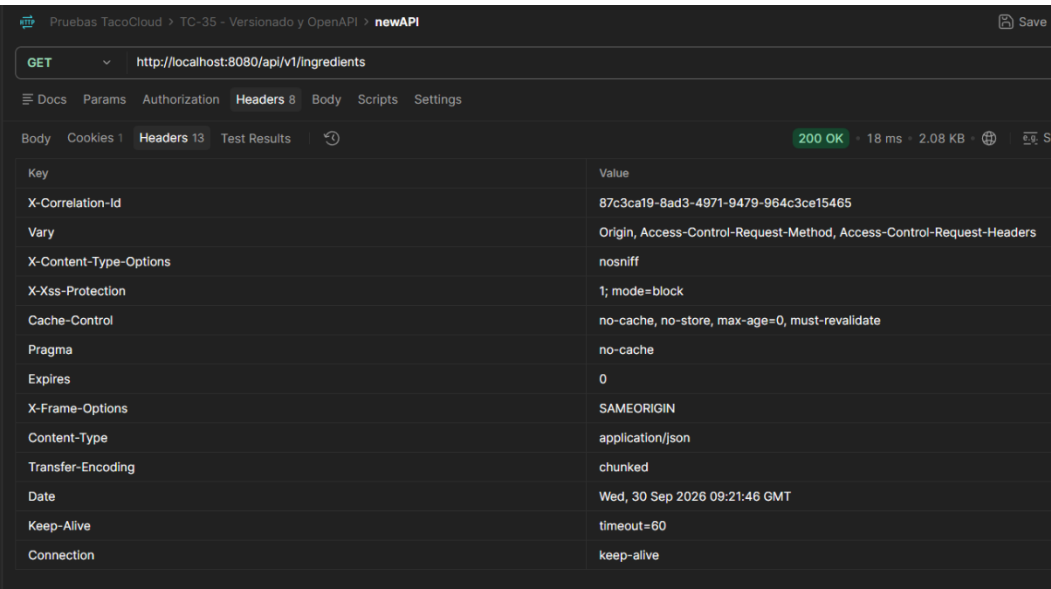
- `ApiVersionAliasFilter.java`
- `openapi.yaml`
- `ERROR_CONTRACT.md`
- `SecurityConfig.java`

Test

```
[INFO]
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.Tc35ApiContractTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.355 s - in tacos.Tc35ApiContractTest
[INFO] Results:
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

Postman

Versión nueva de API:



Versión antigua de API:

Pruebas TacoCloud > TC-35 - Versionado y OpenAPI > **oldAPI**

GET http://localhost:8080/api/ingredients Send

Docs Params Authorization Headers 8 Body Scripts Settings Cookies

Body Cookies 1 Headers 15 Test Results 200 OK · 11 ms · 2.14 KB Save Response

Key	Value
X-Correlation-Id	867f9d60-31d1-45a4-8c4a-aa3192e7e8d2
Deprecation	true
Link	</api/v1>; rel=successor-version
Vary	Origin, Access-Control-Request-Method, Access-Control-Request-Headers
X-Content-Type-Options	nosniff
X-Xss-Protection	1; mode=block
Cache-Control	no-cache, no-store, max-age=0, must-revalidate
Pragma	no-cache
Expires	0
X-Frame-Options	SAMEORIGIN
Content-Type	application/json
Transfer-Encoding	chunked
Date	Wed, 30 Sep 2026 09:22:52 GMT
Keep-Alive	timeout=60
Connection	keep-alive

OpenAPI:

Pruebas TacoCloud > TC-35 - Versionado y OpenAPI > **openAPI**

GET http://localhost:8080/openapi.yaml

Docs Params Authorization Headers 8 Body Scripts Settings

Body Cookies 1 Headers 15 Test Results 200 OK · 37 ms · 12.29 KB

Raw Preview Visualize

```
1 openapi: 3.0.3
2 info:
3   title: TacoCloud API
4   version: 1.0.0
5   description: Public v1 contract. Equivalent /api routes are legacy aliases and return Deprecation and Link headers.
6 servers:
7   - url: http://localhost:8080/api/v1
8 security:
9   - basicAuth: []
10 tags:
11   - {name: Ingredients}
12   - {name: Tacos}
13   - {name: Orders}
14   - {name: Favorites}
15   - {name: History}
16   - {name: Kitchen}
17   - {name: Operations}
18 components:
19   securitySchemes:
20     basicAuth: {type: http, scheme: basic}
21   parameters:
22     CorrelationId:
23       name: X-Correlation-Id
24       in: header
25       required: false
26       schema: {type: string, maxLength: 64, pattern: '^[A-Za-z0-9._-]+$'}
27   IdempotencyKey:
28     name: Idempotency-Key
29     in: header
30     required: false
```

TC-36 – Suite de integración que detenga regresiones reales

Se amplió la suite de pruebas para cubrir regresiones de los retos anteriores y los nuevos cambios del Laboratorio 6. Se incorporaron pruebas unitarias, integración con MongoDB mediante Testcontainers, pruebas de seguridad, contrato OpenAPI, concurrencia de idempotencia y una prueba E2E real con RabbitMQ. La suite quedó diseñada para no depender de los contenedores Mongo o Rabbit ejecutados manualmente.

Archivos creados/modificados:

- Tc31CorrelationIdFilterTest.java
- Tc32BusinessMetricsTest.java
- Tc32OutboxHealthIndicatorTest.java
- Tc33OpsAnnouncementServiceTest.java
- Tc33MongoAnnouncementIntegrationTest.java
- Tc34IdempotentOrderServiceTest.java
- Tc34MongoldempotencyConcurrencyTest.java
- Tc35ApiContractTest.java
- Tc29MongoOutboxTransactionTest.java
- Tc30RabbitContractE2eTest.java
- Tc11AuthorizationIntegrationTest.java

Test

```
[INFO] -----
[INFO] T E S T S
[INFO] -----
[INFO] Running tacos.PaymentDataModelTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.403 s - in tacos.PaymentDataModelTest
[INFO] Results:
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
[INFO]
```

```

ue:5]] to localhost:54172
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 5.987 s - in tacos.web.api.InventoryConcurrencyIntegrationTest
[INFO] Running tacos.web.api.KitchenQueueServiceTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.4 s - in tacos.web.api.KitchenQueueServiceTest
[INFO] Running tacos.web.api.OrderApiControllerTest
[INFO] Tests run: 8, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.538 s - in tacos.web.api.OrderApiControllerTest
[INFO] Running tacos.web.api.OrderPricingServiceTest
[INFO] Tests run: 6, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.026 s - in tacos.web.api.OrderPricingServiceTest
[INFO] Running tacos.web.api.OrderServiceTest
[INFO] Tests run: 14, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.157 s - in tacos.web.api.OrderServiceTest
[INFO] Running tacos.web.api.OrderWorkflowServiceTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.027 s - in tacos.web.api.OrderWorkflowServiceTest
[INFO] Running tacos.web.api.OutboxPublisherTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.043 s - in tacos.web.api.OutboxPublisherTest
[INFO] Running tacos.web.api.PaymentMethodControllerTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.126 s - in tacos.web.api.PaymentMethodControllerTest
[INFO] Running tacos.web.api.PaymentMethodServiceTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.014 s - in tacos.web.api.PaymentMethodServiceTest
[INFO] Running tacos.web.api.TacoClassificationServiceTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.001 s - in tacos.web.api.TacoClassificationServiceTest
[INFO] Running tacos.web.api.TacoControllerTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.195 s - in tacos.web.api.TacoControllerTest
[INFO] Running tacos.web.api.Tc12SensitiveDataMigrationControllerTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.006 s - in tacos.web.api.Tc12SensitiveDataMigrationControllerTest
[INFO] Running tacos.web.api.Tc19CatalogValidationTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.012 s - in tacos.web.api.Tc19CatalogValidationTest
[INFO] Running tacos.web.api.Tc20TacoOfTheDayServiceTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.023 s - in tacos.web.api.Tc20TacoOfTheDayServiceTest
[INFO] Running tacos.web.api.Tc21FavoriteServiceTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.006 s - in tacos.web.api.Tc21FavoriteServiceTest
[INFO] Running tacos.web.api.Tc22RatingValidationTest
[INFO] Tests run: 5, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.044 s - in tacos.web.api.Tc22RatingValidationTest
[INFO] Running tacos.web.api.Tc23OrderHistoryServiceTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.025 s - in tacos.web.api.Tc23OrderHistoryServiceTest
[INFO] Running tacos.web.api.Tc24ReorderValidationTest
[INFO] Tests run: 4, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.043 s - in tacos.web.api.Tc24ReorderValidationTest
2026-09-30 02:42:06.005 INFO 28592 --- [main] o.t.d.DockerClientProviderStrategy : Loaded org.testcontainers.dockerclient.M

```

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[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.52 s - in tacos.web.api.Tc29MongoOutboxTransactionTest
[INFO] Running tacos.web.api.Tc32BusinessMetricsTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.117 s - in tacos.web.api.Tc32BusinessMetricsTest
[INFO] Running tacos.web.api.Tc33MongoAnnouncementIntegrationTest

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```

ue:5]] to localhost:54245 because the pool has been closed.
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 2.375 s - in tacos.web.api.Tc33MongoAnnouncementIntegrationTest
[INFO] Running tacos.web.api.Tc33OpsAnnouncementServiceTest
[INFO] Tests run: 2, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.102 s - in tacos.web.api.Tc33OpsAnnouncementServiceTest
[INFO] Running tacos.web.api.Tc34IdempotentOrderServiceTest
[INFO] Tests run: 3, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.112 s - in tacos.web.api.Tc34IdempotentOrderServiceTest
[INFO] Running tacos.web.api.Tc34MongoIdempotencyConcurrencyTest

```

```

ue:5]] to localhost:54245 because the pool has been closed.
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 1.981 s - in tacos.web.api.Tc34MongoIdempotencyConcurrencyTest
2026-09-30 02:42:15.426 INFO 28592 --- [lionShutdownHook] org.mongodb.driver.connection : Closed connection [connectionId]lo

```

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[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 144, Failures: 0, Errors: 0, Skipped: 0
[INFO]

```